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A Project on

AstraCare - AI Based Women Health Tracker

Submitted by
Preet Jain – 261727
Tuhin Maji – 261725
Atharv Desai - 261707

Under the Guidance of
Ms. Rimsy Dua

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B.Sc. (IT)
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Thakur College of Science and Commerce,
Thakur Village, Kandivali(E),
Mumbai-400101

CERTIFICATE

This is to certify that the project titled "**AstraCare AI - AI Based Women Health Tracker**" undertaken at the "Thakur College of Science and Commerce" by Preet Jain - 261727, Tuhin Maji - 261725, and Atharv Desai - 261707, in partial fulfillment of B.Sc. IT degree Examination, has not been submitted for any other examination and does not form part of any other course undergone by the candidates.

It is further certified that the students have completed all required phases of the project

_____	_____
External Examiner	Internal Examiner
_____	_____
Project Guide	Head of Department

Thakur College of Science and Commerce,
Thakur Village, Kandivali(E),
Mumbai-400101

DECLARATION

I declare that this project titled "**AstraCare AI – AI Based Women's Health Tracker and Predictive Healthcare System**" submitted towards partial fulfillment of the requirements for the award of the Bachelor of Science (Information Technology) degree in the University of Mumbai is our own work done under the guidance of **Ms. Rimsy Dua.**

It is also declared that this project is our original work and has not been submitted in its entirety or in part to any university, institution or examination body for the award of any degree, diploma and/or any academic qualification.

We also declare that all information/data, references and other materials have been duly cited wherever required. We also confirm that to the best of our knowledge, there is no material in this project which violates any intellectual property rights of any person/organisation.

Name of Student	**Signature**
Preet Jain	_____
Tuhin Maji	_____
Atharv Desai	_____

Thakur College of Science and Commerce,
Thakur Village, Kandivali(E),
Mumbai-400101

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ABSTRACT

AstraCare AI: An AI-Based Women's Health Tracking & Healthcare System

Women's health is dependent on many factors such as reproductive health, lifestyle, diet, exercise, sleep and mental health. This can be many times difficult to track using multiple apps and manual ways. AstraCare AI aims to help women by providing with the neccessary information, data and tools to help manage their health.

This application helps women to track their overall health, menstrual cycle and ovulation. It also allows them to monitor their health, diet, exercise, sleep, hydration, mental health, reminders for medication, and digital health records. It also provides an AI Health Assistant and AI Health Intelligence Engine, which will help provide users with the right health information. The Health Dashboard helps users to view their key health information in an organised manner. The Digital Health Twin allows users to see a compilation of their health data.

This application stores and manages health data using cloud services and databases. It also has a wearable device called AstraBand that will be used to collect health and activity data using sensors. This will help expand the application in the future.

The project aim to help women to login, see and understand their health. By combining health tracking, AI assistance and a modular system, this provides a basis for a more personalised and informed health management system. This system is to help increase health awareness and should not be used as a substitute for professional health advice. The wearable device needs to be physically tested and there is potential for improvement in the disease prediction models.

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1. Introduction

1.1 Objective and Scope of the Project

The healthcare industry is changing faster through the integration of Artificial Intelligence (AI), Wearable Technology, Cloud Computing, and Prediction Analytics, but women's health is still left behind. Most wearable devices in the market focus on fitness tracking, daily health records, recovery analysis, etc., but they fail to address the unique physiological and healthcare requirements of women across different life stages. Women's health involves different type of factors like menstrual health, fertility, pregnancy, postpartum recovery, PCOD, menopause, PCOS, hormonal disbalance, stress, ovulation, sleep cycle, and long-term preventive care. So, women need technology that continuously monitors their health and gives intelligent insight about their body, not just small piece of information of daily steps and calorie burn, etc.

AstraCare AI is a health monitoring platform based on artificial intelligence technologies which is based on the intelligent wearable named AstraBand, mobile application and cloud-based healthcare system. The smart band device continuously takes different medical metrics including heartbeat, HRV, blood oxygen saturation (SpO2), ECG, skin temperature, sleep patterns, physical activity and motion data. Then, this information is sent via Bluetooth Low Energy (BLE) to the mobile application where it undergoes the analysis with artificial intelligence algorithms which help to get personalized healthcare recommendations and risk prediction.

The main purpose of this project is to build up a complete health ecosystem which would provide the possibility of continuous health monitoring, risk prediction, intelligent decision making and personalized healthcare guidance for women. In contrast to usual fitness applications, AstraCare AI focuses on preventive healthcare due to the use of continuous information gathering from the wearable and analysis with the help of artificial intelligence algorithms to find out health risks in advance.

Moreover, it helps to improve cooperation between patients and doctors through provision of all necessary information in one platform, including secure medical records, AI ready medical summaries, consultations with doctors, appointments scheduling and emergency alerts. Thus, the combination of smart wearable hardware with intelligent software provides the continuous health monitoring ecosystem.

Scope of the Project

The scope of AstraCare AI goes beyond wearable health monitoring. AstraCare AI incorporates a smart wearable band called AstraBand AI, artificial intelligence a cloud and a mobile health platform into one intelligent health ecosystem. The project aims to offer health monitoring, analytics powered by AI, predictive healthcare and personalized help for womens health. This section explains the proposed system in detail. Shows the main parts, how it works and the scope.

AstraCare AI is supported by a wearable band named AstraBand AI. AstraBand AI constantly watches signs and gathers data from many sensors. AstraBand AI collects heart rate, heart rate variability, SpO2, ECG, temperature, sleep data, activity level, movement, recovery status and stress level. The data from AstraBand AI travels over Bluetooth to the application.

Besides the data from AstraBand AI users can enter their details, symptoms, health history, medication, allergy, lifestyle, diet, hydration, mental health, family medical history, fertility status, pregnancy information, menopause status, lab results and other health details. All of this information helps to build a health twin for each women allowing continuous health check and predictive analytics of their health.

The data gathered from users is checked, processed and saved in the database. The cloud data trains the AI health analytics engine, which produces insights finds patterns calculates risk scores tracks health trends and creates models, with different algorithms.

Using the information from AstraBand AI and the data entered by the users, AstraCare AI will offer a variety of health services such as :

- Health and wellness tracking.
- Predictive healthcare.
- AI-powered ovulation and menstrual cycle tracking.
- Pregnancy, post-partum, and menopause support.
- Tracking of stress, sleep, and recovery.
- Personalized health and wellness coaching.
- AI-generated reports.
- Health twin.
- Doctor portal.
- Notification and emergency support.
- digital health wallet for storing health information and reports.

The ecosystem of AstraCare AI will be connected, with end-users being able to communicate with the wearable health band, the mobile application, the cloud, the AI engine, and the doctors. The proposed system architecture is scalable, and the platform can be connected to other health service providers such as hospitals, electronic health record (EHR) systems, diagnostic labs, smart medical devices, and other sensors.

The overall scope of AstraCare AI is to provide an integrated and intelligent women's health platform that offers a variety of services, including predictive analytics and a personalized health dashboard backed by AI, continuous health monitoring with the Astra Band AI wearable band, and a health ecosystem to support preventative healthcare and commercialize as a start-up healthcare platform.

1.2 Theoretical Background

Artificial Intelligence, gadgets, cloud computing and mobile health are changing how we do healthcare today. Traditional health care struggles to keep track of patients all the time and to give treatment. New wearable devices help by watching health all the time and using AI to predict and treat disease.

AstraCare AI will build a system that mixes a smart wearable (AstraBand AI) and AI to make an intelligent health system for women. The wearable will gather real-time data like heart rate, heart rate variability, blood oxygen, ECG, body temperature, sleep patterns, activity and movement. The data travels to the app via Bluetooth Low Energy. Goes to the cloud.

AI and machine learning will be tools in this system looking at sensor data and personal health info. In this health network AI will check health status make predictions, spot problems suggest improvement plans and give smart health advice. Unlike health apps that only show numbers or graphs AstraCare AI will look at the data and give useful preventive suggestions.

This health system uses cloud computing to give security, easier sync and quick access to personal health records. Both patients and doctors can see data whenever they need. The system will add a Doctor Portal, Digital Health Wallet, AI Clinical Report Generation, AI Personal Health Coach, Predictive Health Analysis and Digital Health Twin. These parts make monitoring, coaching and predicting health problems easier from a distance.

This project shows how AI and mobile health tools can join together to make preventive care for women. AstraCare AI will use wearables, AI and cloud computing to give a fuller health framework, for women improving monitoring, disease prediction and treatment advice.

1.3 Problem Definition

Over recent years health technology has advanced drastically with the inclusion of wearable technologies, AI and digital health applications. Unfortunately, most current commercial wearable technology focuses purely on fitness tracking. This is due to the lack of smart healthcare monitoring systems that have the ability to meet all the needs of female patients. Current wearable fitness bands track physiological parameters. For example, Heart Rate, Steps Per Day, Calories Burned and Sleep, however there is no healthcare analytics, forecasting or female specific healthcare.

Throughout each stage of a woman's life they will experience unique physiological changes relating to adolescence, menstrual periods, fertility, pregnancy, post-natal and menopause. These all impact on her heart, hormones, stress levels, sleep, body temperature and overall health. However, most wearable technologies do not consider these differences in their healthcare algorithms. Therefore they only provide general health insights and many potential signals will go unnoticed.

Traditionally in medicine, regular appointments and visits to hospital are required. This will only give partial information regarding someone's health and may cause delays to diagnosis and prevention. This will reduce the effectiveness of treatment.

Nowadays mobile healthcare applications are becoming increasingly popular, which require many manual inputs. This includes symptoms data, lifestyle information, menstrual periods and medications information. This can often be incorrect and inconsistent.

However with the rapid development of AI technologies this can be overcome by analysing the data from the device. This will extract meaningful health insights, through identifying patterns, anomalies, forecasting and recommendations. There is very little solutions that use AI and continuous physiological data to help improve women's healthcare.

To overcome this issues the proposed AI driven healthcare system, AstraCare will include two components. AstraBand AI wearable device and an AI powered mobile healthcare application. The wearable device will be able to monitor several key health factors including Heart Rate (HR), Heart Rate Variability (HRV), Blood Oxygen Saturation (SpO2), Electrocardiogram (ECG), Body Temperature, Sleep, Physical Activity and Motion. It will also allow users to input extra information relating to their health for example Menstrual Cycle, Symptoms, Health History and Medications.

This data will be sent to the cloud and analysed using the AI Health Intelligence Engine. This will include pre-processing data, extracting features, predicting, detecting anomalies and health assessments. This will produce intelligent health reports, preventative healthcare recommendations, personal health coaching, Digital Health Twin modelling, clinical summaries, emergency alerts and doctor support.

Overall the proposed AstraCare AI will bridge the gap between fitness tracking and preventative healthcare by incorporating wearable technologies, artificial intelligence, cloud computing and healthcare management specific to women..

1.4 Limitations

AstraCare AI would offer an integrated solution for women health, tracking, and AI analysis and prediction, there are some significant limitations are as follows:

1) This project is at prototype stage and would require further development to optimise it for real world applications.
2) The accuracy of the AI depends on the quality and quantity of data provided. The more relevant, complete and credible data there is the better the accuracy will be.
3) The insights and predictions given by the AI will not replace medical knowledge.
4) The smart band smould be weared continuously straight for 24x7 so the continuously monitoring feature works properly
5) The accuracy of the sensors depends on the variety of factors which affects the health.
6) A stable internet connection is required for AI and notifications.
7) The AI analytics and features may not be very useful in real world scenarios due to the availability of relevant data sets.
8) The security and encryption within the system will only provide a certain amount of security. Other factors such as cybersecurity should also be taken into consideration.
9) The system is mainly based around the proposed AstraBand design. Other features may be required for other wearable trackers.
10) This feature will not replace emergency medical services. It may just provide guidance.
11) The health assistant will provide relevant and useful information and suggestions however should not be seen as an alternative to a suitably qualified medical professional.
12) To implement this system on a large scale the system will require optimisation, cloud, databases, security, auditing and monitoring.

2. Requirement Analysis

2.1 User Requirements

User requirements identify the expectations and demands of the different parties associated with the AstraCare AI system. Requirements define what services should be provided by the system to accomplish continuous health monitoring, healthcare analysis, secure medical records management, and personalized healthcare. The following requirements are sorted according to main business functions of the system.

2.1.1 Stakeholders

The AstraCare AI ecosystem has several stakeholders who interact with the system while receiving intelligent healthcare services. Each stakeholder has different activities and responsibilities depending on their area of work and authority level

The main stakeholders are:

- **Women Users (Patients):**
  Users of AstraBand AI and the AstraCare AI mobile application who use the system to monitor their health status, get AI health reports, healthcare information, and advice. In addition, the users track their menstrual cycle, pregnancy status, and get reminders for health checkups.

- **Healthcare Professionals (Doctors):**
  Doctors who access the health reports of the patients, give medical advice, prescribe medication, and monitor patient progress through the Doctor Portal.

- **System Administrator:**
  The system administrator oversees the proper functioning of the AstraCare AI system. Some of the activities include managing users, managing wearables, managing the cloud, managing AI services, monitoring reports, and managing the security system.

- **AstraBand AI Wearable:**
  The AstraBand AI wearable scans and collects health data from users and uploads the information to the AstraCare AI ecosystem.

- **AI Health Intelligence Engine:**
  The AI engine is the central processing system that analyzes health data, predicts future health status, generates health reports, and recommends health options to the users.

2.1.2 Functional Requirements

The functional requirements define the business functions to be performed by the AstraCare AI system. They define the expected performance of the software system, wearable devices, artificial intelligence technology, cloud computing system, and health care services.

FR-1 User & Security Management:

This module would allow the user to do the following:

- Register and authenticate by using either a mobile number and OTP verification or email ID and password authentication. Forgot Password feature would allow for account recovery.
- Users can create, edit and manage their profile. This would include demographic information, profile picture, blood group, height, weight, emergency contact details, lifestyle information and many more.
- Medical information would include allergies, diseases, medication, family medical history, past medical issues and any other health related information required for AI analysis.
- Role based access rights for patients, doctors and administrators. All users would have access to features and healthcare information that correspond to their role.
- Users would be able to manage their profile, this would include managing devices that can be paired with the application, accounts, preferences, language preferences, notifications and profile synchronization on one page.
- All user account information including registration details, login history, wearable pairing history, profile changes and activity logs..

FR-2 Smart Wearable Device Management

- Enable connectivity of the new proposed AstraBand wearable to the AstraCare AI application using Bluetooth Low Energy (BLE). Secure connectivity, device authentication, disconnection, and synchronization will be among the functionalities supported by the system.
- Gather data that can be read by the proposed smart wearable device which includes heart rate (HR), heart rate variability (HRV), blood oxygenation level (SpO_2), ECG readings, skin temperature, movement, sleep data, and physical activity. Availability of these capabilities is conditioned upon the availability of respective sensors.
- Display important information related to the smart wearable device including battery level, charger status, firmware version, connectivity status, and the latest synchronization status.
- Send data collected from the wearable device to the application and also save it in the health records of the user when cloud synchronization capability is included in the system.

FR- 3 AI Health Intelligence Engine

- Retrieve and aggregate the health information shared by the user, which may come from connected devices.
- Look at the health indicators supported, for example, heart rate (HR), heart rate variability (HRV), oxygen saturation (SpO2), electrocardiography (ECG), skin temperature, sleep, physical activities, information relating to stress, and information relating to recovery, wherever applicable.
- Compute or visualize the wellness indicators, if any, derived out of those data. The explanation of how a certain score is obtained should be provided in the report. Nothing in the report should suggest any estimation as a medical measure.
- Detect any trend or pattern within the information of the patient. Detection of disease-related pattern needs proper data, model, and evaluation.
- Give general wellness insights and suggestions with respect to the patient records. It must never try to make diagnosis or replace healthcare professionals.
- Compare with the past data using historical records, wherever possible.
- Prediction and clinical decision support are considered future work unless the models have been made, validated and verified already.

FR- 4 Management of Women's Health

- Enable users to input dates of menstrual cycle and symptoms, predict the upcoming period date and see estimates of ovulation and fertile window dates. They are only predictions and may not be exact for all users.
- Offer an option to input pregnancy data, symptoms, appointment information and other health needs related to pregnancy in case if this function is implemented in the application.
- Make it possible for users to input recovery postpartum information like symptoms, wellbeing and feeding schedule.
- Enable users to input menopause symptoms and see general health information and suggestions regarding women's health.
- Keep track of reproductive health history like menstrual history, pregnancy history, fertility history, contraception history and gynaecological data based on the application's field of use.
- Give users reminders about periods, estimated ovulation, appointments, medications, vaccinations and preventive screening where reminder functions are implemented.
- Use the data that is input by the user to give reproductive health summaries and trends.
- Tell users to consult with a licensed healthcare professional in case of alarming symptoms. The application should not be used to diagnose reproductive health problems.

FR-5 AI Medical Intelligence Assistant

- Answer general questions about health using plain language.
- Give a simple explanation of supported health readings, test results that are uploaded by the user, health reports and commonly used medical terminology. It should be obvious for the user that the explanation is informative but not diagnostic.
- Offer general educational material on nutrition, exercise, stress management, sleep, healthy habits and lifestyle.
- Help users understand their input information and make recommendations on when to consult with healthcare professional.
- Provide basic emergency instructions and prompt users to contact emergency services in case if an emergency can be assumed. The assistant should not delay users from contacting emergency services.
- If multilingual capability is implemented then offer support in multiple languages.
- Personalise responses based on the saved preferences of the user where this feature is available.

FR-6 Intelligent Health Monitoring

- Display the supported health information such as HR, HRV, SpO_2, ECG, skin temperature, sleep, activity, information associated with stress and recovery. The real-time reading requires support from the tested data source.
- Analyze the available readings for any oddities and changes in them. Limits set up as alerts shall be carefully defined and should not be used for medical diagnoses.
- Provide summaries of the supported health information based on the readings collected by the system such as daily activity, sleep records and others.
- Show related health information together where it helps users in understanding the readings, but should not mislead into thinking that the system is capable of diagnosing a health condition.
- Keep the record of past health information as it helps users to go through them for comparison and trend analysis.
- Alert the user when there is crossing of predefined threshold by any readings or entries.
- Sync the records with cloud where cloud sync is allowed and tested.
- Provide the health information via dashboard or application screen.

FR-7 Lifestyle and Wellness Management

- Log the available daily activity information, such as number of steps, distance, exercise, sedentary time, and calories where this data is available from an available data source or input from the user.
- Allow the user to log meals, water intake, vitamins, and other nutritional information.
- Log sleep periods and allow the user to view sleep-related information where available. Do not claim detailed sleep stage information where the data source does not have such data.
- Log stress-related information and make general suggestions to help reduce stress where this information is supported.
- Maintain user inputted medicine, supplements, vaccination, and treatments where these features are supported.

- Make general suggestions based on the logged information regarding exercise, nutrition, hydration, and sleep.
- Allow the user to view his or her habits and any changes over time.
- Allow the user to set wellness goals and follow up on them.

FR- 8 Reports from AI, Health Analytics

- Produce summaries based on health data, activity entries, and wearable data where wearable integration is in place.
- Produce summaries of daily, weekly or monthly nature as appropriate based on reporting capabilities offered by the app.
- Visualize recorded information using dashboards, graphs and trends where it is possible.
- Generate summaries and recommendations for the user's general health based on information available in the system.
- Compare the most recent information with historical data for comparison purposes.
- Enable download, print or sharing features for reports only if they are in place.
- Provide earlier reports access where storing reports history is supported.
- Enable the choice of the kind of reports or health parameters where report customization is supported.

FR-9 Doctor Portal

- Grant access to the doctor portal to authorized and verified healthcare providers where doctor portal is implemented.
- Provide patient information in the portal only if the necessary permissions are granted and allowed by the access rules of the system.
- Present dated health indicators and trends available in the system, e.g., HR, HRV, SpO_2, ECG, skin temperature, sleep and activity, where supported.
- Make possible for doctors to view health summaries as additional information to consider. AI summaries cannot be the substitute for the doctor's conclusions.
- Support notes for consultation, diagnosis, treatment plan and prescription only if corresponding forms and access permissions are implemented.
- Regard messaging and following up and video/audio consultation as proposed features, otherwise treat them as implemented ones.
- Store history of consultations where the application supports it.
- Make possible for doctors to accept or modify and then accept the report to make it an accepted clinician-approved report.

FR- 10 Appointment and Teleconsultation Management

- Enable searching for a doctor and appointment making, changing or cancelling when appointment management is provided.
- Store information on the appointments, like date, time, doctor, type of consultation and appointment status.
- Send reminders regarding the appointment and follow-up if the notification functionality is provided.
- Consider video, audio and chat consultations as the future features unless these features were implemented and tested.

- Enable doctors to manage their schedule and appointment requests if this functionality is provided.
- Store the notes and other data from the consultations and relevant records only if the proper forms and safe storing are provided.
- Enable the patients to share selected records with a doctor using the appropriate and secure way.
- Provide the post-consultation summary only if this functionality is present and the appropriate information is provided by the healthcare professional.

FR-11 Notifications and Smart Alerts

- Inform the users on the configured reminders or recorded values that seem to be abnormal based on the alert rules which were developed and tested.
- Implement reminders for medication, drinking water, activities, menstruation tracking, appointments, vaccinations, and other reminders where enabled.
- Show connection, battery, sync, and update alerts in case wearable connection is available.
- Provide wellness reminders and suggestions in case this function is supported.
- Alerts for emergencies and automated notification of contacts shall be mentioned as implemented after their respective trigger rules and process are tested.
- Enable users to customize notification categories, frequency, time, and silent time where possible.
- Maintain a list of notifications if application supports such feature.
- Employ only the implemented notification types such as in-application, push, email, or SMS. The other types shall be mentioned as future work.

FR- 12 Digital Health Wallet

- Provision a secure repository for storing the relevant documents including test results, prescriptions, vaccine history, and discharge documents if this feature is implemented.
- Store all available health profiles, wearables data, and consultations documents in one place where this functionality is supported.
- Provide users with options of uploading, viewing, downloading, tagging or organizing documents where such functionality is supported.
- Let users share certain documents through the process which is secure and implemented.
- Maintain updates or history of the documents where such capability is supported by the system.
- Apply the AI technology in categorizing documents only where such capability is implemented.
- State that the documents will be stored for lifetime as a design objective rather than promise of availability unless it is guaranteed by some backups or other means.
- Integration with the hospital records, EHR, labs, or insurance systems needs to be stated as future work unless such integrations are there.

FR- 13 Emergency and Safety Management

- Give the ability for users to see or input the necessary emergency information if this feature is included in the application.
- Inform about a suspicious reading only if the data source, threshold, and alerting mechanism are established and validated.
- Notify the emergency contacts only if contact information is provided, there is the consent of the user, and the process is tested.
- Let the users save their emergency information such as allergies, medications and other information that is considered by the user to be saved.
- Provide safety information in general but advise the users to call the appropriate authorities and the medical practitioner in case it is necessary.
- Keep the records of emergency events only if the system records these events.
- Integration with ambulances, hospitals or emergency responders is the future work until this feature is not implemented.
- Create the emergency summaries only based on verified records and do not make any implication that the system has already made any contact with responders.

FR-14 Administration and System Management

- Give the ability for administrators to manage user and doctor accounts and allocate necessary permissions according to access roles.
- Manage wearable registration and device configuration if the feature of wearable devices is implemented.
- Configure available system settings such as notifications preferences and health-information types.
- Monitor the availability of services, API, database and other connected parts of the system using actually configured monitoring tools.
- Manage available educational, wellness information, questionnaires, recommendations templates.
- Prepare administrative summary for the information stored by the system.
- Maintain audit records for the sensitive operations if the audit logging feature is implemented.
- Integration with hospitals, EHR systems, labs, insurance carriers and other services is the future work unless integration is finished.

FR -15 AI Personal Health Coach

- Make recommendations regarding fitness, diet, hydration, sleep, stress management and recovery on the basis of available user information and the capabilities of the app.
- Change suggestions according to changes in user information and goals if such adaptiveness capability has been implemented.
- Assist personal goals and challenges if these capabilities are supported.
- Assist users in reviewing their achievements and routines without promoting this as a medical consultation or guarantee of better health.
- Educate users in the field of preventive health care on the basis of available information and inform that professional opinion is required.

- Take into account the preferences and goals of the user if the application allows personalization.
- Communicate with the user via the health assistant if such capability has been implemented.

FR-16 AI Clinical Documentation and Smart Report Generator

- Generate reports on the basis of available health information, activity entries and consultations if this capability is implemented.
- Generate structured reports for the periods supported by the app.
- Summarize recorded trends including HR, HRV, SpO_2, ECG, temperature, sleep and activity only if the information is available.
- Make health observations and provide general suggestions with clear limitations. Don't suggest that an AI report proves any diagnosis.
- Make it possible to print or share the report only if these actions can be performed and the user controls access to them.
- Customize reports according to selected categories if this capability is supported.
- Store previous reports if the possibility of storing the report history has been implemented.
- Report reduction in clinical paperwork as an expected outcome and not as a result measured by healthcare professionals.

FR- 17 AI Predictive Health and Risk Assessment

- Analyze recorded health and lifestyle data to see if there are any changes over time where the necessary recordkeeping is done.
- Predicting development of cardiovascular diseases, stress-related issues, sleep-related disorders, hormonal problems, dehydration, fatigue or related to reproduction health problems can be considered a research goal if models have not yet been developed.
- Comparison of old and recent records should show what changes might be interesting to discuss with healthcare professional.
- Risk or wellness score calculation should be done only if their formulas or models were defined and documented. It should be noted that no medical diagnosis can be made using those scores.
- Provide general recommendations to find additional information or to consult a professional when needed.
- Avoid saying that your model learns to predict more accurately since it is not implemented yet.
- Development of advanced machine learning algorithms, disease prediction algorithms and clinical decision support are future work unless those models are already made and tested on appropriate data sets.
- If predictions are developed, provide explanations on their main components and describe possible model limitations.

FR-18 AI Digital Health Twin

- Aggregate all the user's health information in the view. Digital health twin creation would require modeling not just normal dashboard.
- Simulation of future health states like results of certain life style changes should be considered future work.
- Comparing new measurement with old one and showing the difference without automatic labeling of the change as some disease.
- Long term forecasting of changes in user's health requires model of future development and enough of the quality data; it is described as planned for now.
- Personalized wellness recommendations should be offered if there is a way for system to create them. Provide recommendations about the treatment should not be provided.
- Providing view for healthcare professionals is possible if there is clinician portal.
- Updating user's health profile according to new entries or data from connected devices is possible if it is implemented in system.
- Precision medicine integration, integration with hospital, remote patient monitoring or integration with other healthcare platforms are future work unless they are done.

2.1.3 Non-Functional Requirements

The Non-Functional Requirements set forth the quality requirements, operability, performance, security, and technical constraints of the AstraCare AI ecosystem. The requirements below will ensure the system's effective, safe, reliable, and high-performing healthcare application and the user-friendly experience it provides to its customers

NFR-1 Performance

- The application will need to be fast when opening pages, viewing health data and using the functions.
- If there are functions for integrating with wearables, the readings should be shown quickly.
- AI summaries and recommendations should be created quickly considering the amount of data being processed.
- The transfer of data between the wearable, application, cloud and AI should not cause any lag in the application.
- The application should still be fast when more data is being processed relating to health and users.
- Older health data and summaries should be easy to view as the size increases.

NFR-2 Reliability & Availability

The system shall comply with the following reliability requirements:

- Deliver uninterrupted health status monitoring services with limited down time.
- Ensure continuous and reliable synchronization between AstraBand AI, mobile applications, the cloud, and AI platforms.
- Avoid data loss during synchronization, application updates, and other unexpected operational disruptions.

- Re-establish connectivity and resume operations after temporary disruptions in between the wearable device and the mobile application.
- Ensure uninterrupted healthcare record accessibility and AI-driven insights availability during connectivity.
- Achieve uninterrupted health record processing and synchronization while maintaining data integrity.

NFR-3 Security & Privacy

The system shall meet the following security and privacy requirements:

- Prevent unauthorized access to sensitive health information and data at all times throughout the data lifecycle.
- Ensure that only verified and authorized users can access specific health-related information according to their permissions.
- Establish secure data exchange mechanisms to protect information traveling between AstraBand AI, the mobile application, the cloud, and authorized end-users.
- Protect individual privacy while collecting, analyzing, and displaying physiological data collected by the wearable device and AI-driven health insights.
- Store private information, medical records, and health reports using industry-leading security measures
- Adhere to all healthcare-related data security and privacy regulations in all jurisdictions where the system operates.

NFR-4 Scalability

- The design of the system should ensure it can accommodate more users and health records without having to be redesigned from scratch.
- The design architecture should ensure that it is possible to add new sensors, features based on AI technology, and healthcare services during the development process of the application.
- It should be easy to scale cloud computing capabilities depending on how many users are registered or how much information needs to be stored.
- Modules can be developed in a way that does not interfere with the functionalities that were already implemented.
- Database and Storage design should ensure scalability as far as health records and reports are concerned.

NFR-5 Usability

- The interface should be intuitive to use for people of various technological backgrounds.
- Menu elements, buttons and navigation through the app should follow a standard pattern.
- The presentation of information related to health and AI should be clear, with proper names and visuals where relevant.
- Language options and accessibility capabilities, if present in the designed application, should make sense.
- The most frequent tasks should take as little steps as possible.
- Notifications and error handling messages should make sense to the user.

NFR-6 Maintainability

- Modules should be designed so that individual feature updates and testing become easier to perform.
- Updates to one module should affect other modules as little as possible.
- Updates to applications should keep users' existing records unchanged. Updates to wearable firmware occur only if there is an actual wearable and a corresponding update mechanism.
- Useful logs of errors/diagnostics should be created in places where logging is necessary; at the same time, it is vital to protect any user health data privacy.
- Any artificial intelligence models and health calculation formulas should be updated via a controlled procedure only.
- Future updates should retain access to existing records or provide a safe way to transfer them to another format.

NFR-7 Compatibility

- It should run in all supported browsers, devices, and OS'es (the full list will be determined by the final scope of development).
- Only Android or other mobile platform compatibility can be mentioned if the application was written, tested for this purpose.
- When a wearable is involved, the way it communicates should be compatible with the application as well as the hardware of supported devices.
- Connectivity to hospitals, EHR's, laboratories, insurance providers and others will be considered future work unless actually implemented.
- External systems, in case of connectivity to them, should exchange data in defined formats.

NFR-8 Safety and Regulations Compliance

- AI-generated data should be provided generally and not serve as a substitution for professional diagnosis and treatment
- The application should warn the user to consult professionals in the field in case symptoms or conditions require it.
- The health records must be managed properly in terms of confidentiality, accuracy, and completeness of the data.
- The project should be reviewed for compliance with privacy regulations and other legal regulations regarding personal data protection and wearable devices usage in accordance with medical software legislation in a particular region before its real-life usage.
- The critical operations and actions should be trackable using auditing logs if there is auditing enabled.

NFR-9 Data Protection and Security Measures

- The application needs a proper method of backing up the stored health records and reports as well as the user profile.
- Cloud synchronization must be applied to decrease data loss only in case the application is based on cloud technologies.

- The system must provide some process of recovering the stored health records after the problem of service or user account or device changing.

2.1.4 Assumptions

The assumptions necessary for successful development and operation of the AstraCare AI are the following:

- The proper use of the AstraBand AI device in daily activities to make sure that the proper data is collected during physiological processes.
- The establishment and maintenance of the BLE connection between the AstraBand AI device and AstraCare AI mobile application in case of synchronization.
- The filling of the personal, medical, and lifestyle information at the moment of account and profile creation.
- The consideration of the insights made by the artificial intelligence before making the decisions about your healthcare.
- The availability of the smartphone supporting AstraCare AI app with internet connection to synchronize and to provide cloud and AI-based healthcare services.
- The availability and proper functioning of the cloud infrastructure, AI services and backend servers to monitor and analyze your health.
- The availability and compatibility of third-party healthcare services, wearables firmware, and any future integration.
- The insights of the AI are meant to help you in making decisions about your healthcare, not replacing the professional medical diagnosis and treatment by doctors.

2.1.5 Constraints

The operation of the AstraCare AI system is limited by the following constraints:

- The reliability of insights into user's health condition based on AI analysis will depend on the accuracy and completeness of physiological data collected with the help of the AstraBand AI and supplied data.
- The capability of performing health monitoring will depend on sufficient amount of power available in the battery of the device and its charging.
- Some functions related to health care will depend on the permanent internet connection for synchronization with the cloud, AI processing, and reporting.
- The range of connection for Bluetooth technology may limit the ability to synchronize in real-time mode between the AstraBand AI and the application.
- The measurements of physiological data may depend on the environmental conditions, wrong position of sensors, movement, and wrong usage of wearables.
- Recommendations from AI are intended for prevention purposes and cannot serve as a replacement for professional medical consultations.
- Any integration with hospitals, electronic health records, laboratories, insurance companies, and teleconsultations services in the future will depend on compatibility of APIs and regulations.

- The product will be developed within the framework of the academic final year project, which will allow the further commercial implementation.

2.2 Feasibility Study

2.2.1 Technical Feasibility

This project is technically feasible as all the required hardware, software, AI and development technologies are available.
The wearable device has low power consumption embedded hardware that can collect a range of physiological data such as Heart Rate (HR), Heart Rate Variability (HRV), Blood Oxygen Saturation (SpO 2), Electrocardiogram (ECG), Skin Temperature and body movement. The data can then be securely sent to the mobile application via Bluetooth Low Energy (BLE). The data is then analysed using advanced health analytics by using AI models in the cloud.
The technologies chosen (Flutter, Python, FastAPI, PostgreSQL, TensorFlow, PyTorch and Cloud Computing) give a scalable and reliable development environment. These technologies are commonly used in healthcare and AI, meaning the project can be maintained and scaled.
This project is therefore technically feasible.

2.2.2 Economic Feasibility

The AstraCare AI project is economically viable as software will be developed using mostly the open source languages, frameworks, AI libraries and cloud technologies to reduce licensing costs.

Whilst the wearable will be developed with specific electronics (biosensors, microcontrollers, batteries and PCB's) the initial prototype can be developed within a reasonable academic project budget. The commercial production of the wearable may even reduce the cost of the electronics and they can be produced in bulk.

As the aim of the project is to improve preventative healthcare and reduce long term medical costs by predicting diseases early and continuously monitoring, there are significant economic benefits.

Therefore, the project is economically viable.

2.2.3 Operational Feasibility

Feasibility of the proposed system on an operational level is justified as the AstraCare AI system was developed using the interface which is very intuitive and convenient for use by women of any age and level of technical skills.

The users will be able to monitor their health in real time using the AstraBand AI wearable device without requiring them to have specialized medical knowledge. Healthcare providers will be able to generate AI based reports and analyze physiological trends and health predictions via the doctor portal in order to make clinical decisions.

Several healthcare processes will be automated (health monitoring, report generation, reminders, health predictions, and personalized recommendations).

Thus, the project is operationally feasible.

2.2.4 Schedule Feasibility

AstraCare AI project could be considered to have schedule feasibility since the development process of this project was subdivided into the following clearly defined stages: requirement analysis, system design, selection of hardware, software development, development of Artificial Intelligence models, wearable devices integration, testing and deployment and documentation.

With the aid of modular development approach, it would be possible to develop simultaneously software, integrate wearable hardware, implement artificial intelligence, develop database and prepare documentation, therefore reducing the development time of the project.

Consequently, the suggested project timeline is quite realistic for the specified academic period set for the development of the final year project.

Thus, the project is schedule feasible.

2.2.5 Legal Feasibility

The AI-based AstraCare project was developed with consideration of the issues related to user privacy, healthcare ethics and data security.

Personal healthcare data, data from wearables sensors and medical records are to be stored safely and accessible only to the authorized users based on their role. The recommendations offered by the Artificial Intelligence are supposed to help users and healthcare professionals and not to replace professional medical diagnosis and treatment.

The architecture of the project can be customized to satisfy the healthcare requirements for electronic health record systems.

Thus, the project is legal.

2.2.6 Overall Feasibility Conclusion

Therefore, considering the results of all the analyses mentioned above, one can say that the AstraCare AI project is feasible enough to start its realization. The system is quite efficient in its combination of wearable technology, cloud computing, AI, and software engineering techniques to build an innovative healthcare service for women's health monitoring.

2.3 Details of Hardware and Software Used

2.3.1 Hardware Used:

Sr. No.	Hardware Component	Selected Component	Purpose
1	Microcontroller Unit (MCU)	Nordic nRF5340	Manages the overall operation of the wearable device, sensor communication, data processing, and Bluetooth Low Energy (BLE) connectivity.
2	Optical Biosensor	MAX86141	Continuously monitors Heart Rate (HR), Heart Rate Variability (HRV), Blood Oxygen Saturation (SpO_2), and Photoplethysmography (PPG) signals.
3	ECG Sensor	MAX30003	Captures Electrocardiogram (ECG) signals for cardiac health monitoring.
4	Motion Sensor	BMI270	Tracks physical activities, sleep patterns, body movements, and posture.
5	Temperature Sensor	TMP117	Continuously monitors skin temperature for physiological analysis.
6	Power Management IC (PMIC)	nPM1300	Manages battery charging, power regulation, and overall power consumption.
7	Flash Memory	Winbond W25Q256JV (256 Mb NOR Flash)	Stores firmware, device configurations, and temporary sensor data.
8	Battery	250–300 mAh Lithium-Polymer Battery	Provides power for continuous wearable device operation.
9	Wireless Communication	Bluetooth Low Energy (BLE 5.3)	Enables secure wireless communication between the wearable device and the AstraCare AI mobile application.
10	Vibration Motor	Linear Resonant Actuator (LRA)	Provides haptic feedback for health alerts, reminders, and notifications.
11	Charging System	Magnetic Charging System	Supports safe, reliable, and convenient battery charging.
12	Printed Circuit Board (PCB)	Custom Multi-layer PCB	Integrates and connects all electronic hardware components.
13	Wearable Enclosure	IP67 Water-Resistant Smart Band	Protects internal electronic components while providing durability and comfort for daily use.

2.3.2 Software Used:

Sr. No.	Software / Technology	Purpose
1	Flutter	Mobile Application Development
2	FastAPI	Backend Development
3	Python	Backend & Artificial Intelligence Development
4	C/C++	Embedded Firmware Development
5	PostgreSQL	Database Management
6	TensorFlow	Artificial Intelligence Model Development
7	PyTorch	Deep Learning Model Development
8	Scikit-learn	Machine Learning Algorithms
9	Google Cloud Platform (GCP)	Cloud Computing & Data Storage
10	REST API	Communication between Mobile App and Server
11	Android Studio	Mobile Application Development
12	Visual Studio Code	Source Code Development
13	SEGGER Embedded Studio	Firmware Development
14	Git & GitHub	Version Control
15	Firebase Cloud Messaging (FCM)	Notification Service
16	React.js/Next.js	Frontend

3. SYSTEM PLANNING

AstraCare Artificial Intelligence project has been designed to last for 28 weeks (7 months). This project has been divided into phases in order to execute it in a systematic way, optimize the use of resources, deliver the outputs within the given timeframe, and integrate the components of wearable hardware, artificial intelligence, cloud computing, and software. The detailed description of these activities is given in the subsequent sections.

3.1 List of Activities

Task ID	Activity	Description
T1	Requirement Gathering	Gather the project requirements, identify the user needs, and determine the objectives of the project.
T2	Literature Survey	Review the existing health monitoring systems, wearable technologies, and Artificial Intelligence techniques.
T3	Feasibility Study	Carry out a feasibility study, including technical, economic, operational, legal, and schedule feasibility.
T4	System Analysis and Requirement Specification	Formulate the Software Requirement Specification (SRS).
T5	System Design	Specify the software architecture, hardware architecture, database, and user interfaces.
T6	Hardware Selection and Design	Determine the hardware components and develop the wearable architecture.
T7	Software Development	Undertake the development of the mobile application, backend server, Application Programming Interfaces (APIs), and database.
T8	Artificial Intelligence Development	Develop, train, validate, and implement Artificial Intelligence models within the healthcare system.
T9	Hardware Integration	Integrate the wearable hardware with the mobile application and backend server.
T10	System Integration	Integrate the hardware, software, cloud services, APIs, and Artificial Intelligence components into a unified system.
T11	Testing and Validation	Test and validate the developed system. Additionally, perform system maintenance to resolve the bugs detected during the validation process.

Task ID	Activity	Description
T12	Documentation, Finalization, and Demonstration	Prepare the project documentation, finalize the project, and demonstrate the completed system to the stakeholders.

3.2 Timeline for Each Activity

Task ID	Activity	Duration	Timeline
T1	Requirement Gathering	2 Weeks	Week 1 – Week 2
T2	Literature Survey	2 Weeks	Week 3 – Week 4
T3	Feasibility Study	2 Weeks	Week 5 – Week 6
T4	System Analysis and Requirement Specification	2 Weeks	Week 7 – Week 8
T5	System Design	3 Weeks	Week 9 – Week 11
T6	Hardware Selection and Design	2 Weeks	Week 12 – Week 13
T7	Software Development	5 Weeks	Week 14 – Week 18
T8	Artificial Intelligence Development	3 Weeks	Week 19 – Week 21
T9	Hardware Integration	2 Weeks	Week 22 – Week 23
T10	System Integration	2 Weeks	Week 24 – Week 25
T11	Testing and Validation	2 Weeks	Week 26 – Week 27
T12	Documentation, Finalization, and Demonstration	1 Week	Week 28

3.3 Gantt Chart

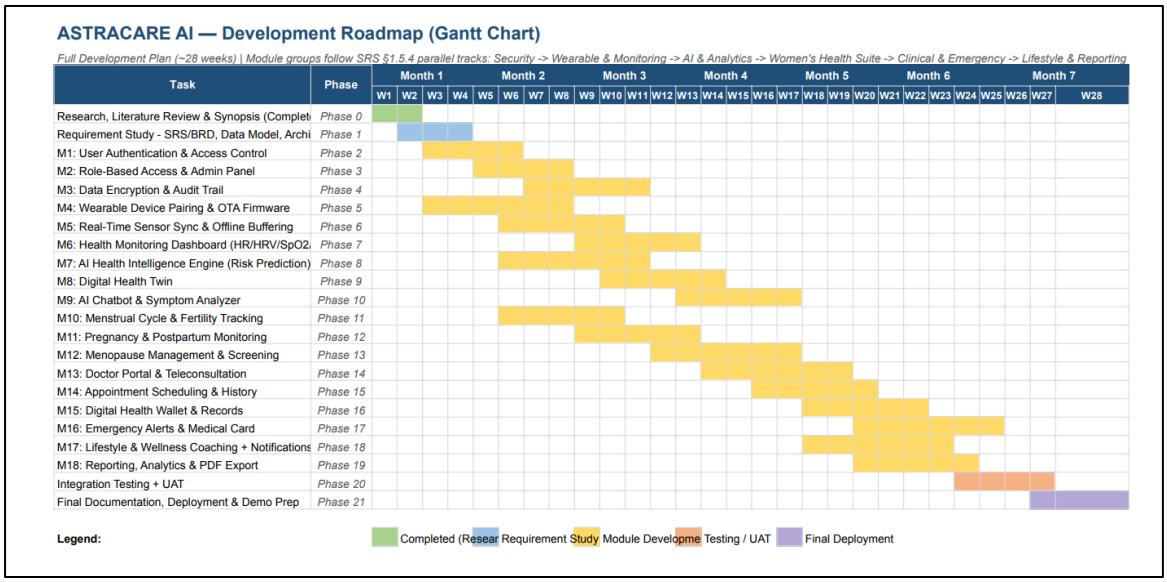


Figure 2.1: Gantt Chart illustrating the project schedule for the development of the AstraCare AI system over a period of 28 weeks

4. SOFTWARE AND HARDWARE MODULES

4.1 Introduction

The AstraCare AI system consists of two main components which comprise the hardware component and the software component. The hardware component is tasked with the job of gathering the physiological and activity data with the help of sensors attached to the AstraBand AI device. The software component, on the other hand, is tasked with the task of analysis of the gathered data with the help of artificial intelligence, healthcare data management, and healthcare services provision.

4.2 Hardware Modules

4.2.1 Hardware Overview:

The AstraBand wearable AI device is built in such a way that it constantly monitors the physiological parameters of the user and sends all collected data to the AstraCare AI platform where the data is further processed and analyzed using artificial intelligence. The hardware is created using the latest technologies in the field of electronics by using low power consuming high performance electronic components to provide the best sensing, processing capabilities, wireless communication, and power consumption.

4.2.2 Hardware Components

Sr. No.	Component	Model	Application
1	Microcontroller Unit (MCU)	Nordic nRF5340	Manages the overall operation of the wearable device, controls connected sensors, processes sensor data, and handles wireless communication.
2	Optical Biosensor	MAX86141	Measures Heart Rate (HR), Heart Rate Variability (HRV), Blood Oxygen Saturation (SpO_2), and Photoplethysmography (PPG) signals.
3	ECG Sensor	MAX30003	Acquires Electrocardiogram (ECG) signals for continuous cardiac monitoring.
4	Motion Sensor	BMI270	Tracks physical activity, step count, sleep patterns, posture, and body movement.
5	Temperature Sensor	TMP117	Continuously monitors the user's skin temperature.
6	Power Management IC (PMIC)	nPM1300	Regulates battery power, manages battery charging, and optimizes power consumption.

Sr. No.	Component	Model	Application
7	Flash Memory	Winbond W25Q256JV (256 Mb NOR Flash)	Stores firmware, configuration settings, and temporary health data collected by the wearable device.
8	Battery	250–300 mAh Lithium-Polymer (Li-Po)	Provides power to the wearable device for continuous operation.
9	Wireless Communication	Bluetooth Low Energy (BLE 5.3)	Enables secure wireless communication between the wearable device and the AstraCare AI mobile application.
10	Vibration Motor	Linear Resonant Actuator (LRA)	Provides haptic feedback for health alerts, reminders, and notifications.
11	Charging System	Magnetic Charging	Allows safe, reliable, and convenient charging of the wearable device.
12	Printed Circuit Board (PCB)	Custom Multi-layer PCB	Integrates and interconnects all electronic components of the wearable device.
13	Wearable Enclosure	IP67 Water-Resistant Smart Band	Protects the internal hardware components while providing comfort and durability for daily wear.

4.2.3 Hardware Requirements

HR-1: Low Power Microcontroller

The wearable product must have a low power microcontroller that will facilitate the management of sensors, data processing and communication features.

HR-2: Optical Biosensor

The wearable device must monitor the following parameters of the user - Heart Rate (HR), Heart Rate Variability (HRV), SpO2 level and Photoplethysmography (PPG) waveforms by utilizing optical biosensor.

HR-3: ECG Monitoring

The band should measure Electrocardiogram (ECG) waveforms to help in cardiovascular activity tracking.

HR-4: Body Temperature Monitoring

The band shall measure the body temperature of the user using the digital thermometer.

HR-5: Movement and Sleep Activity Measurement

The wearable band shall measure body motions, physical activities, sleep and posture of the user using motion sensors.

HR-6: Wireless Communication

The band shall have wireless connectivity capability that enables the transfer of data from wearable device to the AstraCare AI application using Bluetooth Low Energy (BLE) communication protocol.

HR-7: Rechargeable Li-Po Battery

The wearable device shall have rechargeable Lithium-Polymer (Li-Po) battery that will allow for health tracking during regular daily activities.

HR-8: Flash Memory

The wearable product must have flash memory to enable storage of configuration software/firmware and data temporarily.

HR-9: Vibration Feedback

The wearable product must generate vibration signals as a notification mechanism for various activity/health events.

HR-10: Magnetic Charging

The wearable product shall have a magnetic charger which is used as a convenient mechanism for the battery charging.

HR-11: Ergonomic Chassis

The product must have a light-weight ergonomic chassis.

HR-12: Water and Dust Resistant Housing

The product must have housing protection to prevent dust and water from accessing the electronics within its enclosure. The product must have IP67 protection rating against water and dust exposure.

HR-13: OTA Firmware Update

The wearable product should be capable of implementing secure Over-The-Air (OTA) firmware update procedure.

HR-14: Data Protection

The wearable product must ensure the confidentiality and integrity of the data collected by onboard user band’s biosensors during wireless communication of band and system.

4.2 Software Module

4.2.1 Software Overview

It is important to highlight that the main part of the AstraCare AI is the software. The software manages user data, analyse physiological data given by the wearable device, perform the AI process, store healthcare data and provide health services. It is composed by the application, cloud, AI, databases and wearable device. All of them are managed by the software to provide a secure, reliable, intelligent and user-friendly health monitoring system.

The software of the application has been developed using a modular design, so every module has different functions and interacts with others to manage the system efficiently and effectively.

4.2.2 Software Modules Summary

Module ID	Software Module	Description
M1	User & Security Management	It manages the user registration, authentication, authorization, profiles, and security.
M2	Women's Life Stage & Reproductive Health	It deals with the management of menstruation, fertility, pregnancy, postpartum, menopause, and other related functions.
M3	AI Health Intelligence Engine	The module conducts Artificial Intelligence-based health analysis to provide predictive healthcare.
M4	AI Based Health Assistant (Chatbot)	It provides AI-based health advisory and assistances to the users.
M5	Health Dashboard & Analytics	The dashboard display health information, band data, and health analysis.
M6	Nutrition & Hydration	It handles the monitoring of user diet, water intake, and nutrition.
M7	Fitness & Physical Activity	It manages the physical activities and exercise of the user.
M8	Mental Health & Wellness	It focus mainly on providing support and assistance regarding the user's mental health, wellness, and stress management.
M9	Wearable Device Management (Band)	The module deals with managing band, synchronization, pairing, firmware update, and other related functions.
M10	Health Reports & Medical History	It manages and keeps digital copies of health reports and medical history.

Module ID	Software Module	Description
M11	AI Disease Prediction & Risk Assessment	This module uses Artificial Intelligence to predict and manage disease risks.
M12	Medication & Health Reminder	This module handles medication reminders, fluid intake reminders, and other health-related reminders.
M13	Emergency & SOS	It focuses mainly on health-related emergency responses, emergency SOS, and managing emergency contacts.
M14	Doctor's Console	The console is used to manage and view patients' reports and health data.
M15	Admin Management	It manages the administration panel of the application.
M16	Notifications	The module deals with notifications, announcements, reminders, and other user alerts.
M17	Digital Health Twin	It manages the creation and maintenance of a digital twin for health monitoring purposes.
M18	Cloud Management	The application uses this module to synchronize and securely store health data.

M1 – User & Security Management

The purpose of the User & Security Management module is the supervision of the user registration, user authentication, authorization, profile management, and all-in-all security. In this module, users will be able to register their accounts using either a mobile phone number or email address with the help of OTP process. This module offers secure login, password recovery, multi-factor authentication. Moreover, the user session security is handled by JWT. Users can have full control over their personal profiles including managing medical data, emergency contacts, privacy, etc. Moreover, the role-based access control is used in order to provide secure access to healthcare data for patients, doctors, and administration.

M2 – Women's Life Stage & Reproductive Health

The Women's Life Stage & Reproductive Health module allows tracking of various reproductive phases in woman's life which include menstruation, fertility, pregnancy, post-pregnancy recovery, and menopause. The module helps with the process of menstrual cycle tracking, ovulation and fertile days predictions. Moreover, this module is capable of offering the individualized healthcare recommendations based on the obtained information. In case of pregnancy, the module offers services for pregnancy tracking, fetus' development

tracking, appointments' reminder, and health advice. In the period of post-pregnancy recovery, the module assists in monitoring and postpartum health tracking. As for the menopause, the module allows tracking the symptoms and giving health recommendations. The data received by this module will be analyzed by the Artificial Intelligence Engine.

M3 - AI Health Intelligence Engine

The AI Health Intelligence Engine constitutes the core module of the AstraCare AI system, which is capable of analyzing the health information provided by wearable devices and others sources and getting insights on the current state of health in the form of intelligent health care actions. This module analyzes various physiological data, which includes Heart Rate (HR), Heart Rate Variability (HRV), SpO_2, ECG, skin temperature, activity level, sleeping and reproductive data using artificial intelligence and machine learning methods. Based on the analyzed data from the previous modules, the system determines the health trends and abnormalities and gives recommendations about health actions required. The module offers the early diagnosis of issues such as menstrual abnormalities, stress-related diseases, cardiovascular diseases, risks of pregnancy, and various other female-specific health problems. By combining the patient history and the new data from the wearable device, the AI Health Intelligence Engine obtains enhanced prediction capability.

M4 - AI Health Assistant (Chatbot)

AI Health Assistant (Chatbot) tool provides the user with a tool allowing for an advanced dialogue regarding health-related concerns. Through this tool, a user may convercate about their health-related topics with the AstraCare AI using natural language. In particular, this application enables the user to get an answer to their questions on menstruation, pregnancy, nutrition, exercise, mental wellbeing, medication schedule, and other health queries according to health data of the user and AI analysis. Moreover, this module will help the user to analyze health report provided by AI and interpret AI analysis results. Recommendations will be provided to users taking into account their personal health data as well as recommendations on preventive measures. Additionally, this application will notify the user on time for taking medicines, attending appointments, being hydrated, doing exercises, and seeing a doctor if necessary.

M5 - Health Dashboard & Analytics

The Health Dashboard & Analytics tool allows the user to monitor their health data collected via the AstraBand AI wearable and all the other software modules. The health data includes not only live but also stored data: heart rate (HR), heart rate variability (HRV), blood oxygen saturation (SpO_2), electrocardiogram (ECG), skin temperature, physical activity, sleep, nutrition, hydration, and reproductive health data. They are shown in a readable format for users' benefit. This module generates information about the current health state of the user in form of insights, recommendations, trends, risks assessed with help of Artificial Intelligence. Also, daily and weekly summaries of health performance and wellness score will be generated.

M6 - Nutrition & Hydration Management

Nutrition & Hydration Management is the application that provides information concerning food and liquid consumption of the user and the nutritional and fluid balance. It allows sticking to healthy nutritional and hydration regimen. In this module, users will record the food and beverages that they have eaten. Analyzing users' data – age, activity level, health characteristics, connected devices, and artificial intelligence, the system will generate for the user recommendations regarding diet and hydration regimen. According to those recommendations, reminders for drinking water and eating food of specific type will be generated, thus helping the user to achieve the needed nutritional balance. The application also finds any unhealthy eating habits.

M7 - Fitness & Physical Activity Management

The Fitness & Physical Activity Management module helps users to develop a habit of being physically active by constantly tracking their physical activities throughout the day using the AstraBand AI wearable device. The details about the number of steps, walking distance, burnt calories, active time period, exercise, intensity of the movement etc. are collected automatically and then analyzed by the system to determine the overall fitness level of the user. The users have the option to set personal fitness goals according to their age, lifestyle and health condition. In addition, with the help of wearable sensor data and Artificial Intelligence analysis, the module identifies the behavioral pattern in terms of activity and sedentary behavior and makes suitable suggestions for physical activity to ensure better health of the user. The users will be able to get progress reports and performance summaries to motivate them to achieve their health goal and stay active. The module plays a significant part in ensuring the preventive care of the user.

M8 - Mental Health & Wellness Management

Mental Health & Wellness Management module deals with monitoring the state of emotional health of the user and promotion of healthy behavior via the analysis of indicators related to mental well-being of the user. These indicators include stress level, quality of sleep, physical activity, and user input on his/her mental wellness. By applying AI techniques, patterns of user behavior which indicate that the user is experiencing stress, fatigue, or emotional imbalance can be discovered, and wellness recommendations can be given to the user. Therefore, each user receives the recommendations on how to perform relaxation, breathing techniques, meditation, how to improve his/her sleep, and other daily activities to boost overall mental well-being.

M9 - Wearable Device Management Module

Wearable Device Management module is meant to handle the establishment and the maintenance of communication link between the AstraBand AI wearable device and the AstraCare AI mobile application. This module will manage the pairing process, BLE connectivity, synchronization, firmware updates, and other aspects of the device management to guarantee proper functioning of the wearable device. This module constantly collects the physiological data that is collected with the use of the embedded sensors and forwards this data to the mobile application in order to process it and conduct the Artificial Intelligence analysis. This module also monitors the battery level,

connectivity, storage space, firmware versions, and other aspects of the operation and notifies users about required maintenance or updates. In case of any communication breakdowns, this module is able to reconnect and synchronize the health records.

M10 - Health Reports & Medical History

Health Reports & Medical History is the module which is used to store all information about the user's health records during the whole use of the AstraCare AI system. It stores the digital copies of physiological parameters collected using the wearable device as well as laboratory reports, diagnoses, prescriptions, vaccinations, allergies, medication, and any other medical information. The system automatically generates the detailed reports of the health of the person using the wearable sensor data and Artificial Intelligence analysis which helps users to track their progress in health condition in different time intervals. Users can have access to all historical data at any time, making it possible to compare the current health status with previous and make proper conclusions and medical decisions. Additionally, it is possible to share reports with authorized physicians during medical consultations and keep all the information confidential and private. With this module, a user can build the full digital history of his health.

M11 - AI Disease Prediction & Risk Assessment

AI Disease Prediction & Risk Assessment is one of the main intelligent modules in the AstraCare AI system. Using AI and ML algorithms, this module analyses physiological data, data taken from wearable band, medical history, lifestyle and health records to identify potential risks to the user's health in the future. The system continuously analyses changes in heart rate, ECG, blood oxygen levels, body temperature, physical activity, sleep patterns and women's health data to identify if any of the parameters are abnormal. Users will then be given alerts, a health risk score and advice on how to prevent illness or when to see a physician. It is also possible to predict certain women's illnesses such as irregular periods, fertility issues, pregnancy issues, menopause and general health risks.

M12 - Medication & Health Reminder Management Module

This module helps the user manage their daily healthcare routine by reminding them when to take their medication, drink water, see healthcare professionals, have health check-ups and more. The user can set how often they want to take their medication, how much they need to take, when to see healthcare professionals, when to have vaccinations and health goals. The system will automatically remind the user at set times and record the user's answers. Based on the Artificial Intelligence analysis of the user's health and daily routine, the system can recommend times to set reminders to help the user stick to their recommended healthcare routine and encourage healthy behaviours. When integrated with the wearable device, the system will also remind the user if they have been inactive for too long, to drink water, to go to bed and exercise.

M13 - Emergency & SOS

Emergency & SOS is a module that is aimed at giving help to users in critical health conditions. This is achieved by giving the users a quick way of alerting the emergency contacts and the doctors. When there is an abnormal health situation recognized by the

wearable device or the user decides to use the SOS function, the system alerts the user by creating the alerts that contain all the necessary health information and the location of the user (if it is possible). The module constantly monitors the vital physiological parameters through the wearable device and cooperates with the Artificial Intelligence engine to recognize all medical emergencies that require any actions. Users' emergency contacts will get timely notifications, which will allow them to react as soon as possible. Moreover, the Emergency & SOS module keeps the information about blood group, allergens, chronic diseases and other emergency information. Thus, this module provides fast assistance in critical health situations.

M14 - Doctor's Console

Doctor's Console module is a solution that allows the healthcare professionals to have a secure and easy-to-use interface to access all information about the patient's health in case they have the authorization to do so. With the Doctor's Console module, the doctors are able to see the collected physiological parameters, the medical history of the patient, laboratory tests results, Artificial Intelligence health analysis and assessment of the risks of developing diseases. Moreover, the module helps to understand the condition of the patient through historical trends, health reports and constant monitoring. The healthcare professionals are able to make the medical recommendations and monitor the treatment of patients. The access to patient information is secured by the role-based authorization mechanism.

M15 - Admin Management

The Admin Management module is responsible for handling the management of the entire process and administration of the AstraCare AI application. The module allows administrators to be able to monitor system performance, account management of users, health care data management, configuration of the application and ensure the proper working of the whole process of AstraCare AI application. The module allows the administrators to create, edit, suspend and delete accounts of the users as needed while providing proper access to the patients, doctors and other authorized personnel. The module will also allow the monitoring of system activity, security logs, registering of wearable devices, cloud management and Artificial Intelligence service management in order to ensure reliable working of the platform. Besides, the module will also help in the management of software updates, system maintenance, notification service management and data backup services.

M16 - Notifications

The Notifications module ensures the timely delivery of alerts, reminders, and critical updates to the users during their entire usage of the AstraCare AI platform. The module sends notifications concerning the schedule of medication intake, hydration reminders, health check-ups, state of wearables, appointment reminders, emergency alerts, and Artificial Intelligence-based health recommendations. It constantly tracks all user's activities and health events in order to inform the user in a timely manner about all relevant health information. Notifications are sent to the users via the mobile application and prioritized according to their importance in order to allow the user to react quickly to any critical health situation and continue following his/her regular routine of health care

procedures. The module saves the notification history and user reactions in order to optimize the schedule of reminders in the future. Using this module, the users can follow a healthy lifestyle, be aware of their health status and conduct their health care activities effectively.

M17 - Digital Health Twin

The Digital Health Twin module creates a virtual replica of the user's health status based on information from wearable sensors, the user's medical history, lifestyle, reproductive health records, and Artificial Intelligence recommendations. With each new piece of health data collected, the virtual replica becomes more sophisticated and allows the system to get an idea about the current physiological state and future health tendencies of the user. The module allows monitoring the tendency of health development, comparison of current and past health statuses, and simulation of future health conditions due to changes in the user's lifestyle, nutrition, physical activities, medication adherence, or any other health-related changes.

M18 - Cloud Management

This module manages, synchronizes, and stores the data related to healthcare generated by the AstraCare AI system. Thus, the Cloud Management module guarantees that all the data collected by the wearable gadget, mobile app, Artificial Intelligence software system, additional software modules, and applications are safe and available on request from the cloud. This module is responsible for the synchronization of users' data, such as users' profile, vital indicators, medical reports, personal medical history, and Artificial Intelligence recommendations. Thus, it provides access to the latest version of users' health data. Additionally, the module supports data backups, data recovery, and storage management to make sure that all data are safe and protected from losses or damages to precious healthcare data. Cloud Management guarantees implementation of relevant security measures to protect the data stored in the cloud and provides data encryption and authentication. Finally, it provides users with access to safe data.

5. SYSTEM ANALYSIS AND DESIGN

5.1 Context Diagram

5.1.1 Context Diagram of AstraCare AI

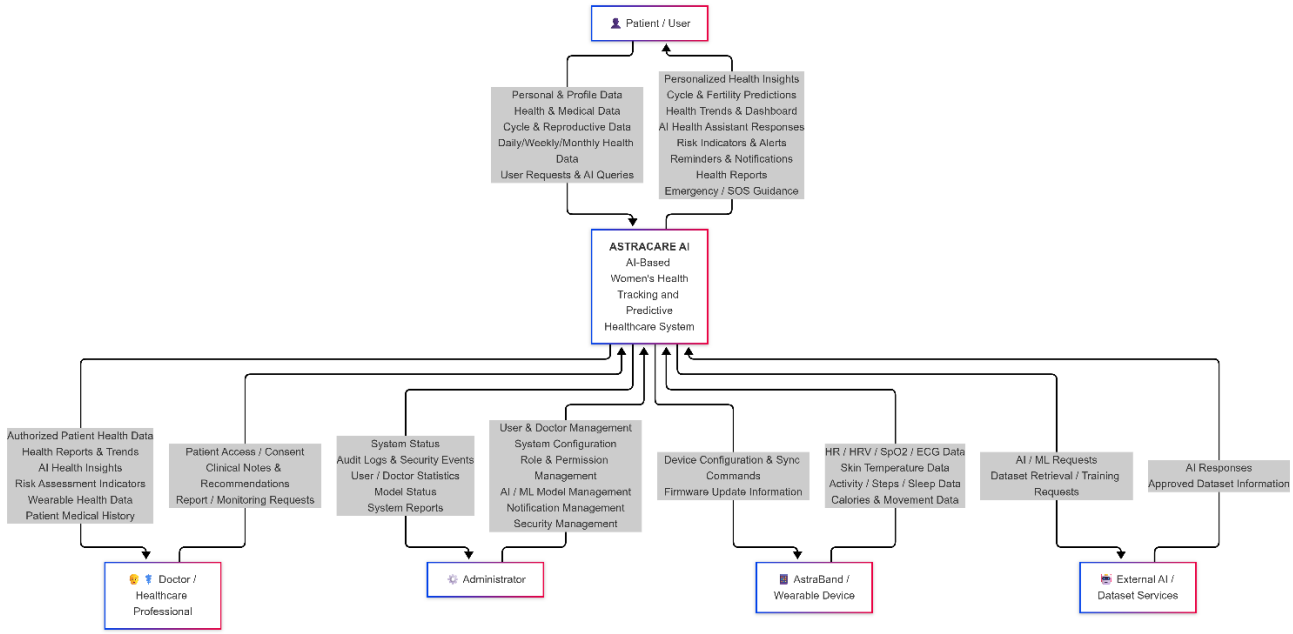


Figure 5.1: Context Diagram of AstraCare AI

The Context Diagram shows interaction at a high level between AstraCare AI and the external entities that can communicate with the system. AstraCare AI is shown as one single main system getting information related to health data, user request, administration, wearable information, and AI inputs, providing health insights, reports, predictions, notifications, and system information in response.

Patient/User is the first entity interacting with AstraCare AI. The patient/user gives the system his/her personal information, medical history, reproductive information and cycle information, as well as information regarding daily, weekly and monthly health. Using this information, the system will give personalized health insights, cycle and fertility predictions, health trends, AI assistant responses, risk indicators, reminders, health reports and emergencies guidance.

Doctor/Healthcare Professional interacts with AstraCare AI to get authorized information about his/her patients, write clinical notes and give recommendations to patients or request reports and monitoring information. AstraCare AI will give authorized health information, medical history, health reports, trends, AI insights, risk indicators and wearable health information to the doctor.

Admin manages the whole system by managing user and doctors, configuring the system, managing permission and roles managing AI/ML models, notifications and security. AstraCare AI will give system status information, audit information,security events, statistics, AI model status and system reports.

AstraBand wearable device bind with AstraCare AI in order to give physiological and activity information like heart rate (HR), SpO2, ECG, heart rate variability (HRV), skin temperatures, activity, steps, sleep, calories, and movements. The system may also give information for device configuration and synchronization or firmware update information.

External AI/Dataset Services help AstraCare AI by giving AI responses and authorized dataset information. AstraCare AI may also give AI/ML requests and dataset retrieval and training requests to these external services.

So the Context Diagram gives an overview of AstraCare AI, its external entity and the main information flows between them, but does not show internal processing and individual software modules of the system.

5.2 Data Flow Diagram (DFD)

5.2.1 DFD Level 0 of AstraCare AI

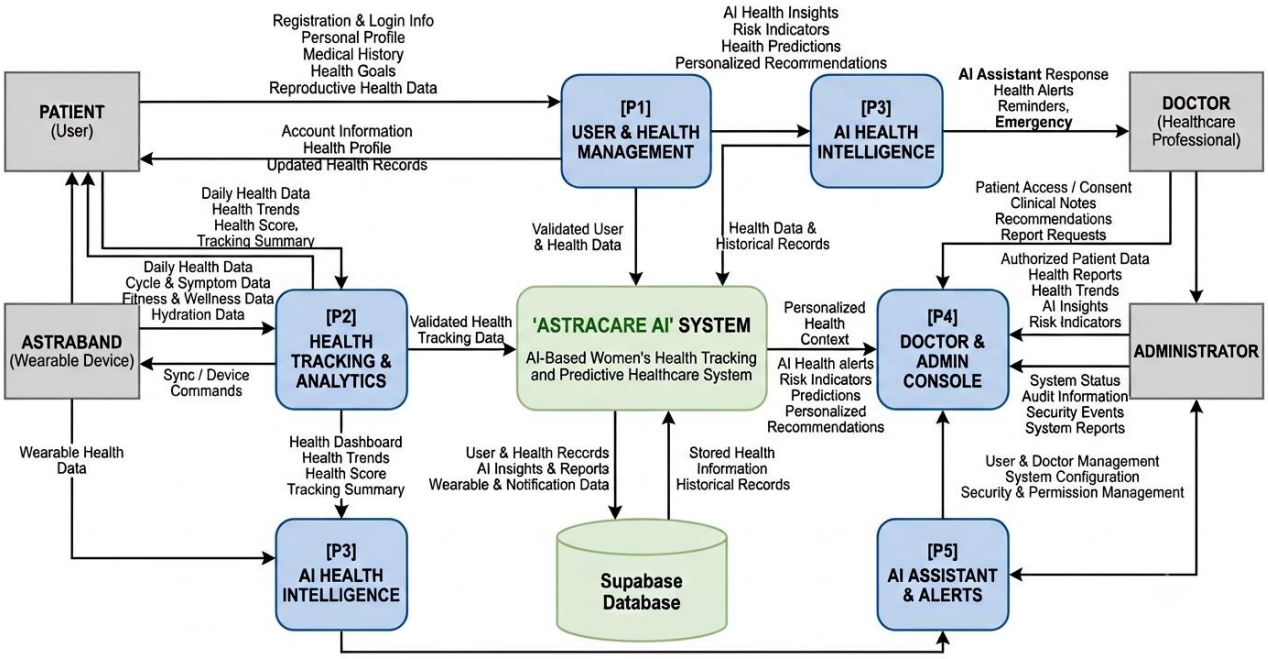


Figure 5.2: DFD Level 0 of AstraCare AI

The Level 0 Data Flow Diagram shows the main flows of information in the AstraCare AI – AI-Based Women's Health Tracking and Predictive Healthcare System. In this diagram, the functional description of the processes of receiving, processing, and storing the information provided by various entities and devices is demonstrated.

The Patient/Users is the main entity of health information supply. The user supplies the system with health information such as registration data, profile, medical history, reproductive health, daily health, symptom, and fitness and wellness information. The information is processed in the User and Health Management and Health Tracking & Analytics processes to create and update the health profile and provide dashboard information, trends, and health scores.

AstraBand wearable device provides the information about health and activity of the user obtained during its use. At this level, it is not shown as several flows of sensor readings,

but rather as the single Wearable Health Data flow. This information is supplied to the Health Tracking & Analytics process where it can be used together with the manual input of health information.

The AI Health Intelligence process receives health records, historical information, and tracking information for analysis and produces such outputs as health insights, risk indicators, predictions, and personal recommendations. They can be further used in other processes of the system to provide the personalized healthcare services.

AI Assistant & Alerts process receives health related queries of the user and, using the information and insights provided by the AI, gives answers, reminders, alerts, and emergency recommendations to the user. The user thus can use the system, asking any health related question based on the health records.

Doctor & Admin Console provides separate functions for the healthcare professionals and administrators. Doctors receive authorized patient information, health trends and reports, and AI insights; Administrators are able to manage the users, doctors, permissions, configuration, and security issues of the system.

Supabase Database is the primary data storage of the system. It stores user information, health data, wearable information, AI results, notifications, and other necessary data of the system. The information stored in this data store can be later used for analysis, report generation, and user services.

Thus, the Level 0 DFD describes the main processes, external entities, data stores, and main flows of information in AstraCare AI. It is an abstract version of the information flow of the system which will be later developed into Level 1 DFD.

5.2.2 DFD Level 1 of AstraCare AI

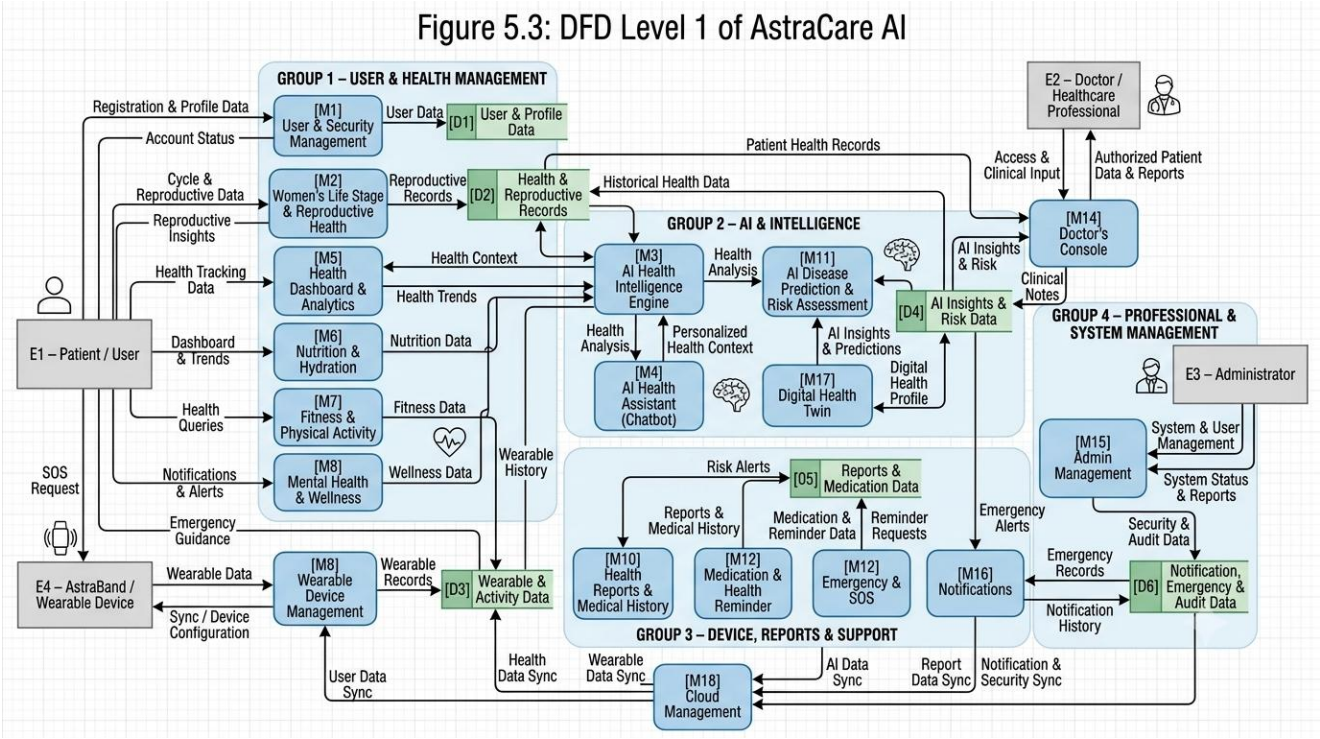


Figure 5.3: DFD Level 1 of AstraCare AI

Level 1 DFD gives the clear picture of internal processes and flow of information in AstraCare AI system. It depicts main processes of the system into modules and their interactions with external entities and data stores.

The Patient/User supplies system with registration information, personal data, medical history, reproductive health data, daily health data, wellness data, health queries, and emergency requests. AstraBand is used to get health and activity information using the data supplied by wearable devices. The processing of this info takes place within module named Wearable Device Management.

The system processes info with help of such modules as Users & Security, Women's Life Stage and Reproductive Health, Dashboard & Analytics, AI Health Intelligence Engine, AI Health Assistant, and AI Disease Prediction & Risk Assessment. Processing this information gives users their personal health information, analytics, risks assessments, trend analysis, and advice and recommendations.

Health Reports & Medical History, Medication & Health Reminder, Emergency & SOS, and Notifications module manage generation of health reports and medical history, management of data related to medication use, reminders, emergencies, and notifications. Doctor's Console helps doctors to access health reports, health history of patients, and trends in their health. The Admin Management manages users and permission, as well as security of system.

This diagram also shows digital health twin and cloud management for personalized health data storage and system data synchronization in the cloud. Data stores are represented by User Profile, Health and Reproductive Records, Wearables, AI Insights, Reports, Medications, Notifications, Emergencies, and Audit Information.

In general, the Level 1 DFD shows information entry into the system, its processing by functional modules, storage and analysis, as well as generation of health reports and alerts, health recommendations and services for patients, doctors, and administrators.

5.3 Entity Relationship Diagram (ERD)

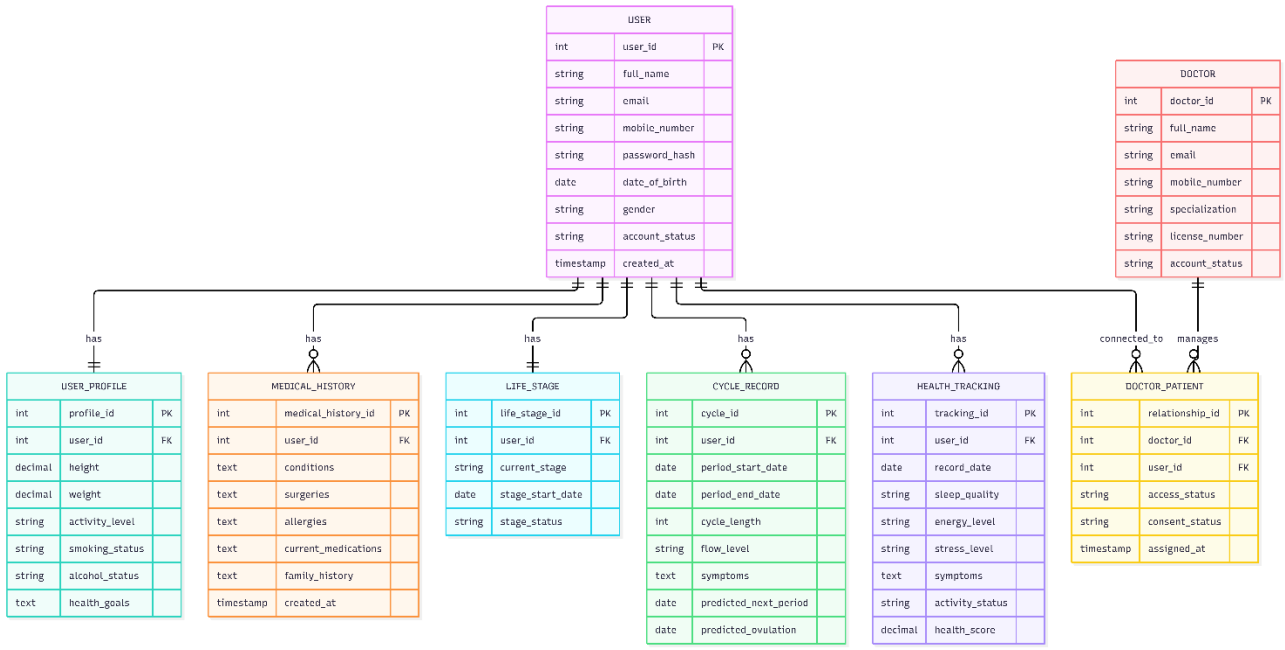


Figure 5.4(a): ERD – User and Health Management

The first part contains all the basic user and health related information for the system. The USER entity is the primary entity and is connected with the USER_PROFILE, MEDICAL_HISTORY, LIFE_STAGE, CYCLE_RECORD, and HEALTH_TRACKING entities. The DOCTOR and DOCTOR_PATIENT entity also represent the relationship between doctors and their patients.

This part consists of all the basic user information, health information, medical history, life stage information, reproductive cycle information, and regular health tracking information.



Figure 5.4(b): ERD - Wearable and AI Intelligence

The second part consists of data from the user's wearable device and AI health intelligence. The entity WEARABLE_DEVICE holds all the data that corresponds to the user's wearable device, and WEARABLE_DATA is the entity storing the measurements made using the device.

The AI entities include AI_INSIGHT, RISK_ASSESSMENT, HEALTH_REPORT, and DIGITAL_HEALTH_TWIN. They store the individual insights, health risks, the reports produced, and the digital version of the user's health, respectively.

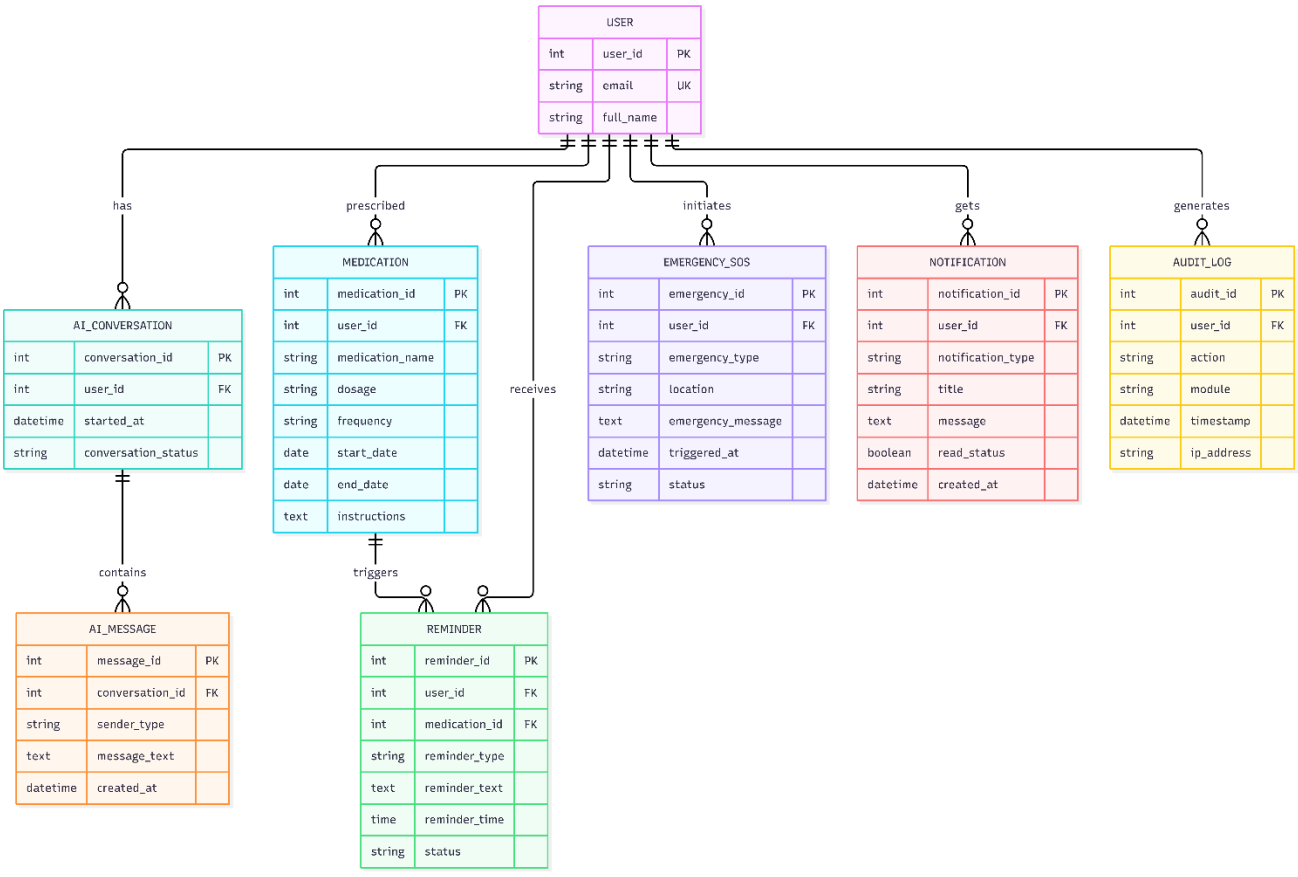
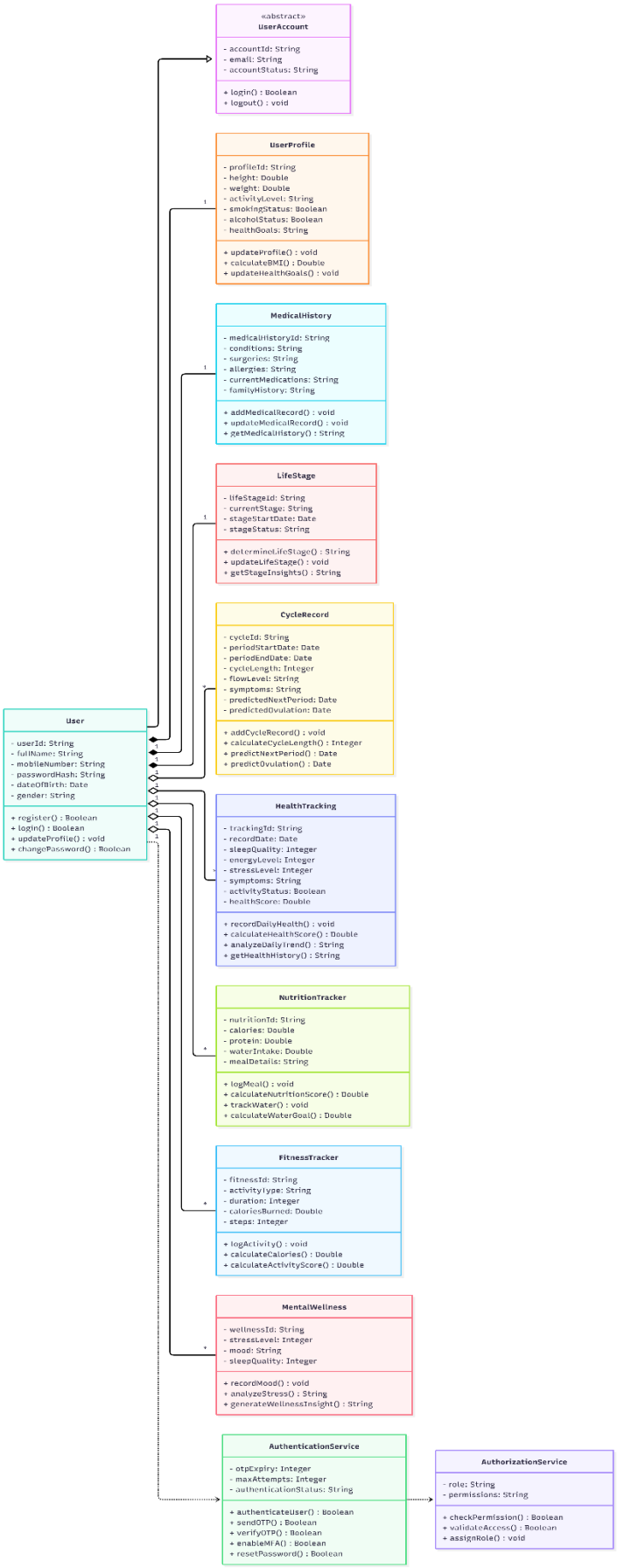


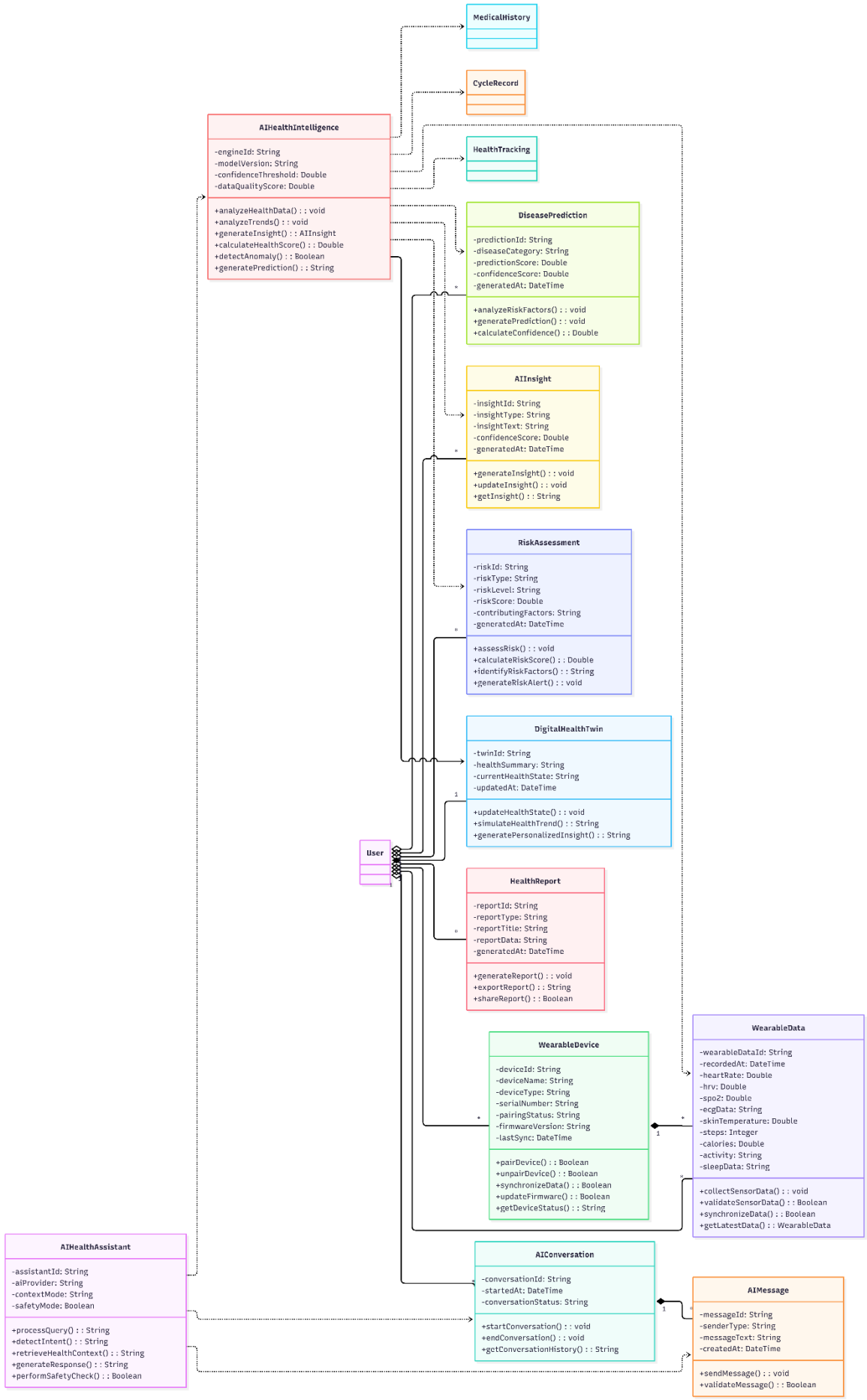
Figure 5.4(c): ERD – Communication, Support and Security

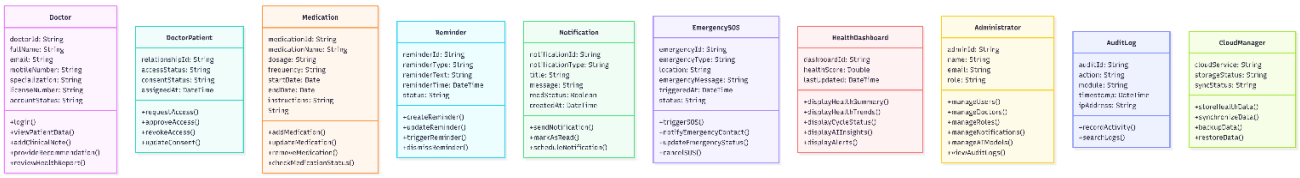
The third category includes information pertaining to communication, medicines, emergencies, notification, and security. AI_CONVERSATION and AI_MESSAGE maintain the records of the conversation that takes place between the user and the AI Health Assistant.

EMERGENCY_SOS deals with all emergency events, whereas NOTIFICATION manages all notifications from the system. The AUDIT_LOG maintains all important events performed by the user and system.

5.4 Class Diagram




MedicalHistory
CycleRecord
HealthTracking
AIHealthIntelligence
-engineId: String
-modelVersion: String
-confidenceThreshold: Double
-dataQualityScore: Double
+analyzeHealthData() : : void
+analyzeTrends() : : void
+generateInsight() : : AIInsight
+calculateHealthScore() : : Double
+detectAnomaly() : : Boolean
+generatePrediction() : : String
DiseasePrediction
-predictionId: String
-diseaseCategory: String
-predictionScore: Double
-confidenceScore: Double
-generatedAt: DateTime
+analyzeRiskFactors() : : void
+generatePrediction() : : void
+calculateConfidence() : : Double
AIInsight
-insightId: String
-insightType: String
-insightText: String
-confidenceScore: Double
-generatedAt: DateTime
+generateInsight() : : void
+updateInsight() : : void
+getInsight() : : String
RiskAssessment
-riskId: String
-riskType: String
-riskLevel: String
-riskScore: Double
-contributingFactors: String
-generatedAt: DateTime
+assessRisk() : : void
+calculateRiskScore() : : Double
+identifyRiskFactors() : : String
+generateRiskAlert() : : void
DigitalHealthTwin
-twinId: String
-healthSummary: String
-currentHealthState: String
-updatedAt: DateTime
+updateHealthState() : : void
+simulateHealthTrend() : : String
+generatePersonalizedInsight() : : String
User
HealthReport
-reportId: String
-reportType: String
-reportTitle: String
-reportData: String
-generatedAt: DateTime
+generateReport() : : void
+exportReport() : : String
+shareReport() : : Boolean
WearableData
-wearableDataId: String
-recordedAt: DateTime
-heartRate: Double
-hrv: Double
-spo2: Double
-ecgData: String
-skinTemperature: Double
-steps: Integer
-calories: Double
-activity: String
-sleepData: String
+collectSensorData() : : void
+validateSensorData() : : Boolean
+synchronizeData() : : Boolean
+getLatestData() : : WearableData
WearableDevice
-deviceId: String
-deviceName: String
-deviceType: String
-serialNumber: String
-pairingStatus: String
-firmwareVersion: String
-lastSync: DateTime
+pairDevice() : : Boolean
+unpairDevice() : : Boolean
+synchronizeData() : : Boolean
+updateFirmware() : : Boolean
+getDeviceStatus() : : String
AIHealthAssistant
-assistantId: String
-aiProvider: String
-contextMode: String
-safetyMode: Boolean
+processQuery() : : String
+detectIntent() : : String
+retrieveHealthContext() : : String
+generateResponse() : : String
+performSafetyCheck() : : Boolean
AIConversation
-conversationId: String
-startedAt: DateTime
-conversationStatus: String
+startConversation() : : void
+endConversation() : : void
+getConversationHistory() : : String
AIMessage
-messageId: String
-senderType: String
-messageText: String
-createdAt: DateTime
+sendMessage() : : void
+validateMessage() : : Boolean



Description:

The **Class Diagram** represents the software-level architecture of AstraCare AI – AI-Based Women's Health Tracking and Predictive Healthcare System, which describes the major class, attribute and method and relation used to realize the system's functional components.

The Class Diagram represents an object-oriented view of the system and shows how different software classes interact. The User class plays a crucial role in the system and is connected with the user profile, medical history, life stage, reproductive health, health tracking, wearable data, AI intelligence, and other supporting services.

Classes connected with security include Authentication Service and Authorization service, which take care about the user authentication, OTP verification, authorization and permission checks. Classes connected with women's health include life stage, cycle, daily health tracking, nutrition, fitness, and mental wellness.

The Wearable Device and Wearable Data classes show how the software system is connected with the AstraBand wearable device and takes care about device pairing, data synchronization, updating, physiological and activity data collection.

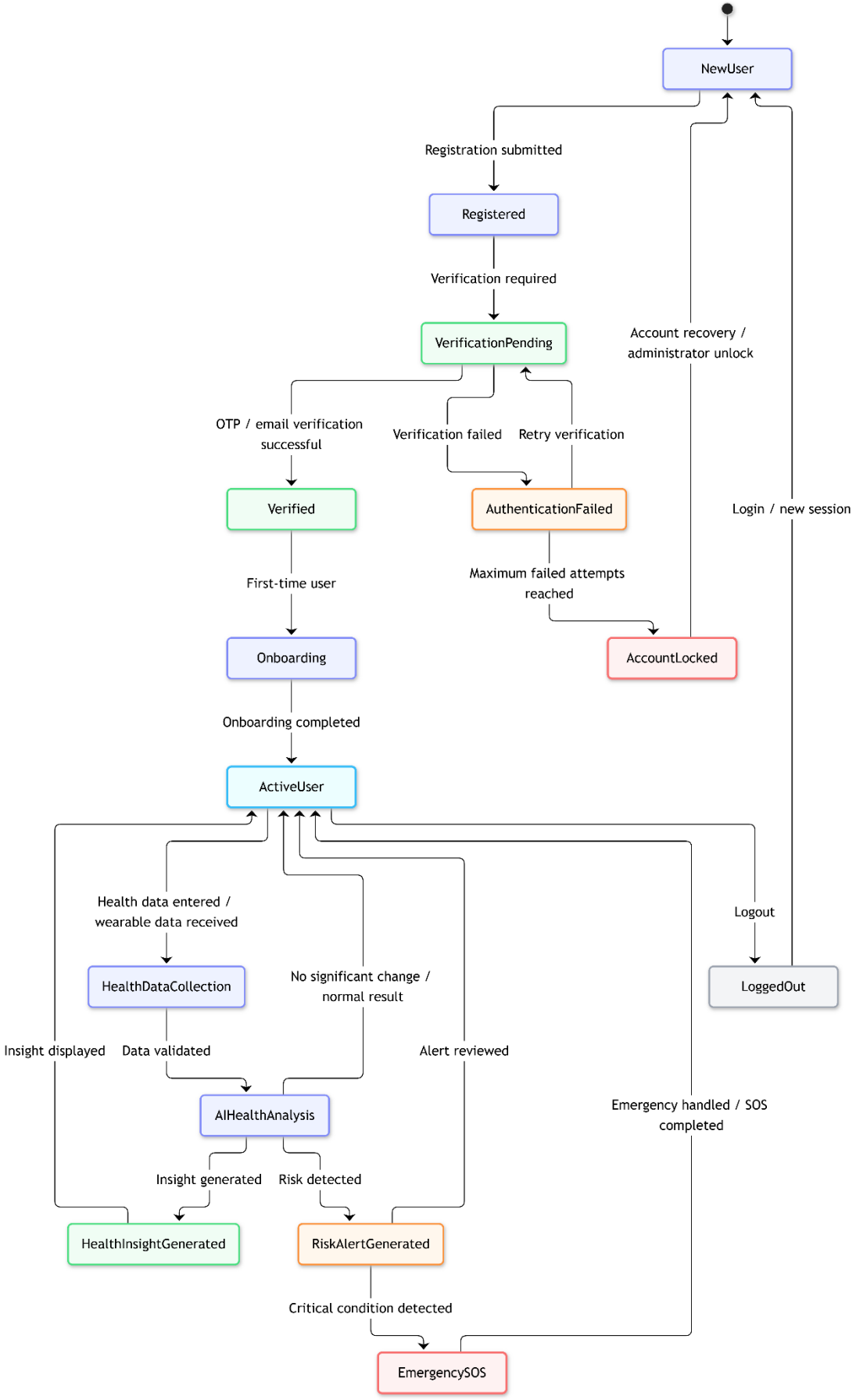
The AI layer includes such classes as AI Health Intelligence, AI Insight, Risk Assessment, Disease Prediction, Digital Health Twin, and Health Report, which helps to analyze the health data, find trends and anomalies, perform risk assessment, make predictions, generate personal insights, make digital representation of the health, and create reports. The AI Health Assistant, AI Conversation, and AI Message classes take care about the user interaction with the AI health assistant.

The system includes classes for healthcare professionals and other supporting services. Classes Doctor and Doctor Patient manage the relationship between the doctor and patients and give an access only to the permitted health information. The Health Dashboard gives health summaries, trends, AI insights, risk information, and reports. The Medication, Reminder, and Notification classes help to manage medications and to schedule reminders and notifications, while Emergency SOS takes care about the emergency-related requests.

The Administrator class manages the system-level functions: users and doctors management, permissions and roles assignment, notifications, and AI model management. The Audit Log class keeps logs of the important system actions for security and accountability reasons, while Cloud Manager helps to safely store, synchronize, back up, and restore health data.

In order to increase readability, the Class Diagram is represented in three related diagrams: User, Security and Women's Health Management; Wearable, AI Intelligence and Health Assistant; and Doctor, Medication, Emergency and System Management.

5.5 State Transition Diagram



Description:

State Transition Diagram explains the different states through which a patient/user goes through while engaging with AstraCare AI system. It shows how the system changes its states in accordance with the events like registration, verification, onboarding, submission of health data, AI analysis, risk alert, emergency, and logout.

The process begins with the New User who registers and reaches the Verification Pending state. After successful OTP or email verification, the user becomes Verified and onboards himself to become Active User.

While operating in regular manner, Active User submits health data or receives data from the wearable device. The system goes through Health Data Collection and after that AI Health Analysis takes place. Based upon the result of analysis, either the Health Insight, Risk Alert or the Active User state comes depending upon whether there is any important change or not. In case of identification of any critical situation, the system may reach Emergency /SOS state.

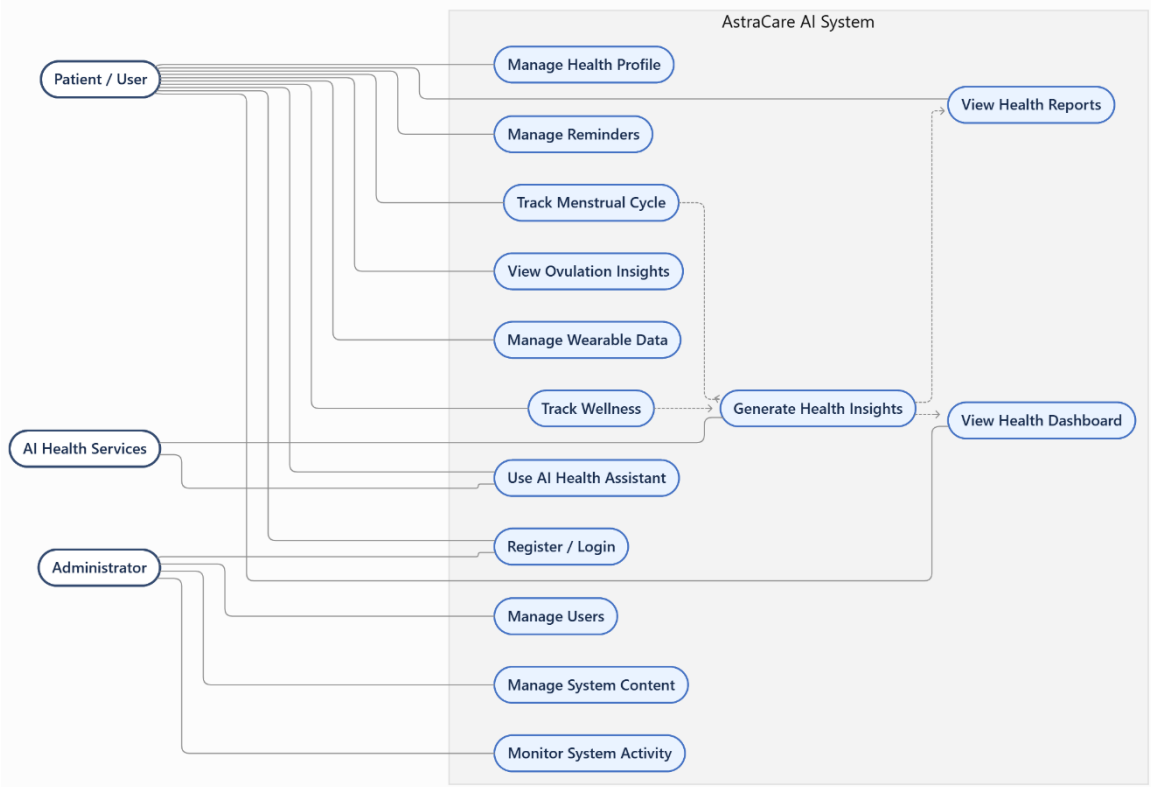
Moreover, the diagram also includes how the authentication failure is handled by the system. The failure of verification results into the Authentication Failed state. Repeated failure may cause Account Locked state. Recovery of account or assistance of administrator helps in getting back to the system again. Lastly, an Active User logs out after which a new session can be started again.

Key States

- **New User** – State prior to registering the account.
- **Registered** – User registration has been completed.
- **Verification Required** – User needs verification of his/her identity/contacts.
- **Verification Completed** – Verification process was successful.
- **Onboarding** – User creates initial profile including health history, goals, preferences etc.
- **Active User** – The core working state of the system.
- **Health Data Input** – Input of health data either manually or from devices.
- **AI Health Processing** – Analysis of collected health data by AI.
- **Insight Created** – Custom health insight created.
- **Risk Created** – Potential health risk detected.
- **Emergency / SOS** – Emergency health state created.
- **Logout** – Logged out state.
- **Auth Failed** – The authentication process was unsuccessful.
- **Account Locked** – Account blocked due to multiple attempts.

Diagram gives us a dynamic picture of the communication between the women and the AstraCare AI system. In contrast to the Class Diagram, which shows classes of software and their relations, and the DFD, which shows the flow of data, State Transition Diagram is concerned with the state of the system/user and the events that affect its state. Thus, it gives us an insight into how the AstraCare AI system operates from registration until session termination.

5.6 Use Case Diagram



Description of the Use Case Diagram

The use case diagram represents the interaction between different actors and the AstraCare AI system.

- **Patient/User:** Registers and logs in, manages personal health information, tracks menstrual cycles and wellness, views the dashboard and health reports, uses the AI Health Assistant, and manages reminders.
- **Administrator:** Manages user accounts, maintains system content, and monitors system activity.
- **AI Health Services:** Supports the AI Health Assistant and generates health insights from available records.
- **System interactions:** Health and wellness records can contribute to generated insights, which may be displayed on the dashboard or included in reports.

5.7 Architecture Design

AstraCare AI architecture, which is an AI-driven system that helps to monitor women health and predict future healthcare needs, reflects the overall structure of the system and relations between its software modules, AI services, wearables, data storage and external services. In order to keep the system clear and understandable, the architecture is split into four interrelated parts.

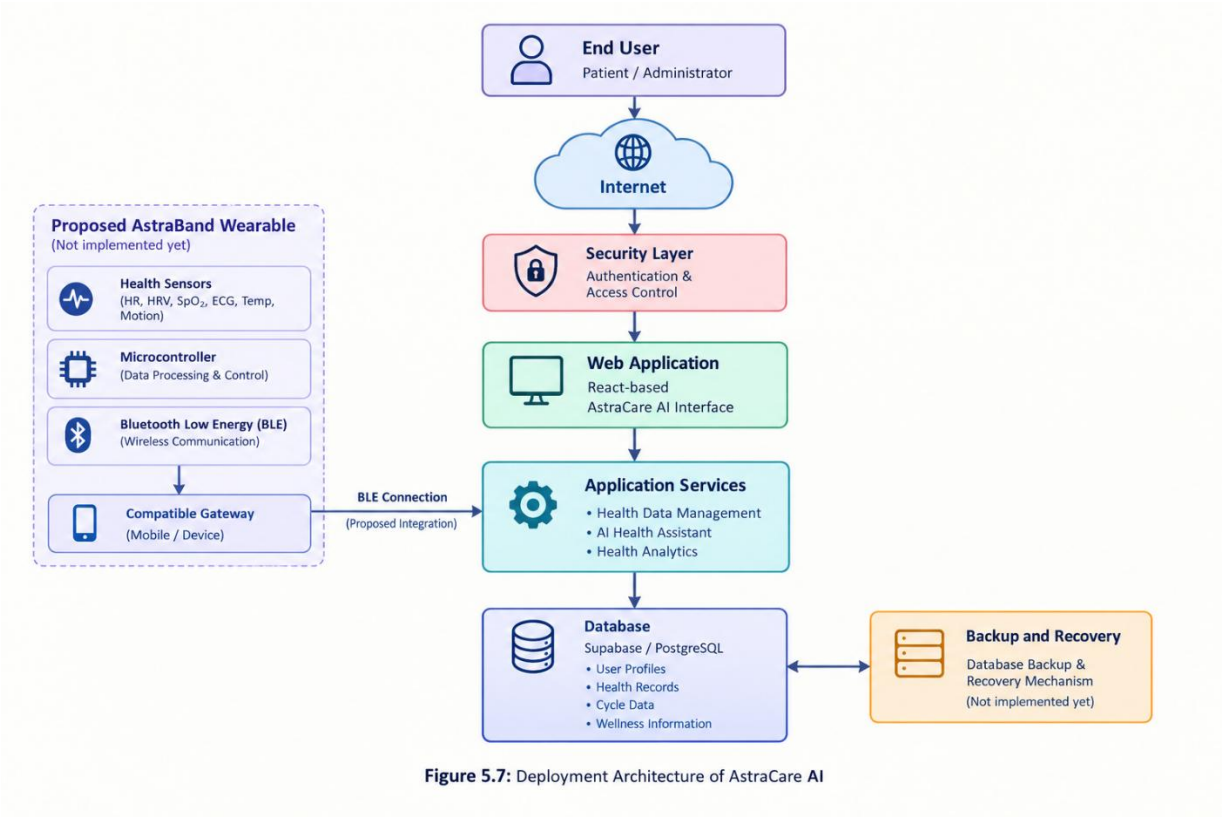


Figure 5.7: Deployment Architecture of AstraCare AI

Description:

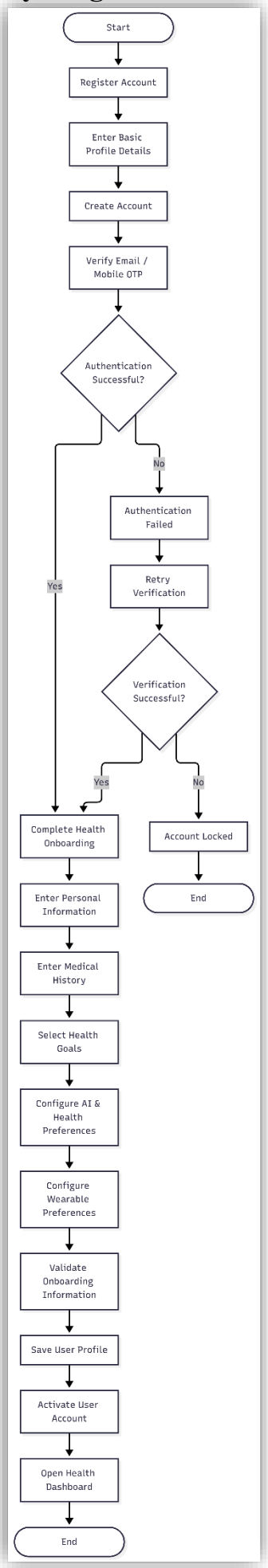
Deployment architecture of AstraCare AI represents the user interaction with the platform as well as data exchange among the web application, backend services and database. End user – either patient or administrator – is able to access the application over the Internet. The security layer is responsible for managing authentication and authorization in order to ensure protection of user accounts and their medical data.

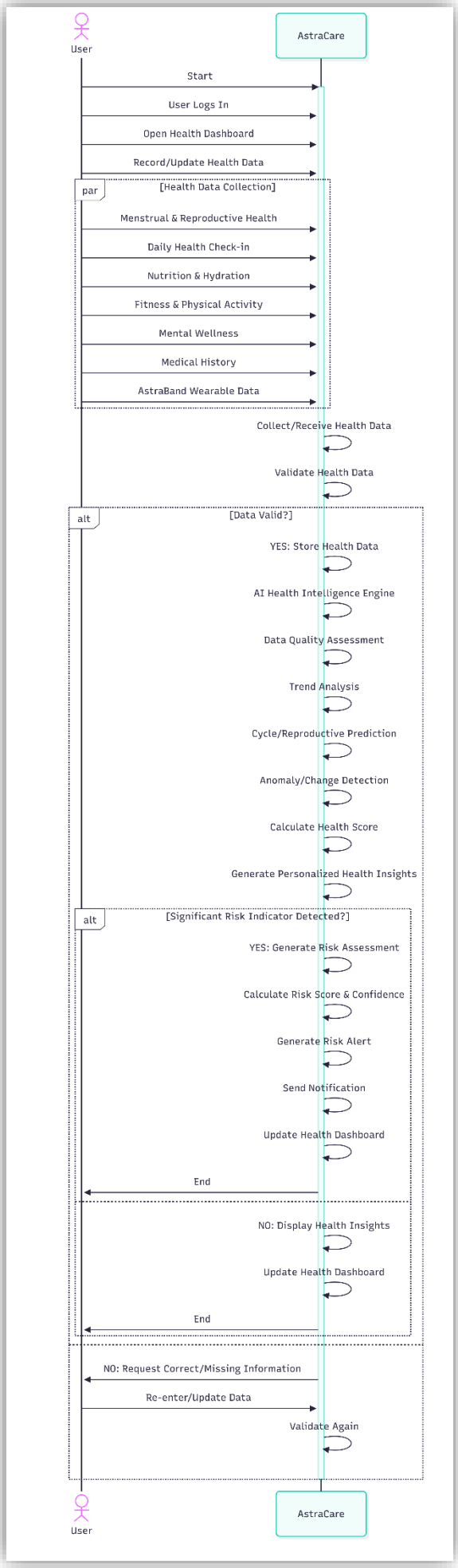
React-based web application provides an interface for accessing health tracking functionality and viewing the corresponding information. Application services represent health data management functionality, AI health assistant and health analytics. Such services have direct connection to the Supabase/PostgreSQL database storing such data as users' profiles, health data, menstrual cycles data and wellness data.

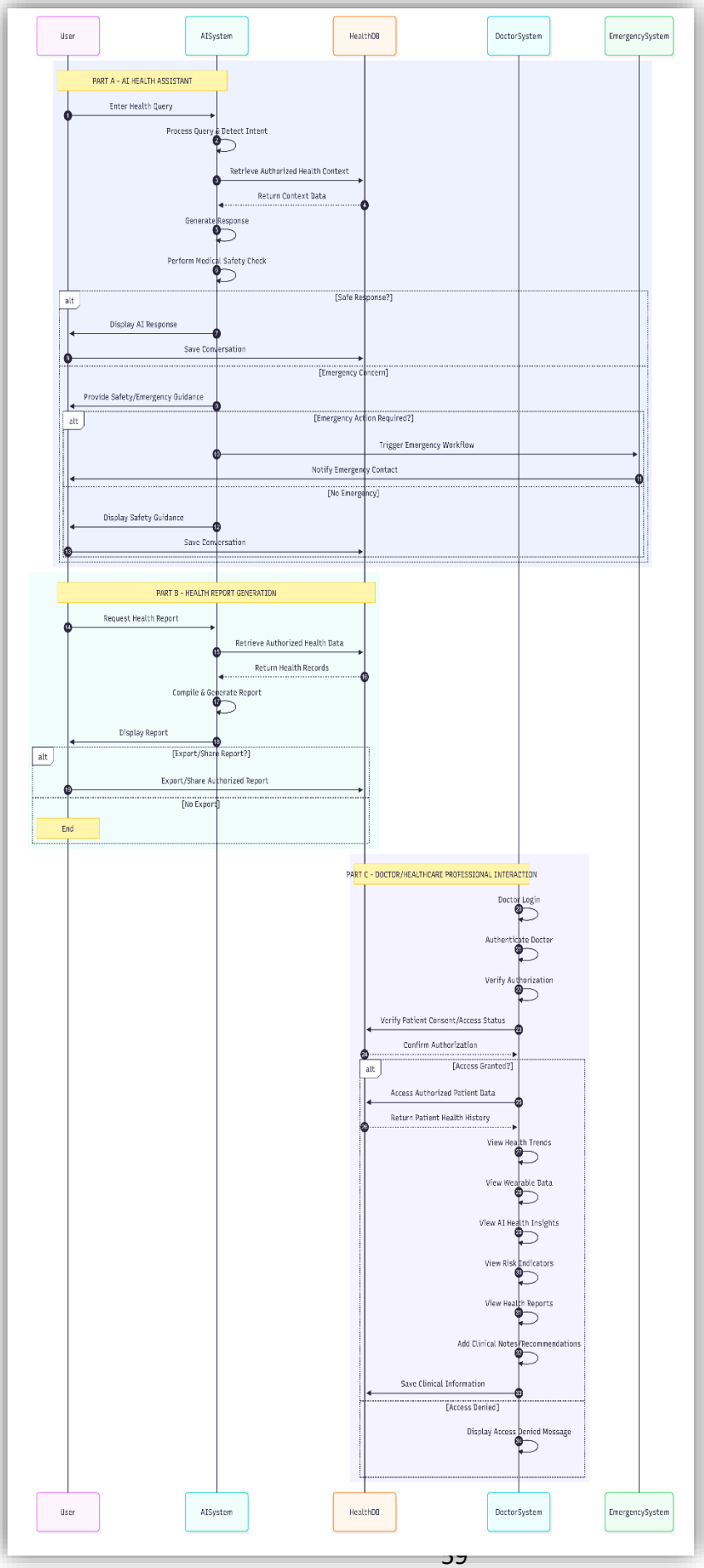
Architecture also includes backup and recovery functionality which should be utilized when necessary for data recovery purposes. Moreover, AstraBand wearable device suggested for implementation should include physiological sensors as well as microcontroller in order to collect and process the relevant data. Data exchange between the wearable and application services is supposed to happen using Bluetooth Low Energy technology and a proper gateway in the further integration.

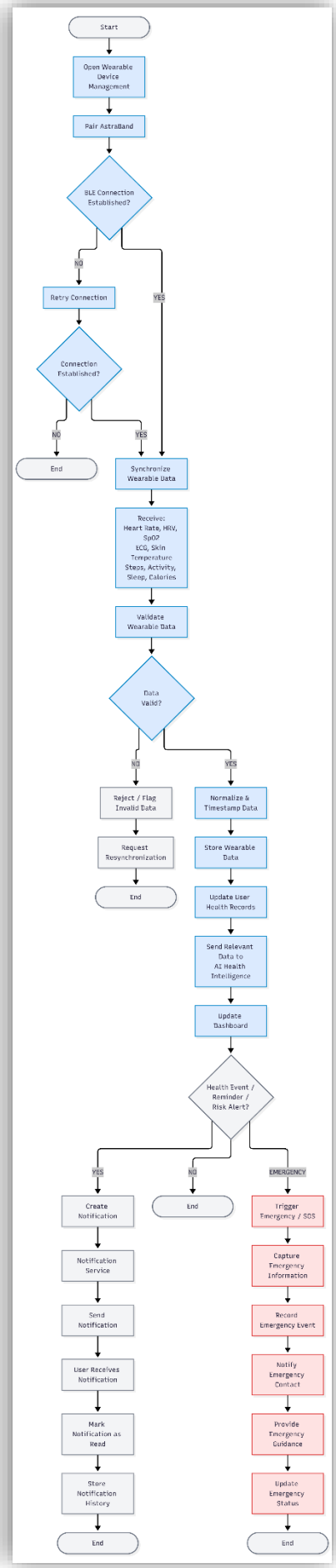
Thus, above-mentioned architecture provides an overview of the key elements and interactions within the system. Integration of the AstraBand and backup/recovery functionality needs to be implemented and tested.

5.8 Activity Diagram






User
AISystem
HealthDB
DoctorSystem
EmergencySystem
PART A - AI HEALTH ASSISTANT
Enter Health Query
Process Query & Detect Intent
Retrieve Authorized Health Context
Return Context Data
Generate Response
Perform Medical Safety Check
alt
[Safe Response?]
Display AI Response
Save Conversation
[Emergency Concern]
Provide Safety/Emergency Guidance
alt
[Emergency Action Required?]
Trigger Emergency Workflow
Notify Emergency Contact
[No Emergency]
Display Safety Guidance
Save Conversation
PART B - HEALTH REPORT GENERATION
Request Health Report
Retrieve Authorized Health Data
Return Health Records
Compile & Generate Report
Display Report
alt
[Export/Share Report?]
Export/Share Authorized Report
[No Export]
End
PART C - DOCTOR/HEALTHCARE PROFESSIONAL INTERACTION
Doctor Login
Authenticate Doctor
Verify Authorization
Verify Patient Consent/Access Status
Confirm Authorization
alt
[Access Granted?]
Access Authorized Patient Data
Return Patient Health History
View Health Trends
View Wearable Data
View AI Health Insights
View Risk Indicators
View Health Reports
Add Clinical Notes/Recommendations
Save Clinical Information
[Access Denied]
Display Access Denied Message
User
AISystem
HealthDB
DoctorSystem
EmergencySystem


Start
Open Wearable Device Management
Pair AstraBand
BLE Connection Established?
NO
YES
Retry Connection
Connection Established?
NO
YES
End
Synchronize Wearable Data
Receive: Heart Rate, HRV, SpO2 ECG, Skin Temperature Steps, Activity, Sleep, Calories
Validate Wearable Data
Data Valid?
NO
YES
Reject / Flag Invalid Data
Request Resynchronization
End
Normalize & Timestamp Data
Store Wearable Data
Update User Health Records
Send Relevant Data to AI Health Intelligence
Update Dashboard
Health Event / Reminder / Risk Alert?
YES
NO
EMERGENCY
Create Notification
Notification Service
Send Notification
User Receives Notification
Mark Notification as Read
Store Notification History
End
End
Trigger Emergency / SOS
Capture Emergency Information
Record Emergency Event
Notify Emergency Contact
Provide Emergency Guidance
Update Emergency Status
End

Description:

The Activity Diagrams illustrate all the key activities performed in the AstraCare AI system such as registration of user, monitoring of their health data, health analysis using the artificial intelligence, interaction using artificial intelligence, reporting of health reports, communication with the doctor, synchronization with wearables, notifications, and emergency services. In order to ensure clarity, the whole workflow is illustrated using four activity diagrams.

The entire process starts with the registration and validation of the account via e-mail or mobile one-time password (OTP). Upon successful validation, the user then proceeds to complete the health onboarding process by giving profile details, health history, health goals, and preferences. Once the onboarding process is complete, the user gains access to the health management services of the system.

In normal use, the user enters data that pertain to menstrual and reproductive health, nutrition, hydration, exercise, mental well-being, and daily health status. Information about physiology and activity levels could also be entered using AstraBand. Validation and storage of the entered data is carried out by the system before forwarding of data to the AI Health Intelligence Engine for analysis. The AI layer is responsible for analysis of trends, calculation of health scores, detection of any anomalies, and provision of personal health insights. An alert or notification is raised upon any risk indicator.

The activity flow also includes the AI Health Assistant, which is made up of the processing of user queries using intent detection, retrieving authorized health context, processing by AI, and safety verification before a response is made. Health reports can be created from the health information stored and are accessible, exportable, or shareable based on the authorization granted. Information about the patients can be accessed by doctors after authentication, authorization, and with patient consent.

The system also handles AstraBand synchronization via BLE. The wearables' data gets processed through validation, time stamping, storing and integration with the patient's health records. Notifications are created for alerts and reminders. During emergencies, the Emergency/SOS workflow handles the situation by recording the event and providing the necessary emergency guidelines.

These three activity diagrams, put together, form an exhaustive diagram of the dynamic workflow of AstraCare AI that explains the flow of information from data gathering to its validation, storing, artificial intelligence analysis, decision support, notifications, professional involvement, and emergency procedures.

5.9 Component Diagram

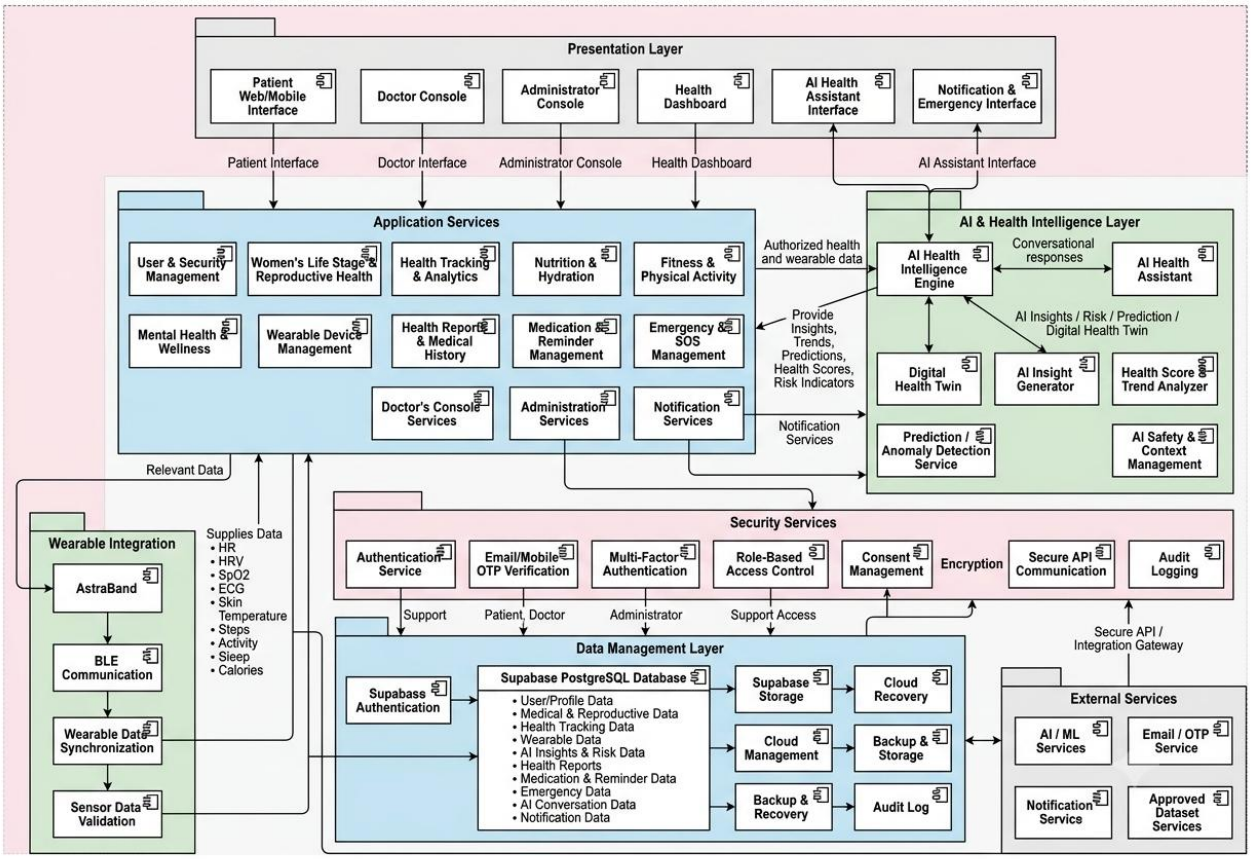


Figure 5.9: Component Diagram of AstraCare AI – AI-Based Women's Health Tracking and Predictive Healthcare System

Description:

Component Diagram highlights the software components used by AstraCare AI along with their dependencies. The System Architecture comprises the following components: Presentation, Application Services, Artificial Intelligence & Health Intelligence, Data Management, Wearable Integration, Security Services, and External Services Component.

The presentation layer includes interfaces for the patients/users, doctors, and administrators like the health dashboard, AI Health Assistant, notifications, and emergency operations. These interfaces interact with the application services which contain all the main functionalities of AstraCare AI.

The AI and Health Intelligence Layer includes the AI Health Intelligence Engine, AI Health Assistant, health-risk assessment and predictions, Digital Health Twin, generating insights, health score analyses, trend analysis, and anomaly detection. These elements use health, reproductive, tracking, medical, and wearable data to generate health insights, predictions, trends, risks, and assistance.

Management of the communication process of AstraBand, BLE connection, synchronization, and validation of sensor data is done by Wearable Integration modules. The gathered data from wearable devices is transmitted to the data management layer and AI for further analysis.

The Data Management Layer consists of Supabase Authentications, PostgreSQL database, Storage, Cloud Management, backup and recovery, and audit logging. These elements enable persistent storage and access control to the health and application data of the system.

The security services span across the entire system and include Authentication, OTP Verification, MFA, Role-Based Access Control, Consent Management, Row Level Security, Encryption, Secure API communications, and Audit Logs. All the external services of AI/ML, OTP, Notification, and Approved Datasets use a secure API/integration gateway as opposed to accessing the internal database.

5.10 Table Design

Table 5.1 — USER

Field	Data Type	Key	Description
user_id	UUID	PK	Unique identifier of the user
full_name	VARCHAR	—	User's full name
email	VARCHAR	—	Registered email address
mobile_number	VARCHAR	—	Registered mobile number
password_hash	VARCHAR	—	Hashed password
date_of_birth	DATE	—	User's date of birth
gender	VARCHAR	—	Gender information
account_status	VARCHAR	—	Current account status
created_at	TIMESTAMP	—	Account creation date and time

Purpose: Stores the basic account and identity information of AstraCare AI users.

Table 5.2 — USER_PROFILE

Field	Data Type	Key	Description
profile_id	UUID	PK	Unique profile identifier
user_id	UUID	FK	References USER
height	DECIMAL	—	User's height
weight	DECIMAL	—	User's weight
activity_level	VARCHAR	—	General activity level
smoking_status	BOOLEAN	—	Smoking status
alcohol_status	BOOLEAN	—	Alcohol-use status
health_goals	TEXT	—	Selected health goals

Purpose: Stores additional personal, lifestyle and health-goal information.

Table 5.3 — MEDICAL_HISTORY

Field	Data Type	Key	Description
medical_history_id	UUID	PK	Unique medical-history identifier
user_id	UUID	FK	References USER
conditions	TEXT	—	Existing medical conditions
surgeries	TEXT	—	Previous surgeries
allergies	TEXT	—	Known allergies
current_medications	TEXT	—	Current medications
family_history	TEXT	—	Relevant family medical history
created_at	TIMESTAMP	—	Record creation time

Purpose: Maintains the user's medical and family-health history.

Table 5.4 — LIFE_STAGE

Field	Data Type	Key	Description
life_stage_id	UUID	PK	Unique life-stage identifier
user_id	UUID	FK	References USER
current_stage	VARCHAR	—	Current women's health life stage
stage_start_date	DATE	—	Start date of the stage
stage_status	VARCHAR	—	Current status

Purpose: Stores the user's current health/life-stage information.

Table 5.5 — CYCLE_RECORD

Field	Data Type	Key	Description
cycle_id	UUID	PK	Unique cycle record
user_id	UUID	FK	References USER
period_start_date	DATE	—	Start of menstrual period
period_end_date	DATE	—	End of menstrual period
cycle_length	INTEGER	—	Length of cycle
flow_level	VARCHAR	—	Menstrual flow level
symptoms	TEXT	—	Recorded symptoms

Field	Data Type	Key	Description
predicted_next_period	DATE	—	Predicted next period
predicted_ovulation	DATE	—	Predicted ovulation date

Purpose: Stores menstrual-cycle and reproductive-health records.

Table 5.6 — HEALTH_TRACKING

Field	Data Type	Key	Description
tracking_id	UUID	PK	Unique health-tracking record
user_id	UUID	FK	References USER
record_date	DATE	—	Date of health record
sleep_quality	INTEGER	—	Sleep-quality rating
energy_level	INTEGER	—	Energy-level rating
stress_level	INTEGER	—	Stress-level rating
symptoms	TEXT	—	Symptoms reported
activity_status	BOOLEAN	—	Whether activity was performed
health_score	DECIMAL	—	Calculated health score

Purpose: Stores daily, weekly and other health-tracking information used for analytics.

Table 5.7 — WEARABLE_DEVICE

Field	Data Type	Key	Description
device_id	UUID	PK	Unique wearable-device identifier
user_id	UUID	FK	References USER
device_name	VARCHAR	—	Device name
device_type	VARCHAR	—	Type of wearable
serial_number	VARCHAR	—	Device serial number
pairing_status	VARCHAR	—	Current pairing status
firmware_version	VARCHAR	—	Installed firmware version
last_sync	TIMESTAMP	—	Last synchronization time

Purpose: Stores information about AstraBand and other supported wearable devices.

Table 5.8 — WEARABLE_DATA

Field	Data Type	Key	Description
wearable_data_id	UUID	PK	Unique wearable-data record
device_id	UUID	FK	References WEARABLE_DEVICE
user_id	UUID	FK	References USER
recorded_at	TIMESTAMP	—	Sensor-record timestamp
heart_rate	DECIMAL	—	Heart-rate data
hrv	DECIMAL	—	Heart-rate variability
spo2	DECIMAL	—	Blood oxygen level
ecg_data	TEXT	—	ECG information
skin_temperature	DECIMAL	—	Skin temperature
steps	INTEGER	—	Step count
calories	DECIMAL	—	Estimated calories
activity	VARCHAR	—	Activity information
sleep_data	TEXT	—	Sleep information

Purpose: Stores physiological and activity data received from AstraBand.

Table 5.9 — AI_INSIGHT

Field	Data Type	Key	Description
insight_id	UUID	PK	Unique insight identifier
user_id	UUID	FK	References USER
insight_type	VARCHAR	—	Type of AI insight
insight_text	TEXT	—	Generated insight
confidence_score	DECIMAL	—	AI confidence value
generated_at	TIMESTAMP	—	Generation timestamp

Purpose: Stores personalized insights generated by the AI Health Intelligence Engine.

Table 5.10 — RISK_ASSESSMENT

Field	Data Type	Key	Description
risk_id	UUID	PK	Unique risk-assessment identifier
user_id	UUID	FK	References USER
risk_type	VARCHAR	—	Type of risk indicator
risk_level	VARCHAR	—	Risk classification
risk_score	DECIMAL	—	Calculated risk score
contributing_factors	TEXT	—	Factors contributing to assessment
generated_at	TIMESTAMP	—	Assessment timestamp

Purpose: Stores AI-generated health-risk indicators and supporting factors.

Table 5.11 — HEALTH_REPORT

Field	Data Type	Key	Description
report_id	UUID	PK	Unique report identifier
user_id	UUID	FK	References USER
report_type	VARCHAR	—	Type of health report
report_title	VARCHAR	—	Report title
report_data	TEXT	—	Generated report information
generated_at	TIMESTAMP	—	Report generation time

Purpose: Stores generated health reports and summaries.

Table 5.12 — DIGITAL_HEALTH_TWIN

Field	Data Type	Key	Description
twin_id	UUID	PK	Unique digital-twin identifier
user_id	UUID	FK	References USER
health_summary	TEXT	—	Current health summary
current_health_state	TEXT	—	Current modeled health state
updated_at	TIMESTAMP	—	Last update time

Purpose: Maintains the user's continuously updated digital health representation.

Table 5.13 — MEDICATION

Field	Data Type	Key	Description
medication_id	UUID	PK	Unique medication identifier
user_id	UUID	FK	References USER
medication_name	VARCHAR	—	Medication name
dosage	VARCHAR	—	Prescribed dosage information
frequency	VARCHAR	—	Medication frequency
start_date	DATE	—	Medication start date
end_date	DATE	—	Medication end date
instructions	TEXT	—	Additional instructions

Purpose: Stores medication information entered or managed by the user.

Table 5.14 — REMINDER

Field	Data Type	Key	Description
reminder_id	UUID	PK	Unique reminder identifier
user_id	UUID	FK	References USER
medication_id	UUID	FK	Related medication
reminder_type	VARCHAR	—	Type of reminder
reminder_text	TEXT	—	Reminder message
reminder_time	TIMESTAMP	—	Scheduled time
status	VARCHAR	—	Reminder status

Purpose: Stores medication and health-related reminders.

Table 5.15 — EMERGENCY_SOS

Field	Data Type	Key	Description
emergency_id	UUID	PK	Unique emergency-event identifier
user_id	UUID	FK	References USER
emergency_type	VARCHAR	—	Type of emergency
location	TEXT	—	Emergency location
emergency_message	TEXT	—	Emergency information
triggered_at	TIMESTAMP	—	Trigger time

Field	Data Type	Key	Description
status	VARCHAR	—	Current emergency status

Purpose: Stores Emergency/SOS events initiated by the user or system.

Table 5.16 — AI_CONVERSATION

Field	Data Type	Key	Description
conversation_id	UUID	PK	Unique conversation identifier
user_id	UUID	FK	References USER
started_at	TIMESTAMP	—	Conversation start time
conversation_status	VARCHAR	—	Current conversation status

Purpose: Stores AI Health Assistant conversation sessions.

Table 5.17 — AI_MESSAGE

Field	Data Type	Key	Description
message_id	UUID	PK	Unique message identifier
conversation_id	UUID	FK	References AI_CONVERSATION
sender_type	VARCHAR	—	User or AI
message_text	TEXT	—	Message content
created_at	TIMESTAMP	—	Message creation time

Purpose: Stores individual messages exchanged through the AI Health Assistant.

Table 5.18 — NOTIFICATION

Field	Data Type	Key	Description
notification_id	UUID	PK	Unique notification identifier
user_id	UUID	FK	References USER
notification_type	VARCHAR	—	Notification category
title	VARCHAR	—	Notification title
message	TEXT	—	Notification content
read_status	BOOLEAN	—	Whether notification was read
created_at	TIMESTAMP	—	Creation timestamp

Purpose: Stores health alerts, reminders, announcements and system notifications.

Table 5.19 — DOCTOR

Field	Data Type	Key	Description
doctor_id	UUID	PK	Unique doctor identifier
full_name	VARCHAR	—	Doctor's name
email	VARCHAR	—	Doctor's email
mobile_number	VARCHAR	—	Doctor's mobile number
specialization	VARCHAR	—	Medical specialization
license_number	VARCHAR	—	Professional license number
account_status	VARCHAR	—	Doctor account status

Purpose: Stores registered healthcare-professional information.

Table 5.20 — DOCTOR_PATIENT

Field	Data Type	Key	Description
relationship_id	UUID	PK	Unique relationship identifier
doctor_id	UUID	FK	References DOCTOR
user_id	UUID	FK	References USER
access_status	VARCHAR	—	Patient-data access status
consent_status	VARCHAR	—	Patient consent status
assigned_at	TIMESTAMP	—	Assignment time

Purpose: Manages the relationship and authorized access between doctors and patients.

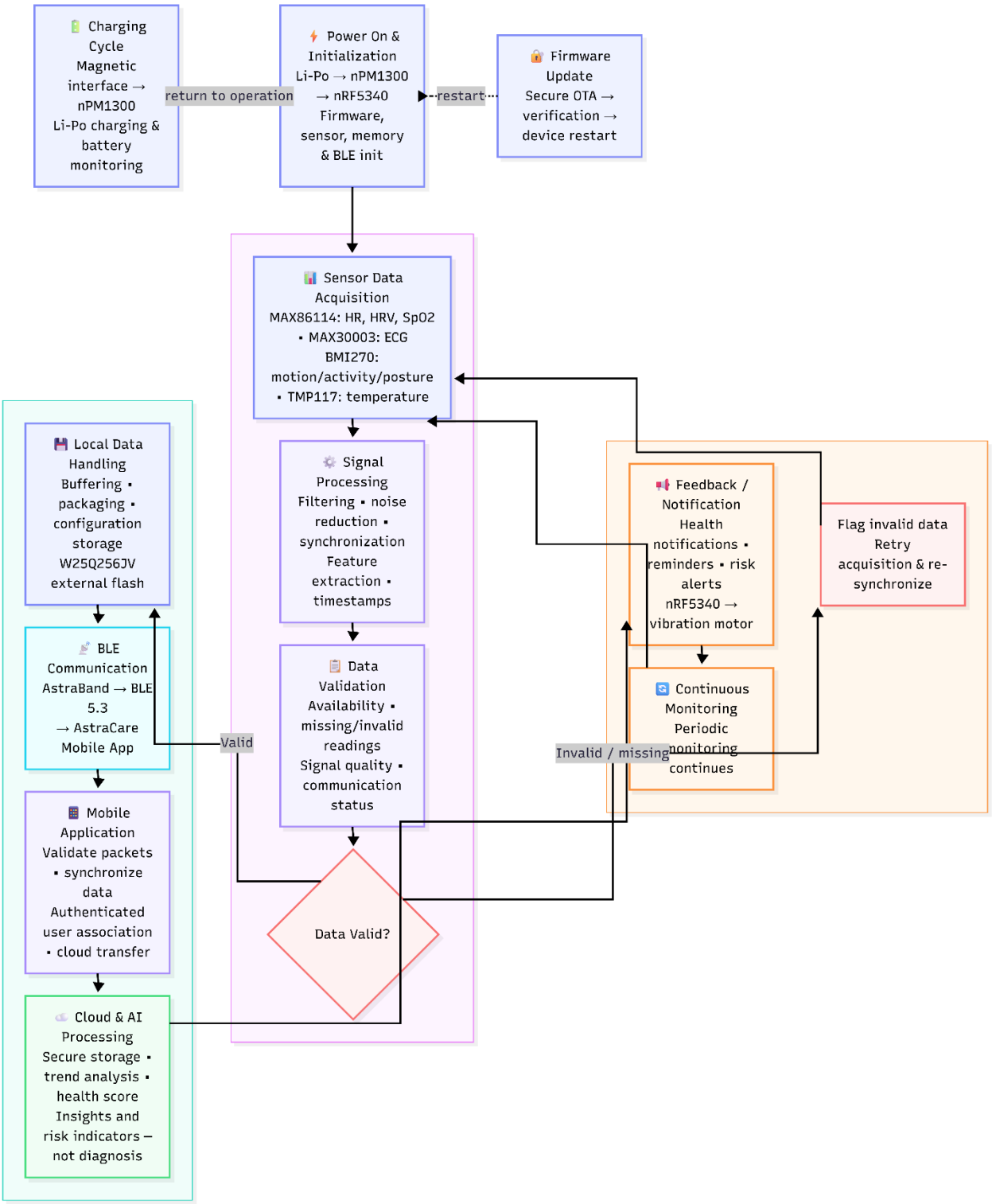
Table 5.21 — AUDIT_LOG

Field	Data Type	Key	Description
audit_id	UUID	PK	Unique audit-record identifier
user_id	UUID	FK	User associated with the activity
action	VARCHAR	—	Performed action
module	VARCHAR	—	Related system module
timestamp	TIMESTAMP	—	Activity timestamp
ip_address	VARCHAR	—	Source IP address

Purpose: Records important system and security activities for auditing and monitoring.

5.2 Hardware System Design

5.2.1 Detailed Life Cycle of AstraBand



Description:

Life Cycle activities for the AstraBand involves the entire process in which the device turns ON, collects biological and physical data of the user, processing and verification of the collected data and then transmits the information to AstraCare AI.

First comes the step of power ON and system initiation. The Li-Po battery powers the system through nPM1300 Power Management System. After which the microcontroller, firmware, sensors, memories and finally the BLE (Bluetooth Low Energy) comes on. Physiological and activity data are measured from the MAX86114, MAX30003, BMI270, TMP117 sensor respectively.

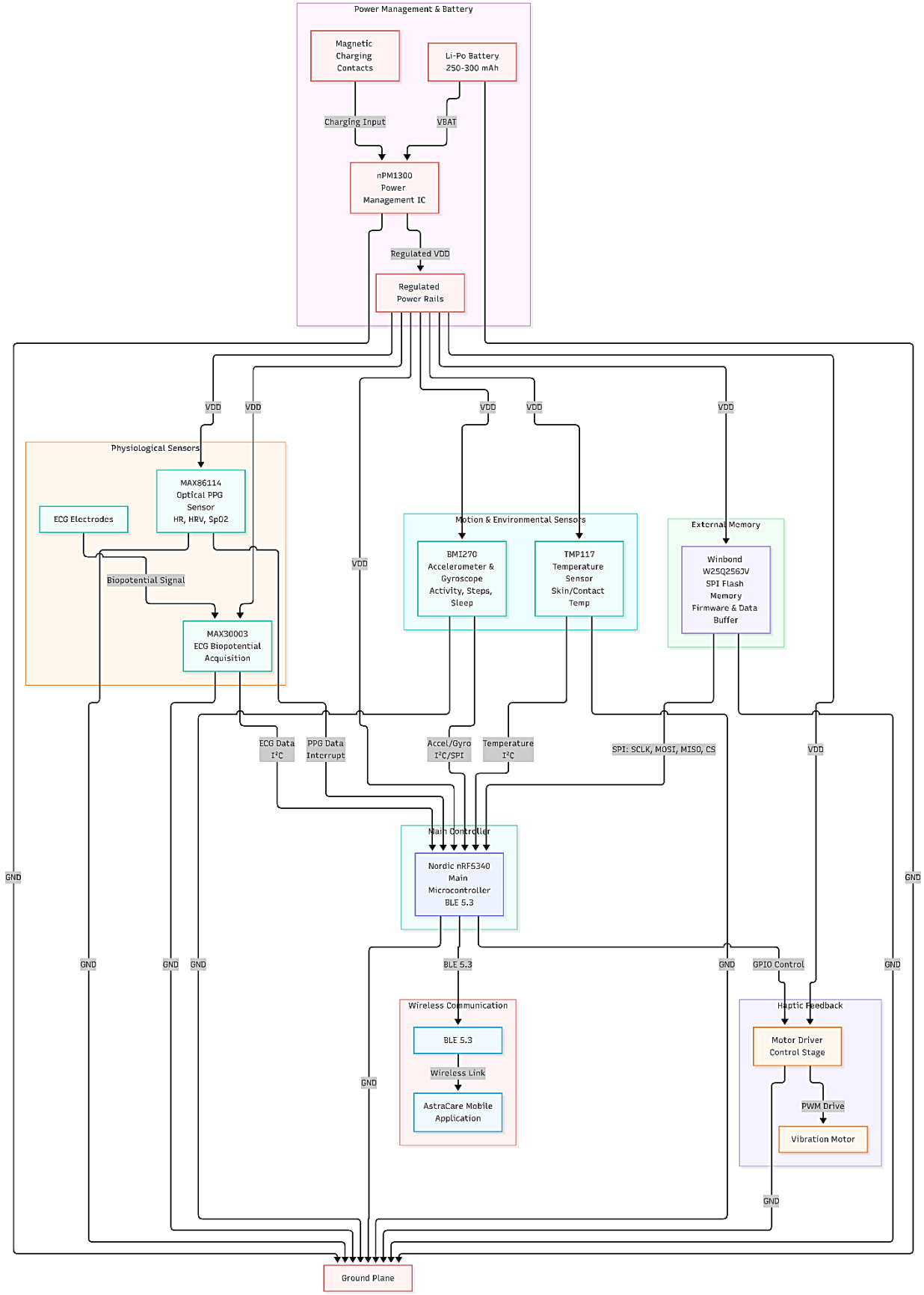
The collected data signals are then processed through a series of filtering, timestamping, synchronization, and feature extraction method in the nRF5340. Thereafter, the data is validated for possible issues such as any form of invalid sample data, loss of data, corrupted data, communication issues, and signal problems. Any invalid data sample collected can be resampled before any processing can take place.

Data that is confirmed to be valid could be buffered for temporary storage through external memory (W25Q256JV). The nRF5340 thereafter transmits the collected data to AstraCare AI mobile application through BLE 5.3. This collected information in the AstraCare AI mobile application gets further verified and synchronized. It will then transmit the verified information to the cloud platform.

At this point, the information in the cloud will be stored and processed using the AstraCare AI Health Intelligence System for calculation of health trends, calculation of health scores and risk indicators respectively. Feedbacks to the user will be done through application notifications and vibrations from AstraBand where applicable.

Lifecycle activities will always be done periodically during normal operating times. Another function includes charge and safe updates to the firmware in the wearable device.

5.2.2 Circuit Diagram of AstraBand



Description:

Circuit diagram is a diagramatic representation of the connection between the main elements that have been employed in the development of AstraBand. nRF5340 functions as a control board and interfaces with physiological and motion sensor, flash memory and vibration feedback circuitry. MAX86114 is used to carry out optical sensing including heart rate, HRV and SpO_2 readings while MAX30003 is used for acquiring ECG readings. BMI270 records information related to motion while TMP117 monitors temperature.

The nPM1300 is responsible for the control of the charging process and power distribution from the 250–300 mAh Li-Po battery to the electronic component. Charging is done through the magnetic contact system. W25Q256JV chip adds some non-volatile flash memory. The vibration motor gives additional tactile feedback.

The nRF5340 chip also support Bluetooth Low Energy 5.3 technology for transmitting wearable devices processed data to the AstraCare application. The diagram thus shows the full electrical interaction of power management, sensing, processing, storage, feedback and wireless communication.

5.2.3 Component-Level Description and Specifications

Sr. No.	Component	Key Specifications	Interface	Function in AstraBand
1	Nordic nRF5340	Dual-core ARM Cortex-M33; BLE 5.3; low-power wireless SoC	I²C, SPI, GPIO, BLE	Main controller; manages sensors, processing, memory, communication and device control
2	MAX86114	Optical PPG sensor; supports heart-rate and SpO_2 measurement	I²C / sensor interface	Captures optical signals for HR, HRV and SpO_2-related measurements
3	MAX30003	Ultra-low-power ECG/biopotential AFE	SPI, ECG electrodes	Acquires ECG/biopotential signals
4	BMI270	Low-power 6-axis IMU; accelerometer + gyroscope	I²C / SPI	Detects movement, steps, activity and posture-related information
5	TMP117	High-accuracy digital temperature sensor	I²C	Measures skin/contact temperature
6	nPM1300	Power-management IC with battery charging and power-management functions	I²C / GPIO	Manages battery charging, power delivery and battery monitoring
7	W25Q256JV	256-Mbit serial NOR Flash memory	SPI	Stores firmware, configuration and temporary buffered data

Sr. No.	Component	Key Specifications	Interface	Function in AstraBand
8	250–300 mAh Li-Po Battery	Rechargeable lithium-polymer battery	Power	Provides portable power to AstraBand
9	Vibration Motor	Compact low-power haptic actuator	GPIO/control via driver	Provides tactile feedback for alerts, reminders and device events
10	Magnetic Charging Interface	Magnetic charging contacts	Power/charging	Provides convenient rechargeable charging connection
11	Custom Multi-Layer PCB	Compact multilayer wearable PCB	Electrical interconnect	Integrates MCU, sensors, power management, memory and supporting circuitry
12	IP67 Enclosure	Dust-protected and water-resistant enclosure	—	Protects the electronics and provides a wearable, ergonomic structure

Nordic nRF5340

nRF5340 is a core processing and communication unit in the AstraBand. It takes care of all the communication operations between the AstraBand and sensors, processes the collected information, and controls the external memory and BLE communication with the AstraCare application.

Its in-built BLE 5.3 feature helps eliminate the use of additional Bluetooth module and thus reduces the size of the wearable device.

MAX86114

MAX86114 is used in optical physiological sensing. This module delivers the optical sensing subsystem necessary for obtaining PPG signals which are used for heart rate and SpO_2 monitoring.

The sensor is properly placed against the user's skin to ensure the quality of optical measurement under normal wear conditions.

MAX30003

The MAX30003 serves as the ECG / Biopotential acquisition module. The device acquires electrical cardiac information using proper ECG electrodes and delivers this information to the nRF5340.

With this component, AstraBand can acquire ECG information along with optical information.

BMI270

BMI270 serves as the motion sensor module of AstraBand. Data from its accelerometer and gyroscope can be used in measurement of activities and movements, which may include counting steps and monitoring movements during sleep.

TMP117

The TMP117 is capable of providing digital temperature readings. In AstraBand, it will be used to measure the temperature of the skin, which can also be analyzed in conjunction with other wearable sensors.

nPM1300

This part of the wearable is controlled by the nPM1300, which takes care of the power management subsystem of the wearable.

In addition to the above tasks, it also assists in battery monitoring and allows efficient operation under a constrained power budget.

W25Q256JV

W25Q256JV is an external Serial NOR Flash memory chip connected to the nRF5340. This gives more non-volatile memory space to store firmware, configuration data, and buffered data.

With the use of external memory, AstraBand can keep necessary data while there is no continuous communication with the mobile application.

Li-Po Battery

It is suggested that a rechargeable Li-Po battery rated 250–300 mAh be used as the portable power supply for AstraBand due to its small size.

The battery will connect to the nPM1300, not powering the components directly.

Vibration Motor

The vibration motor offers haptic feedback without the need for a screen. It is used for notifications, status of the device, and for certain health notifications.

The motor has to be connected using a suitable driver stage instead of connecting it directly to the GPIO of the MCU if its current draw exceeds that of the GPIO.

Magnetic Charging Interface

The charging contacts for magnetic charging are the physical interfaces for charging between AstraBand and its charger. The charging contacts connect to the nPM1300, which controls the rechargeable Li-Po battery.

Custom Multi-Layer PCB

Custom multi-layered PCB assembly brings together all processing, sensing, memory, and power management components in one tiny wearable package.

Points that should be considered during the PCB assembly include:

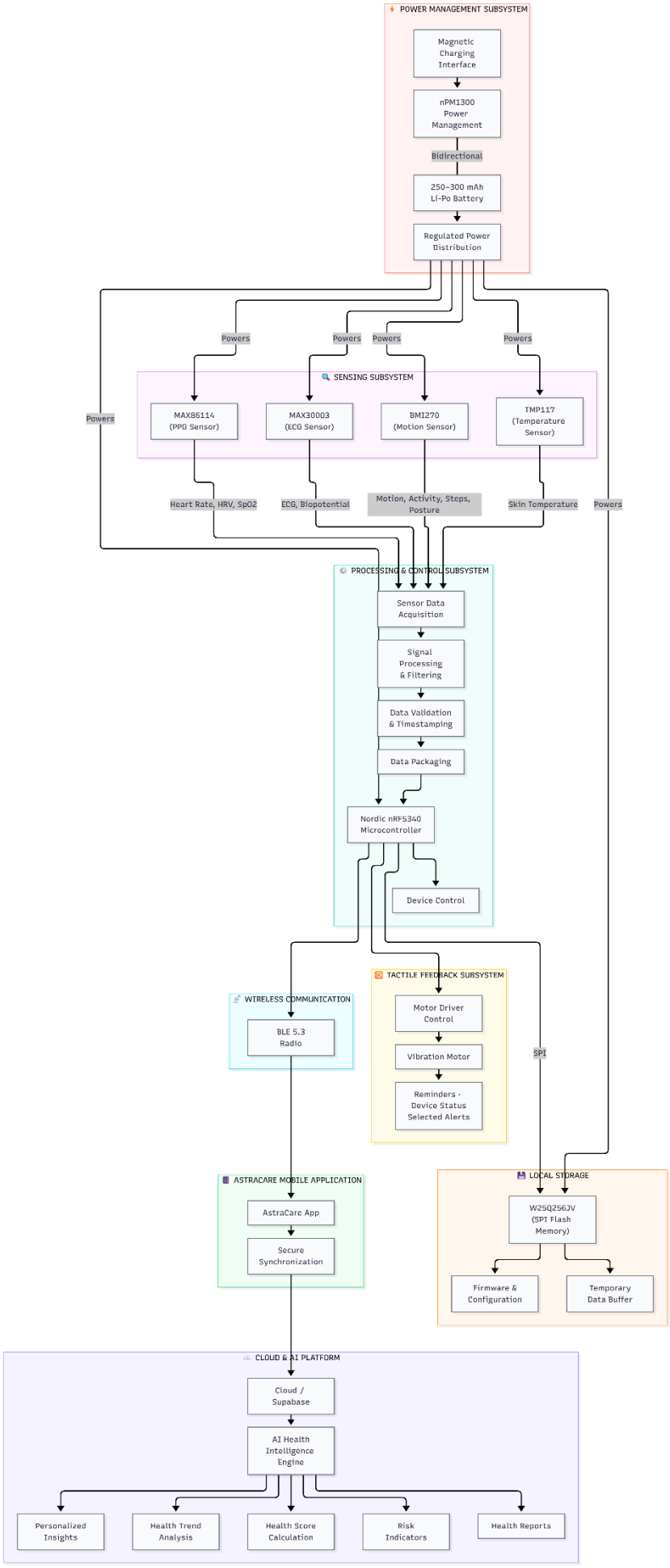
- Sensor location
- Power routing
- Signal integrity
- ECG signal routing
- RF/BLE routing
- Battery placement
- Heat dissipation
- Physical dimensions

IP67 Enclosure

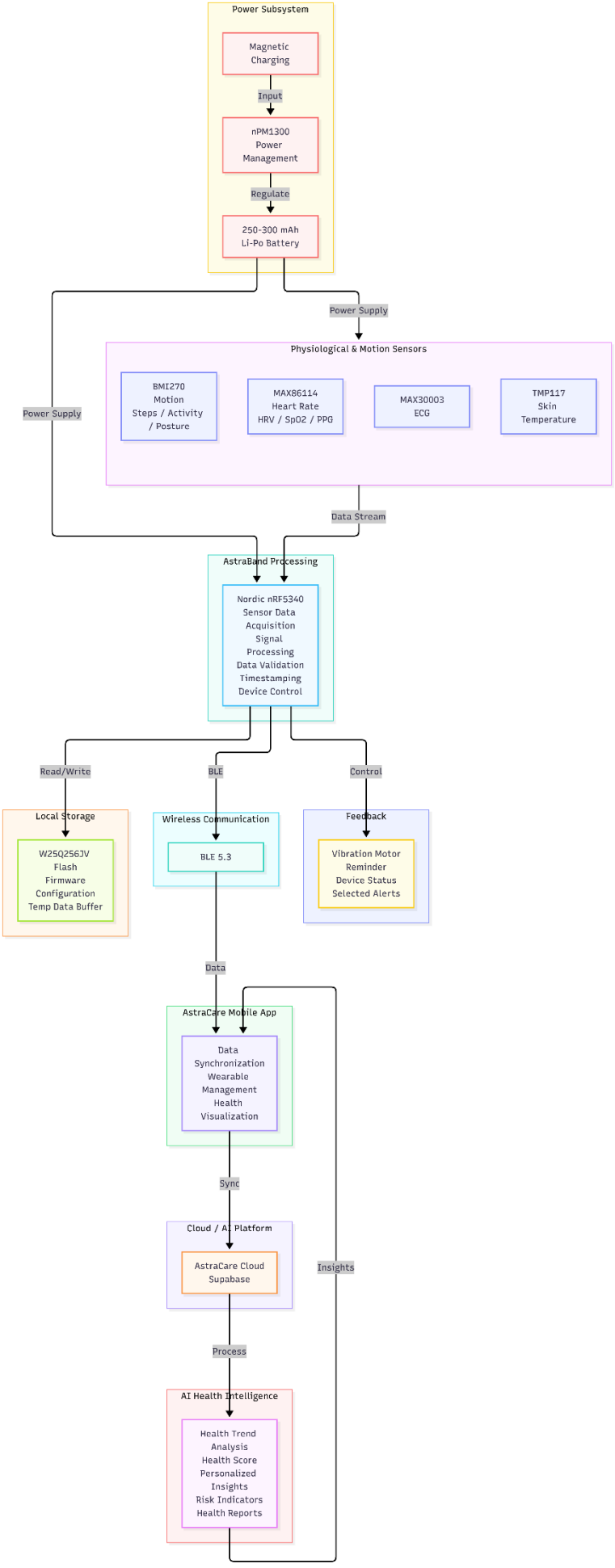
The AstraBand enclosure serves to keep the electronic components protected from exposure to the environment while offering a compact and ergonomically sound wearable enclosure.

The desired IP67 rating offers protection against dust along with protection against temporary immersion. It is important that the enclosure keeps sufficient contact between the sensors and the wearer’s skin.

5.2.4 Hardware Architecture Design



5.2.5 Hardware Block Diagram



Description:

The hardware block diagram is an overview of the wearable AstraBand device. The data from the physiological, motion, and temperature sensors are collected using MAX86114, MAX30003, BMI270, and TMP117 sensors. This data is then passed to the nRF5340 that works as the processor and controller.

The nRF5340 will carry out basic data processing and validation before storing the data in the local memory in W25Q256JV flash memory. The processed data will then be sent via BLE 5.3 to the AstraCare mobile application. This will then synchronize the data from the authorized wearable with AstraCare cloud platform where the AI Health Intelligence Engine will analyze the health trends and create insights, health scores, risks and reports.

The power sub-system includes the Li-Po battery, nPM1300 power management IC and magnetic charging circuit. The nRF5340 is also responsible for controlling the vibration motor that provides tactile feedback to selected device events.

6. System Implementation and Methodology

6.1 Development Methodology

The AstraCare AI project is based on an Agile and Incremental Development Methodology. Given that the system involves software, AI, database, and AstraBand hardware, it will be developed in small increments and step by step in small parts it will be integrated. Each development increment will be tested before proceeding to the next one.

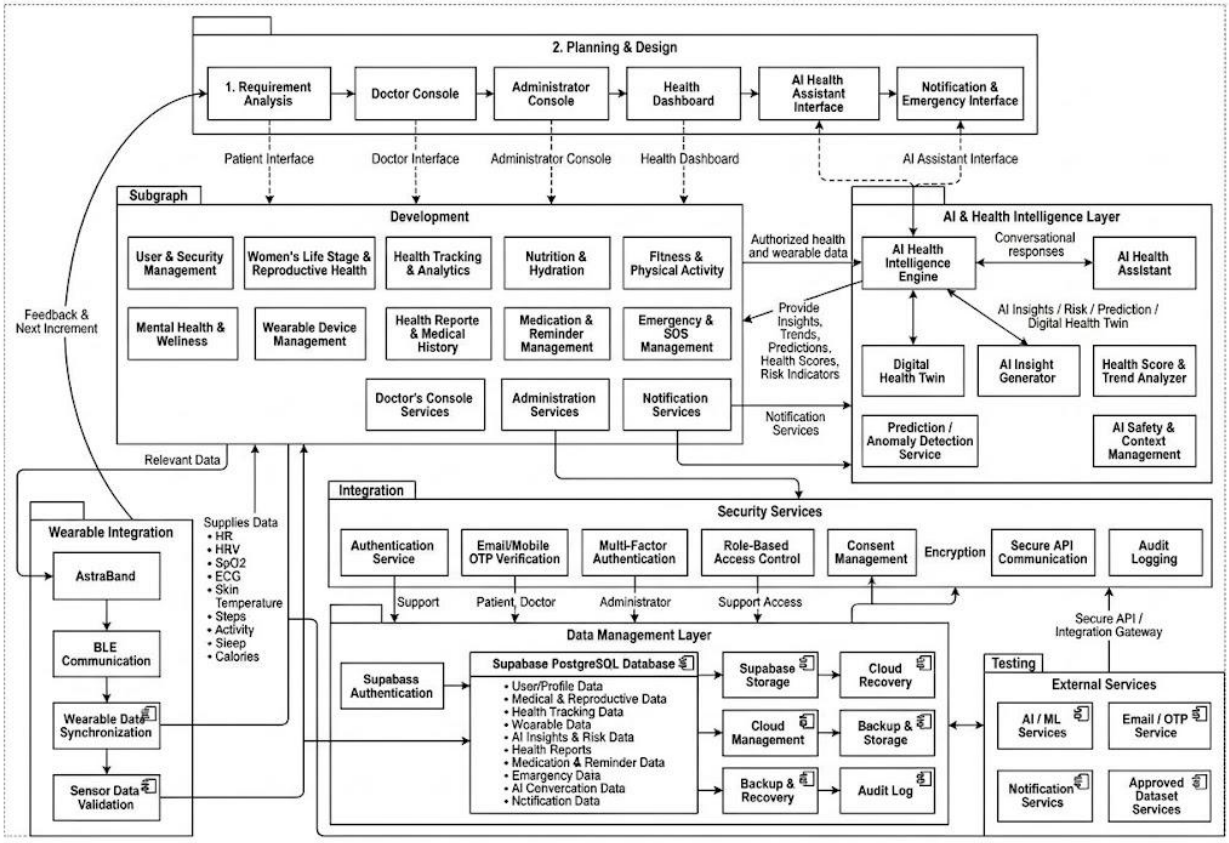


Figure 6.1: Agile and Incremental Development Methodology of AstraCare AI

Development Process

1. Requirement Analysis

The software, hardware, AI, database, security, and user requirement are identified. The systems functional and non-functional requirements are defined.

2. System & Hardware Design

The system architecture, database structure, UML, data-flow diagrams, and AstraBand hardware architecture are designed. The necessary sensors, microcontroller, power-management circuits, memory, communication tools, and other elements are determined.

3. Software Development

The software components like user management, women's health tracking, dashboard, analytics, nutrition, hydration, fitness, mental wellness, AI features, reports, notifications, doctor services, and administration are developed.

4. Hardware Development

The AstraBand hardware board based on the nRF5340 microcontroller is developed. The MAX86114, MAX30003, BMI270, and TMP117 sensors are integrated, and power management, flash memory, battery, charging, vibration circuits are added.

5. AI Integration

The AI Health Intelligence Engine is integrated, which analyzes the validated data and provides health trends, health score, relevant insights, and risk indicators. The AI Health Assistant is created to offer conversational health support, with contextually relevant information.

6. Hardware–Software Integration

The AstraBand and AstraCare App are integrated over BLE 5.3. The data from the AstraBand are received by the AstraCare App, where they are validated, synchronized, and sent to a cloud platform for processing.

7. Testing and Validation

The developed software components, hardware elements, database, BLE, and AI components are tested, and the system's operations are validated.

8. Refinement

The system is refined and improved on the basis of the testing results. The application errors are removed, and the software components, hardware, and AI parts are enhanced.

6.2 Technologies Used

Category	Technology / Component	Purpose
Frontend	HTML5, CSS3, JavaScript	Structure, styling and client-side functionality
Frontend UI	React	Component-based user interface development
Backend / API	Node.js	Server-side application and API development
Database	PostgreSQL	Structured storage of user, health, wearable and application data
Backend Platform	Supabase	Database services, authentication, storage and backend integration
Authentication	Supabase Auth, OTP	User registration, login and identity verification
Authorization	RBAC, Row Level Security (RLS)	Role-based and record-level access control
AI / ML	Python	AI/ML processing and model development
AI Intelligence	AI Health Intelligence Engine	Health trends, scores, insights and risk indicators
AI Assistant	NLP / Conversational AI	User interaction and health-related conversational assistance
Data Processing	Python / Pandas	Data preprocessing and analysis
Data Visualization	JavaScript / Charting Libraries	Health trends, statistics and dashboard visualization
API Communication	REST APIs / HTTPS	Communication between application services and external services
Wireless Communication	Bluetooth Low Energy 5.3	Communication between AstraBand and mobile application

Category	Technology / Component	Purpose
Microcontroller	Nordic nRF5340	Central controller of AstraBand
PPG / SpO_2 Sensor	MAX86114	Heart-rate, HRV and SpO_2-related sensing
ECG Sensor	MAX30003	ECG / biopotential signal acquisition
Motion Sensor	BMI270	Motion, activity, steps and posture-related sensing
Temperature Sensor	TMP117	Skin/contact temperature measurement
Power Management	nPM1300	Battery charging, power management and monitoring
External Memory	W25Q256JV	Firmware, configuration and temporary data storage
Power Source	250–300 mAh Li-Po Battery	Portable power supply for AstraBand
Feedback	Vibration Motor	Tactile alerts and device feedback
Charging	Magnetic Charging Interface	Rechargeable wearable charging
Hardware Design	Custom Multi-Layer PCB	Integration of wearable electronics
Enclosure	IP67 Enclosure	Physical protection of wearable electronics
Development Environment	Antigravity IDE	Software development and AI-assisted development
Version Control	Git / GitHub	Source-code management and collaboration
Security	HTTPS, Encryption, MFA, RBAC, RLS, Audit Logs	Protection of application and health data

Description:

AstraCare AI is a modern web solution that utilizes web technologies, cloud platforms, artificial intelligence, databases, communication means, and hardware components. The application is based on front and back-end, giving a medical platform to users, doctors, and administrators. The front-end provides the user interface for a user, doctor, and admin, while back-end is responsible for managing the application service and API operation.

Supabase and PostgreSQL database help to manage, save, and ensure safety and the applications' proper performance through the implementation of the security features, such as authentication, roles, Row Level Security (RLS), encryption, and audit logging. Besides, there are some AI technologies used that are capable of processing various data using specific algorithms and providing insight, health trend, score, and prediction. This AI health assistant allows communicating through a chatbot while considering health protection measures.

The work of the AstraCare AI system takes place due to the utilization of hardware components like sensors, microcontroller, flash memory, and battery. AstraBand was developed by using Nordic nRF5340 MCU to provide an advanced solution for physiological, motion, and temperature sensors. The system utilizes Bluetooth Low Energy 5.3 for creating a safe connection between wearable devices and mobile application. In addition, the nPM1300 power management solution and Li-Po battery pack are used as a great power supply to the wearable device, while flash memory provides additional storage capabilities.

6.3 Algorithms / Techniques Used

Sr. No.	Algorithm / Technique	Application in AstraCare AI
1	Authentication & OTP Verification	Verifies user identity during registration and login
2	Role-Based Access Control (RBAC)	Controls access for patients, doctors and administrators
3	Row Level Security (RLS)	Restricts database records according to authorized users
4	Data Validation & Preprocessing	Validates health and wearable data before analysis
5	Feature Engineering	Converts raw health information into useful analytical features
6	Trend Analysis	Identifies changes and patterns in health data over time
7	Cycle Prediction	Estimates future menstrual-cycle events using available historical cycle information
8	Anomaly / Change Detection	Identifies unusual changes or deviations in tracked health measurements
9	Health Score Calculation	Combines relevant health indicators into a summarized health score
10	Risk Assessment	Generates risk indicators based on available health information
11	Personalized Recommendation Logic	Generates user-specific health and wellness suggestions
12	Natural Language Processing (NLP)	Understands user queries submitted to the AI Health Assistant
13	Context Retrieval	Retrieves relevant user health information for contextual AI responses
14	Confidence & Data-Quality Assessment	Indicates the reliability of AI-generated insights based on available data

Sr. No.	Algorithm / Technique	Application in AstraCare AI
15	Digital Health State Modeling	Maintains a structured representation of the user's current and historical health state
16	Time-Series Analysis	Analyzes health measurements collected across different dates and periods
17	BLE Data Synchronization	Transfers and synchronizes AstraBand data with the AstraCare application
18	Sensor Data Filtering & Validation	Reduces invalid/noisy wearable readings before application-level analysis

Description:

1. Data Validation and Preprocessing

Health information gathered by the users and AstraBand is initially validated before undergoing the processing by the AI modules. The main aim is to detect and handle missing, invalid, inconsistent or low-quality values beforehand.

2. Feature Engineering

The raw health information is processed into engineered features that hold meaningful information suitable for analysis. Some of the expected features are cycle length, sleep, activity, stress, heart-rate trends and others.

3. Time-Series and Trend Analysis

AstraCare AI focuses on a broader picture by analysing health information over a period of time (more than one entry). As a result, it becomes possible to track changes and see trends in sleep, activity and cycle data as well as wearable measures.

4. Cycle Prediction

Using historical cycle information, it becomes possible to predict future cycle-related events. The prediction process is based on the available information and historical data patterns instead of following the same cycle routine across all users.

5. Anomaly / Change Detection

AstraCare tracks irregular changes in any of the available measurements or followed behaviour. The detected changes can then be used to raise awareness instead of presenting a diagnosed medical condition.

6. Health Score Calculation

The health score is determined from a subset of the available indicators. The information is aggregated and scored to show a quick overview of the current tracked health state. It should be noted that the score is dynamic and may update over time depending on the information availability.

7. Risk Assessment

The risk-assessment module combines relevant health information and detected patterns to come up with risk indicators. Again, these indicator should only be used as an starting awareness before seeking the professional's confirmation.

8. Natural Language Processing

The AI Health Assistant employs NLP and conversation techniques to detect the user's need. After identifying the intent, relevant information or context can be retrieved before providing a practical response.

9. Context-Aware AI Response

The AI Assistant can utilise authorised information such as user profiles, health records, tracking details and relevant AI context to offer better responses compared to a standard chatbot.

10. Confidence and Data-Quality Assessment

Inside the insights provided by the AI, it is always possible to identify the degree of confidence and information quality. In a practical sense, it means that not all information presented is certain or evidence-based.

11. BLE Synchronization

AstraBand employs Bluetooth Low Energy 5.3 to sync information with the AstraCare app. In particular, the wearable measurements are transferred, validated and synced before they are added to the user's health records or analytics.

6.4 Module-wise Implementation

The AstraCare AI is designed as a set of eighteen integrated software modules that operate as a modular healthcare platform. Each module works for a specific healthcare need and is connected through application services, API, database operations, AI services, and wearable-device services, etc. The best part of the platform is that each component can be developed, tested, maintained, and updated separately.

The technology is applied through frontend interfaces, backend services, database systems using Supabase, authentication and authorization mechanisms, AI-integrated health intelligence, wearable device integration, notification systems, reporting tools, and administrative functions.

Module	Module Name	Implementation Description
M1	User & Security Management	Implements user registration, login, email/mobile verification, authentication, profile management, authorization, role-based access, and security controls.
M2	Women's Life Stage & Reproductive Health	Implements menstrual cycle tracking, fertility and ovulation information, life-stage tracking, pregnancy-related information, postpartum and menopause support.
M3	AI Health Intelligence Engine	Processes available health and tracking data to identify trends, generate personalized insights, analyze changes, and provide data-driven health intelligence.
M4	AI Health Assistant (Chatbot)	Provides a conversational interface through which users can ask health-related questions and receive context-aware, evidence-aware wellness information and guidance.
M5	Health Dashboard & Analytics	Displays health summaries, health metrics, trends, cycle information, hydration status, activity information, and other personalized indicators through an interactive dashboard.
M6	Nutrition & Hydration	Allows users to record meals, monitor nutritional information, track water consumption, and view nutrition and hydration-related progress.
M7	Fitness & Physical Activity	Records physical activities, steps, active duration, calories, sleep-related information, and provides activity-related insights based on available health data.

Module	Module Name	Implementation Description
M8	Mental Health & Wellness	Provides daily emotional check-ins, stress-level recording, wellness tracking, reflection/journal input, and mindfulness-oriented support.
M9	Wearable Device Management	Handles AstraBand registration, pairing, synchronization, device information, sensor data transfer, device status, and wearable configuration.
M10	Health Reports & Medical History	Stores medical history and generates/organizes health reports containing relevant user health information and historical records.
M11	AI Disease Prediction & Risk Assessment	Provides a framework for processing health indicators and generating risk indicators and risk scores from available validated data. These outputs are intended as decision-support indicators and not medical diagnoses.
M12	Medication & Health Reminder	Allows users to maintain medication/supplement schedules and receive reminders for medication, hydration, and other health activities.
M13	Emergency & SOS	Provides an emergency interface through which users can initiate an SOS workflow and access configured emergency-support information.
M14	Doctor’s Console	Provides authorized healthcare professionals with controlled access to patient information, reports, trends, medical history, and relevant health insights.
M15	Admin Management	Provides administrative functions for managing users, doctors, system configuration, permissions, notifications, and system-level operations.
M16	Notifications	Manages system notifications, reminders, health alerts, announcements, and other user-facing notifications.
M17	Digital Health Twin	Maintains a longitudinal computational representation of the user's available health information, trends, physiological indicators, and wellness state.
M18	Cloud Management	Manages secure cloud-based storage, synchronization, database operations, backup-related functions, and controlled access to application data.

6.5 Database Implementation

The database implementation of AstraCare AI was achieved by the use of Supabase PostgreSQL database. The database was incorporated in the application to act as a storage system for user data, tracking data, records, AI-generated information and other related information.

Database implementation process entailed development of database tables as per the final database design and application and database integration through Supabase APIs. This involved the development of functions for creation, retrieval, updating, and deletion of records in the application.

Database implementation process comprised of the following main stages:

1. **Creation of Database and Tables**
1. Creation of required PostgreSQL database tables in Supabase project as per final database design.
2. **Application and Database Integration**
   Integration of application frontend and backend with Supabase to facilitate storage and retrieval of dynamic data as opposed to static data.
3. **Insertion and Updating of Information**
   This process entails insertion and updating of information into the database. It comprises of user profiles, health tracking information, cycle records, nutrition information, fitness information, medication and other related information.
4. **Data Retrieval**
   It entails retrieving the required information from the database and presenting it in the relevant dashboards and modules. This also includes retrieval of historical information when required.
5. **Module-Wise Data Management**
   The relevant data is managed in the relevant modules. For example, cycle data is managed in reproductive-health tracking module while medication data is managed in the medication module and reminder section.
6. **Authentication Data Integration**
   The process of authentication was integrated with Supabase authentication to enable application users to access application features by logging into their accounts.
7. **Validation of Data to Be Entered into the Databases**
   The process of data validation is done to ensure that the data to be entered in the database meets certain criteria. This minimizes the amount of wrong data in the database.
8. **Update of Application According to Database Information**
   It involves updating application interface, dashboard, tracking module and other elements based on the information in the database

6.6 User Interface Design

The user interface of AstraCare AI was designed to provide an easy-to-use and responsive experience when managing the health information of women. The UI design of the application is consistent across all modules to make it easy for the users to switch from one feature to another.

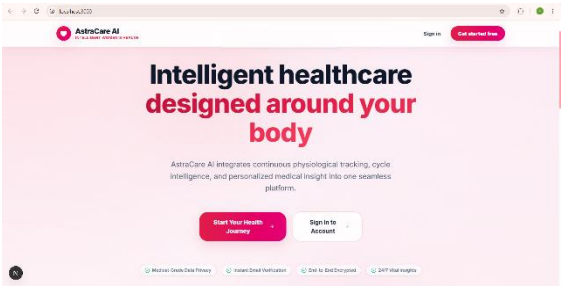
The main UI design principles considered in the implementation of the software are:

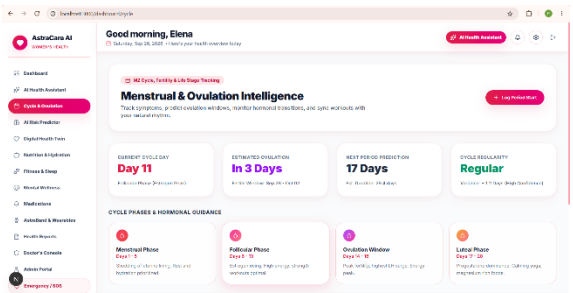
1. **User-Friendly Navigation**
   The application makes it easy for the users to move around between all sections, including Dashboard, Women's Health, Nutrition, Fitness, Mental Wellness, AI Assistant, Reports and Medication.
2. **Dashboard-Based Design**
   The use of the dashboard as the centre of all activities and as an overview of the health state and recent developments is used in the software.
3. **Consistent Visual Design**
   The consistent colours, fonts, buttons, cards and spacing are used by the software to create a uniform design and enhance the navigation. Need affordable essay writing services? Our essay writers are available 24/7 to help you out!
4. **Visualizing Health Data**
   All health data, including the information about the menstrual cycle, activity, sleep, hydration and other metrics can be visualized through the use of proper cards, indicators, progress bars and other visualization tools.
5. **Responsive Design**
   AstraCare AI application is responsive and thus can be used both on desktop and mobile devices.
6. **Easy Data Input**
   The software uses such input controls as simple forms, checkboxes, radio buttons and others to make it easier for the users to enter their health data.
7. **AI Assistant User Interface**
   The health assistant has a conversational UI, allowing the user to ask health-related questions and get contextual answers.
8. **Notifications and Feedback**
   All important notifications, alerts, confirmations and other messages from the system are shown clearly to help the user understand the status of the actions performed by them.
9. **Accessibility and Readability**
   A proper text contrast, readability, legibility, clear labelling and sufficient spacing are used in the software to improve its accessibility.
10. **Consistency in Modules**
    Consistent UI components and patterns are used across all modules of the application.

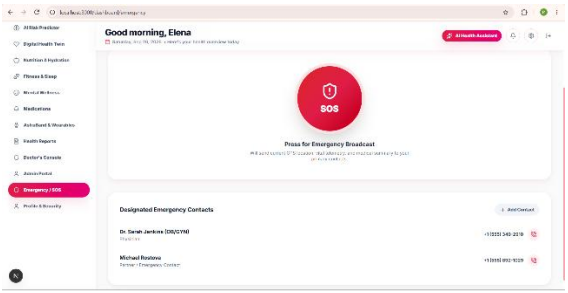
UI Design Flow:

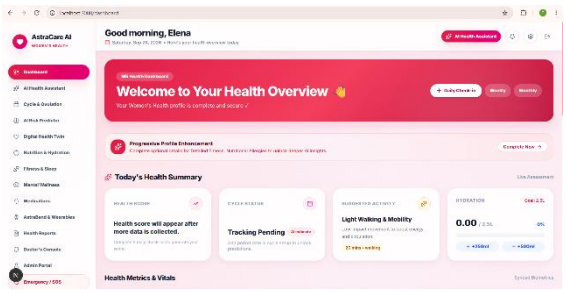
Login / Registration → Onboarding → Health Dashboard → Health Modules → AI Assistance → Reports & Reminders

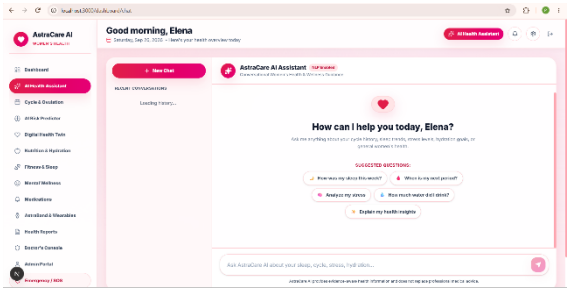
Some of the images of AstraCare AI:

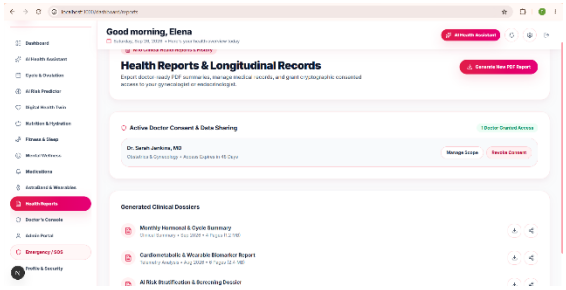












7. System Testing

7.1 Testing Strategy

The strategy of testing of the AstraCare AI – AI-based women's health tracking and predictive healthcare system will be aimed at the verification of correctness, reliability, security, and usability of the software and the AstraBand wearable device. Testing will be carried out on different levels of testing so that the functionality of modules and components as a whole system will be checked.

The critical processes include:

Unit Testing: The functions, components, APIs, database components, and hardware interfaces are tested as separate components to meet the defined requirements.

Integration Testing: The interactions between the software components, services, Supabase, AI services, notification services, AstraBand communication are evaluated to ensure effectiveness.

System Testing: The complete system is tested as a whole. The components that were tested individually and as parts of the system were evaluated together to check if they meet the intended purpose. The features include user management, health tracking, AI analysis, dashboards, reports, reminders, doctor access, and administration.

Hardware Testing: The AstraBand components including nRF5340, MAX86114(MAX30003), BMI270, TMP117, memory, battery-management circuit, vibration motor, and BLE communication are tested.

Hardware–Software Integration Testing: This process focuses on ensuring that the data collected from AstraBand is correctly transmitted, synchronized, validated, stored, and visualized in the AstraCare AI application.

Security Testing: This component includes authentication, OTP, authorization, role-based access control, Row Level Security, protected APIs, consents, and auditing to ensure that the health information of the users is accessible only by the authorized personnel.

AI and Data Validation Testing:

The AI operations which include knowledge, health trends, prediction, anomalies, confidence, and data validation are tested under valid and invalid conditions. The AI output is intended to be health information, and it must not be used as diagnostic information.

Usability Testing:

The testing includes usability of user interface, onboarding, dashboards, tracking, AI assistant, notifications, and navigation capabilities.

Performance Testing:

The testing is done for performance of response time, databases, synchronization process, APIs, dashboards, and wearables data.

Error and Recovery Testing:

Error and recovery testing are carried out on input errors, OTP fail, BLE sync failure, network connection problem, invalid sensors data, and service disruption.

The testing procedure was done using the following process:

Test Planning → Unit Testing → Integration Testing → System Testing → Hardware Testing → Hardware–Software → Security & AI Validation → Performance Testing → User Validation → Final Verification

The testing processes identified above were vital in developing a reliable healthcare system. The application offers complete health tracking and ensures the security of health data. Most importantly, the AI-assisted health data analysis meets the required standards and has effective communication channels with the wearable devices.

7.2 Testable Unit

In relation to AstraCare AI, a testable unit is a particular software module, hardware part, service, or interface which can be independently tested in order to ascertain whether it fulfills its purpose.

Testable units include:

Sr. No.	Testable Unit	Main Testing Area
1	M1 – User & Security Module	Registration, login, OTP, authentication, authorization, profile management
2	M2 – Women's Life Stage & Reproductive Health Module	Cycle tracking, reproductive health records, life-stage information
3	M3 – AI Health Intelligence Engine Module	Health-data analysis, trends, predictions, anomaly/change detection, insights
4	M4 – AI Health Assistant (Chatbot) Module	User queries, intent handling, context retrieval, AI responses and safety handling
5	M5 – Health Dashboard & Analytics Module	Health score, trends, cycle status, sleep, stress, recovery and dashboard data
6	M6 – Nutrition & Hydration Module	Food/nutrition records, water intake, hydration goals and tracking
7	M7 – Fitness & Physical Activity Module	Activity records, exercise tracking and activity-related data
8	M8 – Mental Health & Wellness Module	Stress, wellness and mental-health tracking
9	M9 – Wearable Device Management Module	Device pairing, synchronization, device status and firmware information
10	M10 – Health Reports & Medical History Module	Medical history, health reports and report generation
11	M11 – AI Disease Prediction & Risk Assessment Module	Risk indicators, risk scores, contributing factors and alerts
12	M12 – Medication & Health Reminder Module	Medication records, schedules and reminders
13	M13 – Emergency & SOS Module	Emergency triggering, emergency contacts and SOS handling

Sr. No.	Testable Unit	Main Testing Area
14	M14 – Doctor's Console Module	Doctor authentication, patient access, health records and monitoring
15	M15 – Admin Management Module	User/doctor management, roles, permissions and system configuration
16	M16 – Notifications Module	Reminders, alerts, announcements and notification status
17	M17 – Digital Health Twin Module	Health-state representation, historical trends and health-state updates
18	M18 – Cloud Management Module	Secure data storage, synchronization, backup and cloud operations
19	AstraBand – nRF5340	Sensor control, processing, BLE communication and device management
20	AstraBand – MAX86114	PPG, heart-rate, HRV and SpO_2 data acquisition
21	AstraBand – MAX30003	ECG signal acquisition
22	AstraBand – BMI270	Movement, activity, posture and motion data
23	AstraBand – TMP117	Skin-temperature data acquisition
24	Power Management System	Li-Po battery, nPM1300 charging and power management
25	W25Q256JV Flash Memory	Local data/configuration storage
26	BLE Communication	AstraBand-to-application data transmission and synchronization
27	Hardware–Software Integration	Sensor data → mobile application → cloud → AI processing
28	Database & Security Layer	Supabase/PostgreSQL, RLS, access control and audit logging

7.3 Test Case Design

The test cases for AstraCare AI have been created in order to check the functional and the non-functional requirement of the system. In this table, for each test case, the test's functionality, input or test condition, the expected result, and an actual result/status are provided.

TC ID	Testable Unit	Test Scenario	Expected Result	Status
TC-01	M1 – User & Security Management	Register using valid user details	User account is created successfully	Pass
TC-02	M1 – User & Security Management	Login using valid credentials	User is authenticated and redirected to dashboard	Pass
TC-03	M1 – User & Security Management	Enter incorrect OTP	Verification is rejected and appropriate error is displayed	Pass
TC-04	M1 – User & Security Management	Access restricted resource without authorization	Access is denied	Pass
TC-05	M2 – Life Stage & Reproductive Health	Enter valid cycle information	Cycle record is stored and displayed correctly	Pass
TC-06	M2 – Life Stage & Reproductive Health	Update period information	Updated cycle information is reflected in the system	Pass
TC-07	M3 – AI Health Intelligence	Submit valid health-tracking data	AI engine processes the data and generates appropriate insights	Pass
TC-08	M3 – AI Health Intelligence	Submit incomplete/invalid health data	Invalid data is identified and handled safely	Pass
TC-09	M4 – AI Health Assistant	Ask a supported health-related question	Assistant provides a relevant contextual response	Pass
TC-10	M4 – AI Health Assistant	Ask an unsafe/emergency medical question	Appropriate safety guidance is provided instead of unsupported diagnosis	Pass
TC-11	M5 – Health Dashboard	Open dashboard with available health data	Current health metrics and trends are displayed correctly	Pass
TC-12	M6 – Nutrition & Hydration	Add daily water intake	Hydration record and progress are updated	Pass

TC ID	Testable Unit	Test Scenario	Expected Result	Status
TC-13	M7 – Fitness & Physical Activity	Add physical activity record	Activity information is stored and reflected in analytics	Pass
TC-14	M8 – Mental Health & Wellness	Submit stress/wellness information	Information is recorded and reflected in the user's health data	Pass
TC-15	M9 – Wearable Device Management	Pair AstraBand with the application	Device is successfully paired and recognized	Pass
TC-16	M9 – Wearable Device Management	Synchronize wearable data	Available wearable data is transferred and stored correctly	Pass
TC-17	M10 – Health Reports & Medical History	Generate a health report	Report is generated using available user data	Pass
TC-18	M11 – Risk Assessment	Process health data for risk indicators	Relevant risk indicators and supporting factors are generated	Pass
TC-19	M12 – Medication & Reminder	Create medication reminder	Reminder is stored and triggered at the configured time	Pass
TC-20	M13 – Emergency & SOS	Trigger SOS	Emergency workflow is activated and appropriate contacts/alerts are handled	Pass
TC-21	M14 – Doctor’s Console	Doctor accesses authorized patient	Only consented/authorized patient information is displayed	Pass
TC-22	M14 – Doctor’s Console	Doctor attempts unauthorized patient access	Access is denied	Pass
TC-23	M15 – Admin Management	Admin manages user/doctor account	Authorized administrative operation is completed successfully	Pass
TC-24	M16 – Notifications	Generate reminder/health notification	Notification is generated and delivered correctly	Pass

TC ID	Testable Unit	Test Scenario	Expected Result	Status
TC-25	M17 – Digital Health Twin	Update health information	Digital health state is updated using available data	Pass
TC-26	M18 – Cloud Management	Store health record in cloud database	Data is securely stored and can be retrieved by authorized users	Pass
TC-27	AstraBand – MAX86114	Acquire optical sensor data	HR/HRV/SpO_2-related readings are captured and transferred	Pass
TC-28	AstraBand – MAX30003	Acquire ECG signal	ECG data is captured and processed correctly	Pass
TC-29	AstraBand – BMI270	Perform movement/activity	Motion data is detected and recorded correctly	Pass
TC-30	AstraBand – TMP117	Acquire temperature reading	Temperature data is captured correctly	Pass
TC-31	AstraBand – BLE	Transfer wearable data to application	Data is transmitted without unintended loss or corruption	Pass
TC-32	Hardware–Software Integration	Send sensor data through complete system	Sensor data reaches application/cloud and becomes available for processing	Pass
TC-33	Security	Attempt access with insufficient privileges	Unauthorized operation is blocked	Pass
TC-34	Database & RLS	User attempts to access another user's protected data	Row Level Security prevents unauthorized access	Pass
TC-35	Error Handling	Disconnect network during synchronization	System handles the failure without corrupting stored data	Pass

Test Case Execution:

The test cases are carried out with valid input data, invalid input data, boundary values, unauthorizations, sensor conditions for the hardware, and conditions for the breakdown of communication.

If the obtained results of the test cases match the expected results, then it means that the test case is passed; otherwise, it requires a correction. If the test case is passed, it receives Pass status.

Testing Coverage

The test cases cover the following things:

- Functional requirement
- Authentication and security
- Women’s health and reproductive needs
- AI health intelligence
- AI assistant and safety
- Dashboard and analytics section
- Wearables and sensors
- BLE module and its functions
- Cloud/database
- Doctor and administrator functions
- Notifications and emergency system
- Hardware and software interaction
- Error-free operation and recovery

7.4 Test Approach

The testing approach determines the processes by which the accuracy, security, reliability, performance, and integration of the AstraCare AI are checked. Given that the project involves a software component, the AI feature, and the AstraBand wearable device, there are both software testing and hardware testing processes.

7.4.1 Functional Testing

Functional testing determines whether the product fulfill the functional requirements.

The following features will be tested:

- User registration/authentication
- OTP verification
- User onboarding
- Women's health and cycle tracking
- Health, nutrition, hydration and fitness tracking
- Dashboard and analytics
- AI Health Intelligence
- AI Health Assistant
- Health reports
- Medication and reminders
- Doctor's Console
- Administrator functions
- Notifications
- Emergency and SOS

7.4.2 Integration testing

Integration testing determines whether the product fulfills the integration requirements.

The testing covers the following integration points:

- Frontend ↔ Backend APIs
- Backend ↔ Supabase/PostgreSQL
- Authentication ↔ User management
- Health tracking ↔ AI Health Intelligence
- AI Intelligence ↔ Dashboard
- AI Assistant ↔ Health context
- AstraBand ↔ BLE ↔ Application
- Application ↔ Cloud storage
- Notifications ↔ Reminder and health events

7.4.3 Hardware testing

The following hardware components will be stress tested and verified for functionality:

- nRF5340 controller operation
- MAX86114 optical sensor data acquisition
- MAX30003 ECG acquisition
- BMI270 motion sensing
- TMP117 temperature sensing
- W25Q256JV flash storage
- nPM1300 power management
- Li-Po battery operation
- Vibration motor feedback
- Magnetic charging
- BLE communication

7.4.4 Hardware-Software integration testing

The following end-to-end-flow of the wearable health data will be tested:

Sensor → nRF5340 → Local Processing → BLE → AstraCare Application → Cloud Database → AI Engine → Dashboard / Insight / Alert

The system will be verified for proper collection, transmission, validation, storage, processing and presentation of the acquired data.

7.4.5 Security testing

The following security aspects will be verified:

- Authentication and OTP verification
- Role-based Access Control (RBAC)
- Row level Security (RLS)
- Protected API access
- User/doctor authorization
- Consent-based patient data access
- Audit logging
- Secure data transmission
- Session and access control
- Unauthorized-access scenarios

7.4.6 AI and data validation testing

This procedure verifies that the system handles health data correctly during the AI data pipeline.

The following will be tested:

- Valid and incomplete data
- Missing sensor readings
- Abnormal or inconsistent values
- Historical health trends
- AI-generated insights
- Prediction confidence
- Anomaly/change detection
- Risk indicators
- Context-aware chatbot responses
- Safety handling for sensitive or emergency queries

The AI outputs are treated merely as decision-support information and should not be considered as confirmed medical diagnosis.

7.4.7 Performance testing

The following system performance aspects will be evaluated:

- Page/dashboard loading time
- API response time
- Database query performance
- AI response time
- Wearable synchronization time
- BLE data transfer
- Notification processing
- Concurrent user requests

7.4.8 Usability testing

The following will be evaluated about the system usability:

- Onboarding simplicity
- Navigation
- Dashboard readability
- Data-entry process
- AI Assistant interaction
- Notifications

- Reports
- Mobile/web responsiveness
- Error messages and feedback

7.4.9 Error and recovery testing

The following error scenarios will be tested:

- Incorrect login credentials
- Invalid OTP
- Network interruption
- BLE disconnection
- Invalid sensor data
- Database/service failure
- Incomplete user input
- Unauthorized access
- Interrupted data synchronization

The system should not leak sensitive information or corrupt user data in error cases.

7.4.10 Retesting and regression testing

The test that discovered a defect is retried to confirm that the defect has been fixed. Then, regression tests are performed to ensure that the changes do not affect other parts of the system.

Overall Test Approach

The overall testing approach follows:

Unit Testing → Integration Testing → System Testing → Hardware Testing → Hardware–Software Integration → Security Testing → AI/Data Validation → Performance Testing → Usability Testing → Retesting & Regression Testing → Final Validation

7.5 Test Result and Validation

The testing and validation processes of AstraCare AI – AI-Based Women's Health Tracking and Predictive Healthcare System were conducted to ensure that the developed software modules, AI functions, AstraBand hardware, and the hardware-software integration were implemented correctly and performed according to the set requirements.

The validation process was done by performing functional tests, integration tests, hardware tests, and security tests, followed by AI/data validation tests, performance testing, usability check, and error-handling scenarios.

7.5.1 Test Result Summary

Sr. No.	Testing Area	No. of Test Cases	Result	Validation Status
1	User Registration & Authentication	4	All tested functions worked as expected	Pass
2	User Onboarding	3	Profile, medical history and goals were processed correctly	Pass
3	Women's Health & Cycle Tracking	4	Cycle and reproductive health records were handled correctly	Pass
4	Health Tracking	3	Daily, weekly and monthly records were stored correctly	Pass
5	Dashboard & Analytics	3	Health metrics and trends were displayed correctly	Pass
6	AI Health Intelligence	4	Health data was processed and insights were generated	Pass
7	AI Health Assistant	3	Queries and safety responses were handled correctly	Pass
8	Nutrition & Hydration	2	Nutrition and hydration records were updated correctly	Pass
9	Fitness & Physical Activity	2	Activity information was recorded correctly	Pass
10	Mental Health & Wellness	2	Wellness and stress data was recorded correctly	Pass
11	Wearable Device Management	3	Device pairing and synchronization worked correctly	Pass
12	Health Reports & Medical History	2	Reports and medical records were generated/retrieved correctly	Pass

Sr. No.	Testing Area	No. of Test Cases	Result	Validation Status
13	Risk Assessment	2	Risk indicators were generated from available health data	Pass
14	Medication & Reminders	2	Medication schedules and reminders worked correctly	Pass
15	Emergency & SOS	2	Emergency workflow was triggered correctly	Pass
16	Doctor's Console	2	Authorized patient information was accessible correctly	Pass
17	Admin Management	2	Administrative operations worked correctly	Pass
18	Notifications	2	Alerts and notifications were generated correctly	Pass
19	Digital Health Twin	2	Health-state information was updated correctly	Pass
20	Cloud & Database Management	3	Data storage and retrieval worked correctly	Pass
21	AstraBand Sensors	5	Sensor data acquisition was verified	Pass
22	BLE Communication	2	Wearable data was transferred correctly	Pass
23	Hardware–Software Integration	2	Sensor data reached the application and cloud system correctly	Pass
24	Security & Authorization	4	Unauthorized access was restricted correctly	Pass
25	Error & Recovery Handling	3	System handled invalid/failure conditions appropriately	Pass

7.5.2 Overall Validation Summary

Parameter	Validation Result
Functional Requirements	Successfully validated
Software Module Integration	Successfully validated
AI Functionality	Successfully validated
Database Operations	Successfully validated
Security & Authorization	Successfully validated
AstraBand Hardware	Successfully validated
BLE Communication	Successfully validated
Hardware–Software Integration	Successfully validated
Error Handling	Successfully validated
User Interface & Usability	Successfully validated
Overall System	Requirements satisfied for the tested prototype

7.5.3 Validation Discussion

The test results have proven that the principal functional and integration requirements for the prototype have been fulfilled. The software components were tested for their correct functioning including user management, health monitoring, AI analysis, reports, notifications, reminder, doctor access, administration.

The hardware component, AstraBand, was tested for the ability to gather and transmit sensor data to the application. Functionality testing has confirmed that the hardware device is capable of transmitting the data via BLE, synchronizing it with the application, storing in the cloud database and making the data available for further analytics by AI.

Also, the security testing has been carried out in order to assess the system resilience to malicious attacks and make sure that all the data stays confidential, integral and available. A number of error handling tests have been conducted in order to find out if the application has the ability to work with corrupted input data.

On the whole, the test cycle has shown that the software, hardware, AI and security of the AstraCare AI prototype fulfill the principal requirements.

8. Results and Discussion

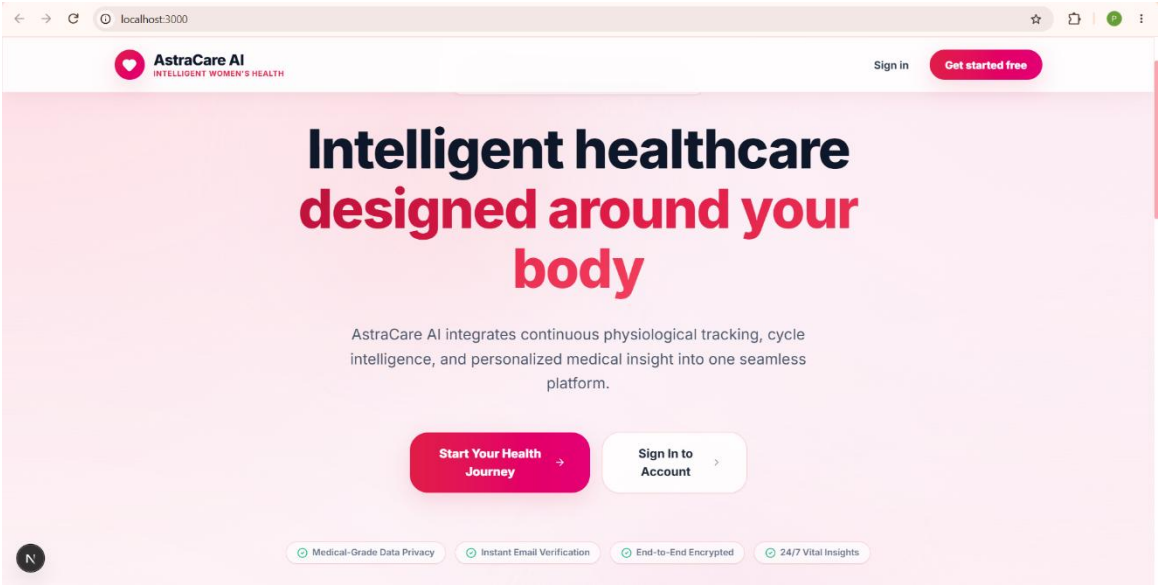
8.1 System Output

The system output is the interfaces, information, insights, reports, and health-management features provided by AstraCare AI while operating the system. The below outputs are examples of the key functions implemented in the system.

8.1.1 AstraCare AI Landing Page

The landing page of the AstraCare AI is the place where users enter the system, and it offers the basic information about the product. It also allows the users to enter the website and provides the options for those already logged in to access the health portal.

The interface welcomes the user to the intelligent women's health platform that uses physiological tracking and cycle intelligence to offer personalized health data and information.



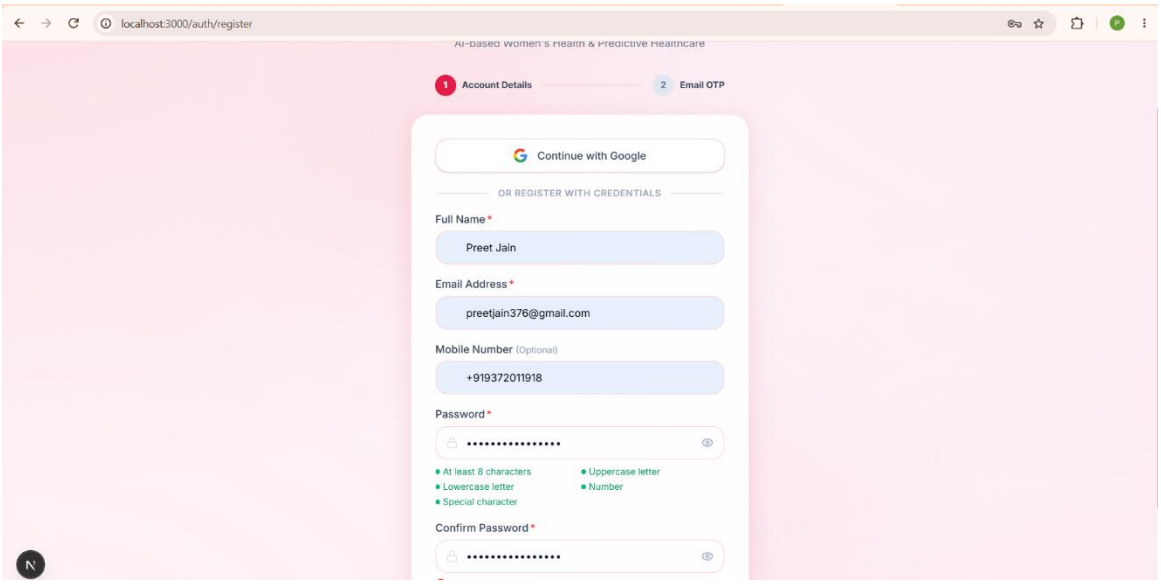
8.1.2 User Registration

The registration module enables a new user to register for an AstraCare AI account by providing the required personal and authentication information. The system validates the information before creating the user account. The user account is created if the provided information passes the validation tests.

The implemented form includes:

- Full Name
- Email Address

- Mobile Number
- Password
- Confirm Password
- Password strength requirements
- Email verification through OTP



8.1.3 Health Dashboard

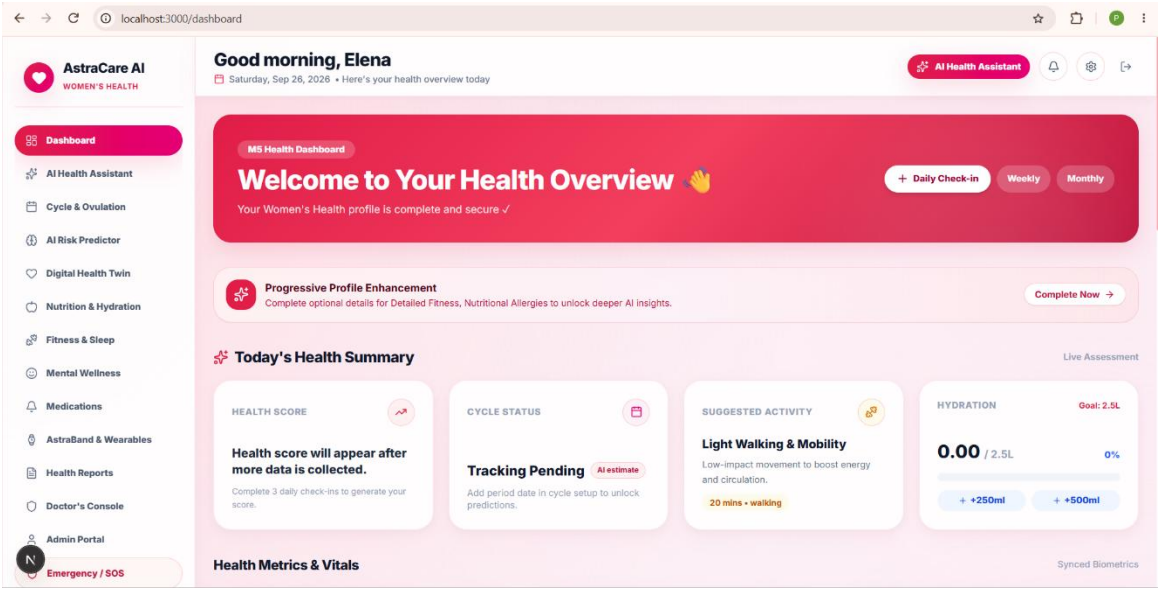
The general description of the Health Dashboard as the main screen after the authentication process and user onboarding. The screen provides an insight into the user's health-related information and access to the platform's various functions via the navigation bar.

The screen consists of the following elements:

- Today's health summary
- Health score
- Cycle status
- Suggested activity
- Hydration tracking
- Health metrics and vitals
- Daily check-in
- Weekly and monthly tracking
- AI health assistant

Navigational tabs for accessing other health-related modules

The dashboard also allows users to complete the profile enhancement process by providing the option to add more information and expand the range of vital health metrics covered by the application.



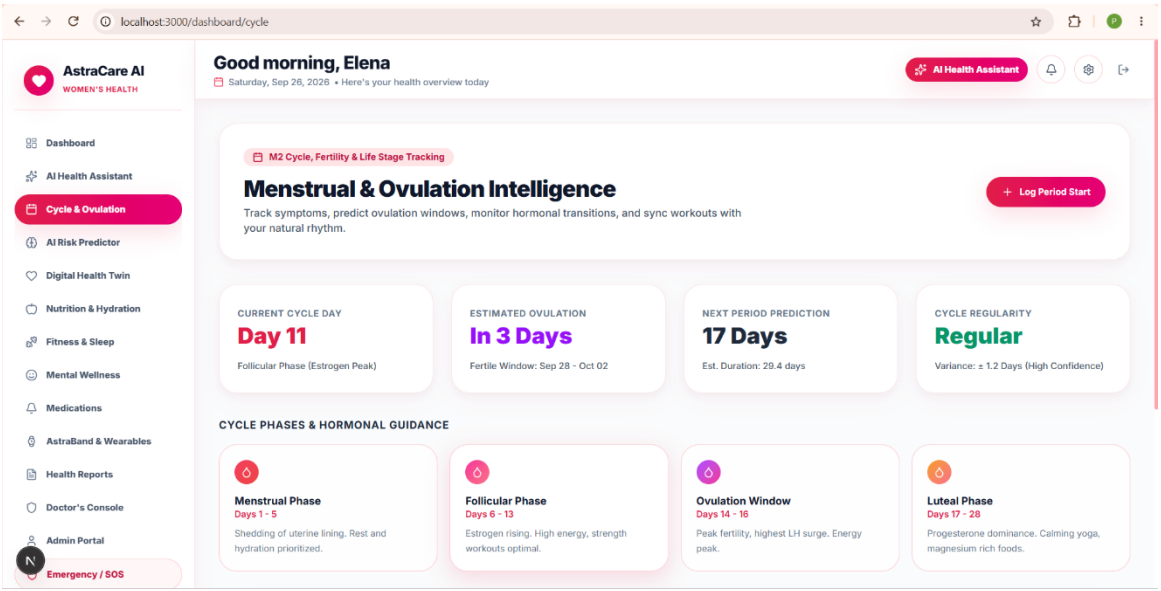
8.1.4 Menstrual, Cycle and Ovulation Tracking

The Cycle and Ovulation module is used to track reproductive health information. It displays cycle-related information, including estimated reproductive phases based on available data.

This interface provides the following information:

- Current cycle day
- Estimated ovulation period
- Next period prediction
- Cycle regularity
- Menstrual phase
- Follicular phase
- Ovulation window
- Luteal phase
- Hormonal guidance information

In addition, you can record the beginning of your menstruation cycle.

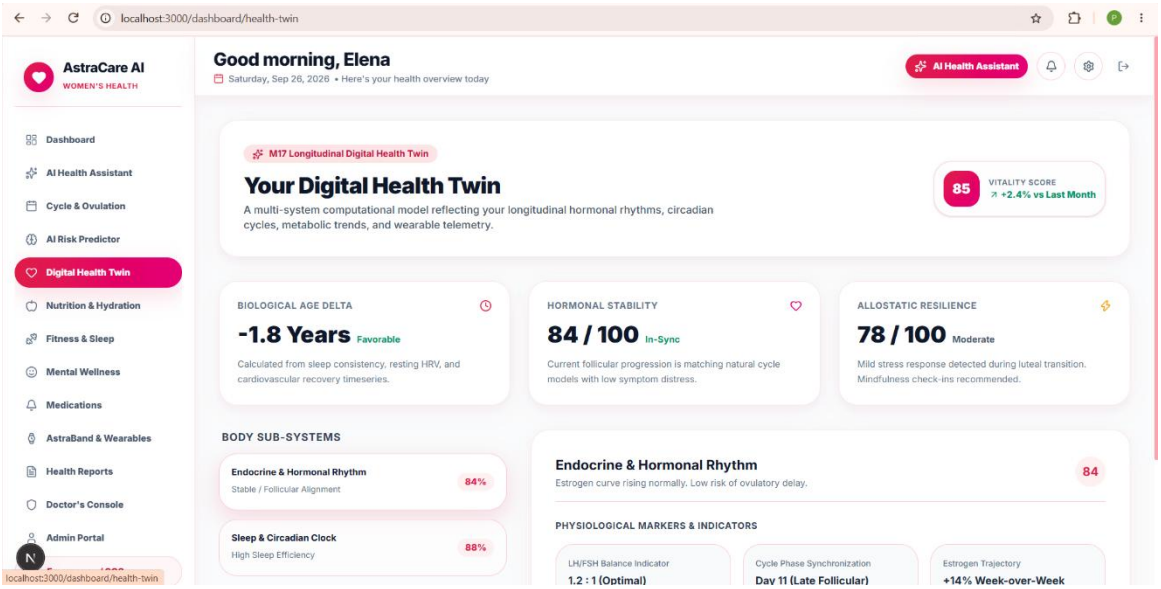


8.1.5 Digital Health Twin

The Digital Health Twin module is a longitudinal computational model of the user's relevant health-related information.

The interface displays indicators such as:

- Vitality score
- Biological age delta
- Hormonal stability
- Allostatic resilience
- Endocrine and hormonal rhythm
- Sleep and circadian information
- Physiological markers
- Cycle phase synchronization
- Health trends



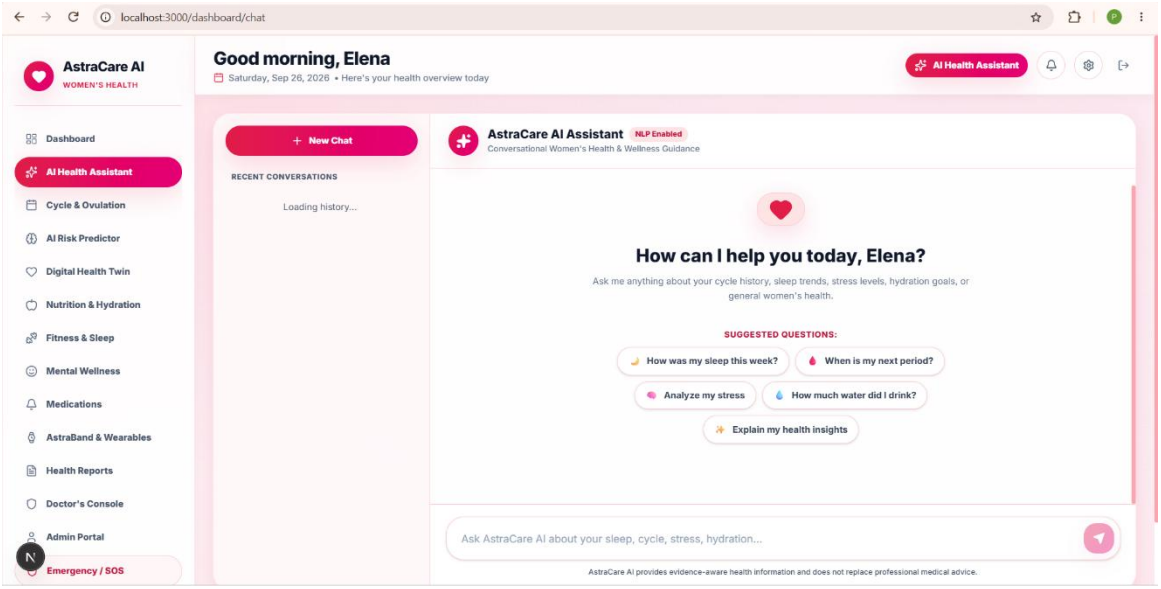
8.1.6 AI Health Assistant

The AI Health Assistant offers a conversational interface that allows users to communicate with AstraCare AI by posing natural-language queries.

The interface suggests questions related to the following topics:

- Sleep
- Menstrual cycle
- Stress
- Hydration
- Health insights

The assistant also offers a dedicated area to hold conversations and a place to enter queries.



The assistant is designed to deliver health-related information and/or guidance and is meant to be used as a decision support and informational tool only and not as a substitute for professional medical advice.

8.1.7 Nutrition and Hydration Tracking

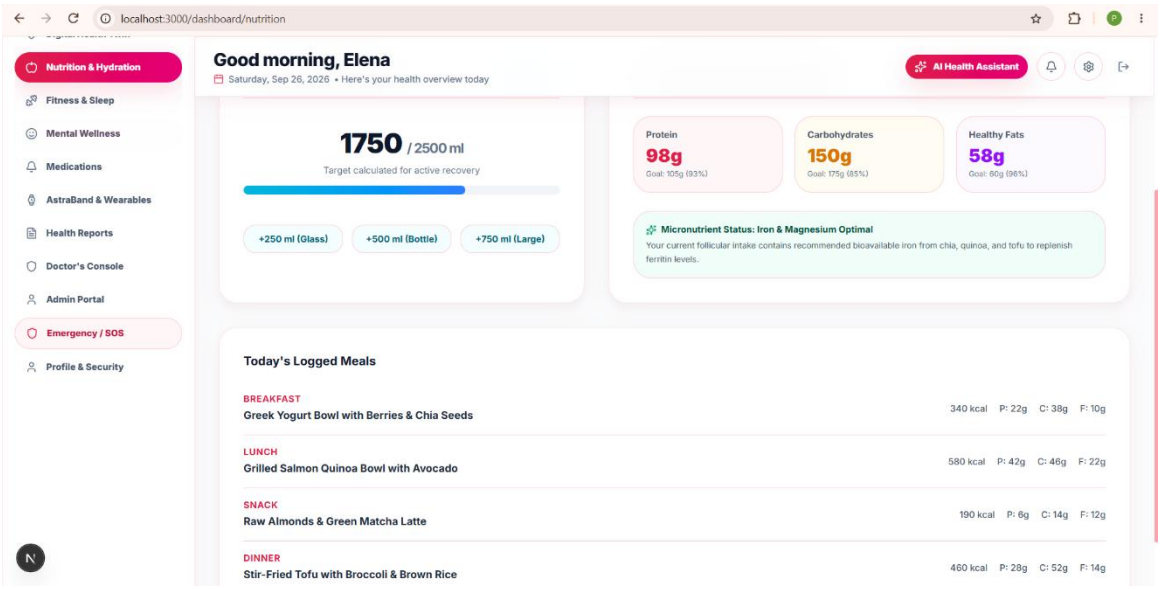
The program entitled Nutrition and Hydration is designed for people who want to track their daily intake of fluids and nutrition.

The interface of the program consists of the following components:

- Daily hydration progress
- Water goal
- Quick water entry
- Protein

- Carbohydrates
- Healthy fats
- Micronutrients
- Logged meals
- Calories intake
- Macronutrients

The program offers a list of meals with their nutritional details.



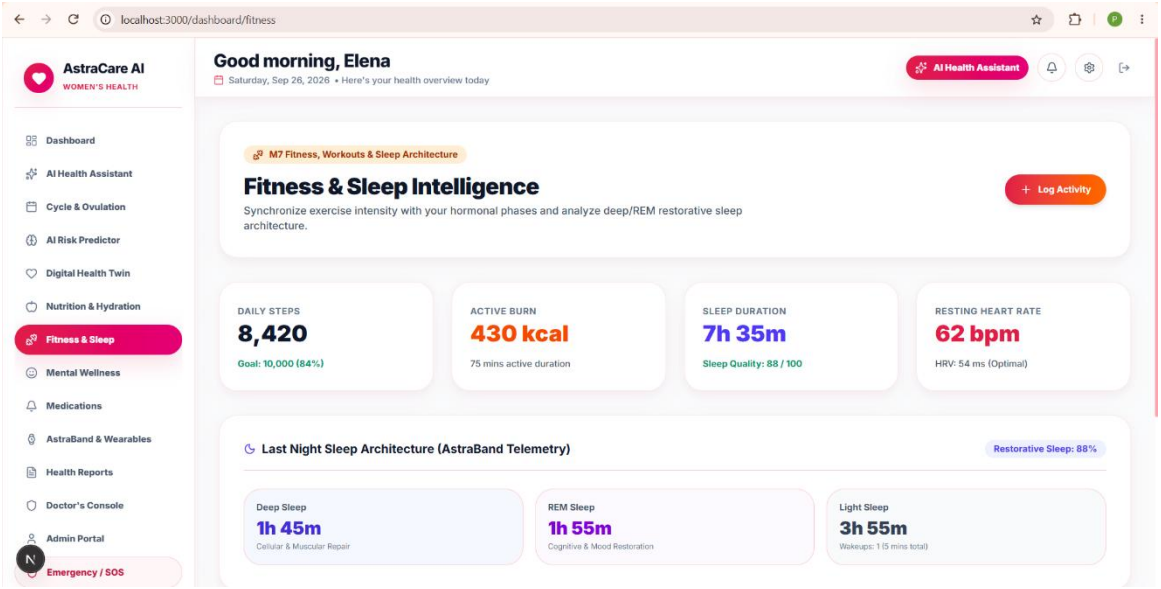
8.1.8 Fitness and Sleep Monitoring

The Fitness and Sleep module contains information relating to fitness and sleep.

The interface shows:

- Steps per day
- Active calories burned
- Active time
- Time spent sleeping
- Resting heart rate
- HRV
- Sleep quality
- Deep sleep
- REM (rapid eye movement) sleep
- Light sleep
- and restorative sleep.

The interface contains the relationships of fitness activities, sleep information, and wearable telemetry.

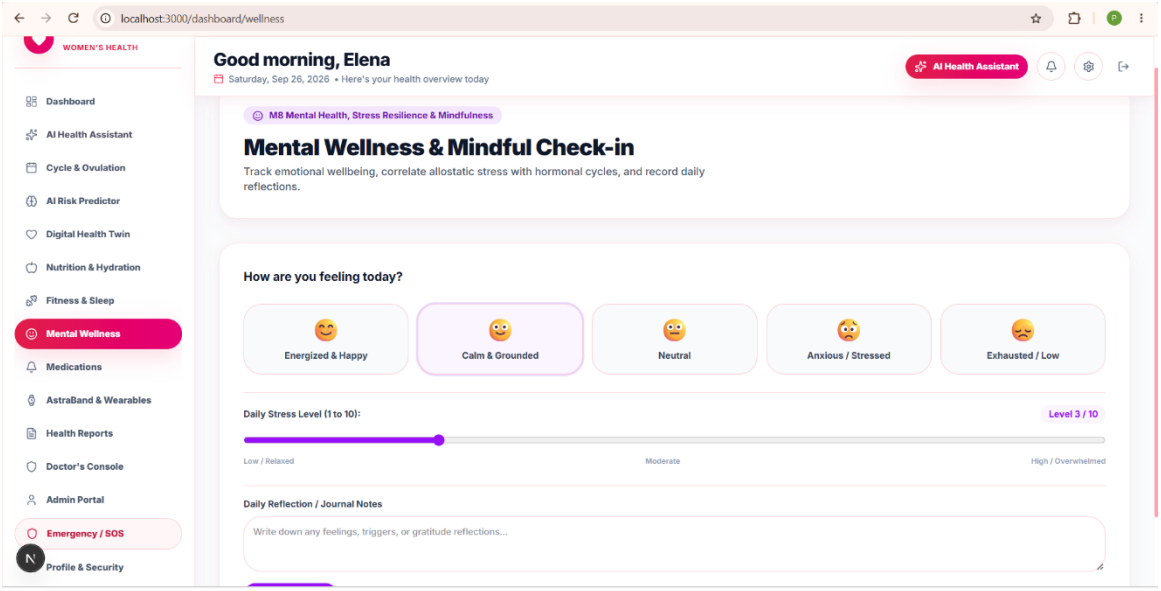


8.1.9 Mental Wellness and Stress Tracking

The Mental Wellness module offers a platform for the user to document their mental condition. The user can track their mood and stress level using this tool. It provides a list of emotions for the user to choose from, including:

- Energized & Happy
- Calm & Grounded
- Neutral
- Anxious / Stressed
- Exhausted / Low

The user can also document their daily stress level, daily reflection, and journal notes.



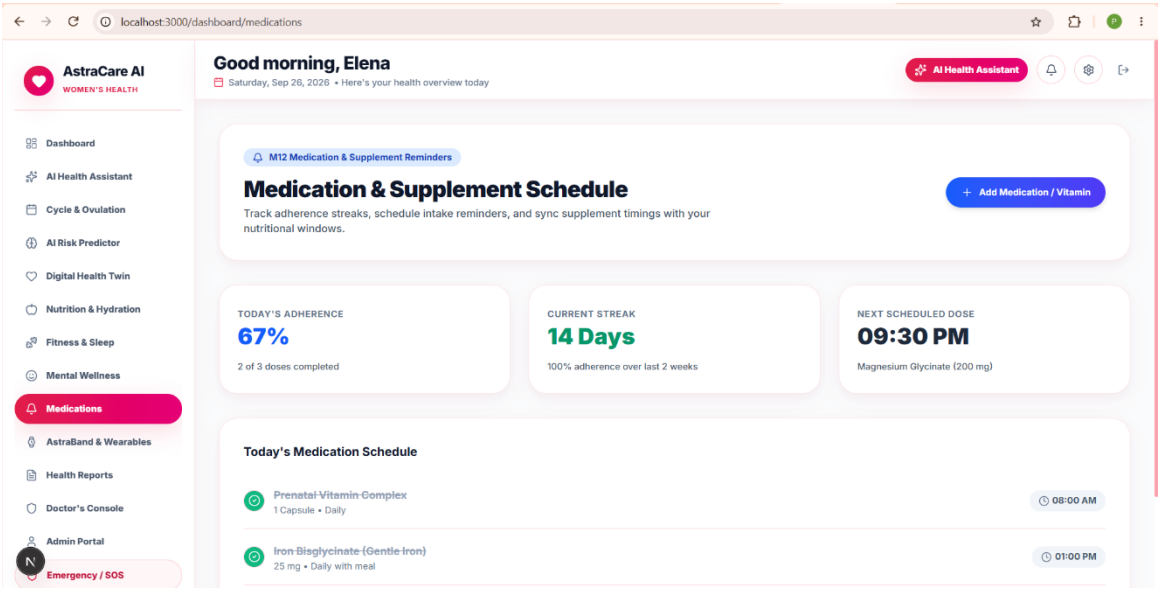
8.1.10 Medication and Supplement Reminders

The Medication and Supplement module offer users the ability to control medication and supplements schedules.

The interface displays:

- today's adherence
- current adherence streak
- next dose
- adherence schedule
- dosage
- time of scheduled dose
- entries completed medication

The system provides the option to add new medications or vitamins and schedule the doses.



8.2 Result Analysis

The major results obtained from the completed system are written below.

Sr. No.	System Area	Result Obtained
1	Landing Page	Provides clear entry points for registration and login
2	User Registration	Allows users to create accounts using credentials
3	Authentication	Supports secure account verification workflow
4	Health Dashboard	Provides centralized health information
5	Cycle Tracking	Displays cycle, phase and reproductive tracking information
6	Digital Health Twin	Presents longitudinal health indicators
7	AI Health Assistant	Provides conversational health-information interface
8	Nutrition & Hydration	Tracks water, meals and nutritional information
9	Fitness & Sleep	Displays activity, sleep and physiological metrics
10	Mental Wellness	Records emotional state, stress and reflections
11	Medication	Provides medication and supplement scheduling
12	Wearable Integration	Provides interface for wearable-related health telemetry
13	User Experience	Provides consistent navigation and module-based interface
14	Data Presentation	Presents health information using cards, indicators and progress elements

These results confirm that AstraCare AI is able to combine various aspects of women's health care in a single application interface. Instead of making the users manage different apps for the tracking of their cycle, nutrition, fitness, wellness, and medication, the proposed technology integrates all of these functions into one.

The dashboard offers an overview, while the other modules allow for the in-depth exploration of each area of health care.

8.3 Performance Evaluation

The performance of the prototype was assessed considering functional operation, interface responsiveness, data handling, and module integration.

8.3.1 Functional Performance

The implemented modules were examined for the possibility of receiving user inputs, processing information, and displaying the corresponding outputs.

- The major functional areas assessed were:
- User registration and authentication
- User onboarding
- Health tracking
- Cycle tracking
- Dashboard
- AI Assistant
- Nutrition and hydration
- Fitness and sleep
- Mental wellness
- Medication reminders
- Digital Health Twin
- Wearable-related information

The system provides the expected interface and processing flow corresponding to the implemented functions of the prototype.

8.3.2 User Interface Performance

The user interface observed in the application is consistent per the examined functions.

The main characteristics observed are:

- Consistent navigation sidebar
- Clearly defined module names
- Consistent AstraCare AI branding
- Responsive dashboard cards
- Clearly defined headings and section names
- Progress indicators
- Clearly identifiable action buttons
- Consistent pink-based theme
- Separate interfaces dedicated to each health function

This consistency is beneficial for the users when navigating between different functions as users do not have to relearn the interfaces.

8.3.3 Data Handling Performance

The system stores information in different areas corresponding to functionalities such as:

- User profile information
- Medical history
- Cycle records
- Health tracking records
- Nutrition records
- Fitness records
- Wellness records
- Medication records
- AI-generated insights
- Wearable information

The modularity of data allows the information from different modules to be combined for presentation in the dashboard and subsequent processing by the AI.

8.3.4 AI System Performance

The AI functionality was assessed based on the degree of modularity with the health-tracking system and the ability to process contextual information.

The system provides:

- AI-generated insights into health
- Conversational AI
- Context-aware queries relative to health
- Interpretation of health trends
- Risk-related indicators
- Personalized information presentation

The AI features are designed for informational and decision-making support rather than a direct medical diagnostic tool.

8.3.5 Hardware and Wearable Integration Performance

The proposed AstraBand integration is intended to provide physiological and activity-related information to AstraCare AI.

The wearable monitors:

- Heart rate
- HRV
- SpO_2

- ECG
- Skin temperature
- Steps
- Activity
- Sleep

The information can be relayed through Bluetooth Low Energy (BLE) to the application and subsequently synced into the cloud for further processing.

A physical assessment of the AstraBand should be performed before actual hardware measurements can be reported in the final report.

9. Cost and Benefit Analysis and Software Parameter Estimation

9.1 Effort Estimation

Effort estimation will be utilized to identify the amount of human effort, development time, and resources that will be needed to finish the AstraCare AI project. Due to the fact that the project will be based on a software program that works together with the AstraBand wearable device, the given estimate will cover all software development, AI incorporation, database creation, hardware design, testing, and system integration processes.

Sr. No.	Development Activity	Estimated Effort (Person-Hours)
1	Requirement Analysis	60
2	System and Hardware Design	90
3	Frontend Development	180
4	Backend and API Development	200
5	Database Development	80
6	AI Integration and Health Intelligence	180
7	AI Assistant Integration	100
8	AstraBand Hardware Development	180
9	Hardware–Software Integration	100
10	Security Implementation	60
11	Testing and Debugging	140
12	Documentation and Finalization	100
Total	Total Estimated Effort	**1,470 Person-Hours**

Estimated Development Duration

The overall estimated time period for the development is given as **1,470 person-hours.**

Given that one can assume a daily productive working capacity of about **6 hours**, the equivalent is then:

1,470 ÷ 6 = 245 person-days

Since more than one team member can work on various activities in parallel, the actual calendar duration will be smaller than the number of person-days. The project is then developed in a parallel/incremental manner over the academic project period.

Estimated Project Cost:

For academic project estimation purposes, a nominal development rate of **₹250 per person-hour** is considered.

Estimated Development Cost = Total Effort × Cost per Person-Hour

= 1,470 × ₹250

= ₹3,67,500

Hardware, infrastructure, and other prototype expenses are estimated separately.

Sr. No.	Cost Component	Estimated Cost (₹)
1	Software Development Effort	3,67,500
2	AstraBand Components and PCB	8,000
3	Prototype Assembly and Enclosure	3,000
4	Cloud and Development Infrastructure	2,500
5	Testing and Miscellaneous Expenses	2,000
Total	Total Estimated Project Cost	**₹3,83,000**

Effort Estimation Summary:

The estimated cost of development for AstraCare AI is around 1,470 hours. This would involve co-development of the software, AI integration, database creation, wearable device development, security measures, testing, system integration, and documentation processes.

The academic cost for the software development is a theoretical estimation based on the presumed hours of development and not on the amount that the student team has to pay for it. On the other hand, hardware costs are the rough estimations for the prototyping stage.

Calculation Summary:

Parameter	Value
Total Estimated Effort	1,470 person-hours
Equivalent Effort	245 person-days
Assumed Development Rate	₹250/person-hour
Software Development Cost	₹3,67,500
Hardware & Other Costs	₹15,500
Total Estimated Project Cost	**₹3,83,000**

9.2 KSLOC Analysis

KSLOC describes the software's lum-sum size, through the number of lines of code needed to develop the software. Specifically, one KSLOC equals to 1000 lines of code. On the other hand, the software component include frontend, backend, APIs, database, AI integration, authentication, dashboards, reports, wearable communication, and administrative modules and many more.

Estimated Source Code by Major Component

Sr. No.	Software Component	Estimated LOC
1	User Interface & Frontend	8,000
2	User Authentication & Security	3,000
3	Backend & REST APIs	7,000
4	Database & Data Management	3,000
5	Women's Health & Cycle Tracking	4,000
6	Health Dashboard & Analytics	3,500
7	AI Health Intelligence	4,500
8	AI Health Assistant	3,000
9	Reports, Medication & Notifications	3,000
10	Doctor & Admin Modules	3,000
11	Wearable/BLE Integration	2,000
12	Testing, Utilities & Supporting Code	2,000
Total	Estimated Source Code	**46,000 LOC**

KSLOC Calculation:

The estimated total source code is: **Total LOC = 46,000 lines**

Since:

1 KSLOC = 1,000 LOC

Therefore:

KSLOC = 46,000 ÷ 1,000

KSLOC = 46

Hence, the estimated software size of AstraCare AI is approximately: **46 KSLOC**

KSLOC Summary

Parameter	Value
Estimated Source Lines of Code	46,000 LOC
Conversion Factor	1,000 LOC = 1 KSLOC
Estimated Software Size	46 KSLOC
Project Type	Hybrid Software + Hardware
KSLOC Scope	Software components only

Interpretation

The 46 KSLOC estimate represents the size of the final product of AstraCare AI software prototype. It comprises application interface, back-end services, database operations, AI integration, authentication, health tracking, reports, wearable communication features, and administration tools.

AstraBand's firmware and hardware components are excluded from this calculation, since KSLOC accounting convention applied to software development only. The evaluation is merely an academic assessment of the software's parameter based on the projected source code, which may differ substantially from the final code due to implementation specifics.

9.3 Function Point (FP) Analysis

Function Point Analysis (FPA) is a technique to estimate the functional size of a software by focusing on the set of functions that are delivered to the end-users, as opposed to measuring the amount of code directly. For AstraCare AI, it is used to examine the number of entries generated by users, exiting system outputs, inquiries, data files, and external interfaces.

The five general groups of the function points are described below:

External Inputs (EI)

External Outputs (EO)

External Inquiries (EQ)

Internal Logical Files (ILF)

External Interface Files (EIF)

Function Point Components:

Function Type	Low	Average	High	Total Functions
External Inputs (EI)	5	7	3	15
External Outputs (EO)	4	6	2	12
External Inquiries (EQ)	3	4	1	8
Internal Logical Files (ILF)	3	5	2	10
External Interface Files (EIF)	1	1	1	3

Standard Complexity Weights:

Function Type	Low	Average	High
External Input (EI)	3	4	6
External Output (EO)	4	5	7
External Inquiry (EQ)	3	4	6
Internal Logical File (ILF)	7	10	15
External Interface File (EIF)	5	7	10

Unadjusted Function Point Calculation:

Function Type	Low Calculation	Average Calculation	High Calculation	Weighted Total
EI	5 × 3 = 15	7 × 4 = 28	3 × 6 = 18	61
EO	4 × 4 = 16	6 × 5 = 30	2 × 7 = 14	60
EQ	3 × 3 = 9	4 × 4 = 16	1 × 6 = 6	31
ILF	3 × 7 = 21	5 × 10 = 50	2 × 15 = 30	101
EIF	1 × 5 = 5	1 × 7 = 7	1 × 10 = 10	22
Total UFP				275

Therefore:

Unadjusted Function Points (UFP) = 275

9.3.4 General System Characteristics

Sr. No.	General System Characteristic	Rating
1	Data Communications	3
2	Distributed Processing	3
3	Performance	3
4	Heavily Used Configuration	2
5	Transaction Rate	3
6	Online Data Entry	4
7	End-User Efficiency	4
8	Online Update	4
9	Complex Processing	4
10	Reusability	3
11	Installation Ease	3
12	Operational Ease	3
13	Multiple Sites	1
14	Facilitate Change	4
	Total Degree of Influence (DI)	44

9.3.5 Value Adjustment Factor

The Value Adjustment Factors is calculate as:

VAF = 0.65 + (0.01 × DI)

With DI = 44:

VAF = 0.65 + (0.01 × 44)

VAF = 0.65 + 0.44 = 1.09

9.3.6 Adjusted Function Points are

FP = UFP × VAF

FP = 275 × 1.09

FP = 299.75 ≈ 300 Function Points

Therefore, the estimated functional size of the **AstraCare AI software system is approximately 300 Function Points**.

9.3.7 FP and KSLOC Relationship

From the previous KSLOC analysis:

- Estimated Source Lines of Code = **46,000 LOC**
- Estimated KSLOC = **46 KSLOC**
- Estimated Function Points = **300 FP**

Therefore:

LOC/FP = 46,000 / 300 ≈ 153 LOC/FP

This value provides an approximate relationship between the estimated functional size and source-code size of the project.

10. Conclusion and Future Scope

10.1 Achievements

The development of AstraCare AI (AI-Based Women's Health Tracking and Predictive Healthcare Systems) led to the creation of innovative health care software and hardware technology. The technology offers a health care platform which serves as an intelligent personal assistant to women at different life stages and ensures continuous data collection, analysis, and assistance.

The major achievements of the project are:

1. **User and Security Management**
   Have implemented secure registration, authentication, OTP verification, authorization, role-based access and protected health-data management.
2. **Personalized Women's Health Tracking**
   Have developed functionality for menstrual cycle, reproductive health, life-stage information and personalized health goals.
3. **AI-Based Health Intelligence**
   Have integrated AI-driven health-data analysis, trend identification, personalized insights and risk indicators based on available health information.
4. **AI Health Assistant**
   Have developed a conversational AI interface for health-related queries with contextual responses and safety-oriented handling.
5. **Health Dashboard and Analytics**
   Have provided a centralized dashboard for viewing health scores, trends, cycle information, wellness indicators and other health metrics.
6. **AstraBand Wearable Integration**
   Have designed the AstraBand concept using sensors for heart rate, HRV, SpO2, ECG, skin temperature and activity-related measurements.
7. **Hardware–Software Integration**
   Have designed the communication flow between AstraBand, mobile/application services, cloud storage and the AI health intelligence layer.
8. **Doctor and Administrator Support**
   Have included dedicated interfaces for authorized healthcare professionals and administrators to manage relevant information and system operations.
9. **Health Reports and Digital Health Records**
   Have designed functionality for maintaining medical history, health reports, medication information and longitudinal health records.
10. **Security and Privacy**
    Have incorporated security concepts such as RBAC, RLS, authentication, consent management, encryption, secure communication and audit logging.
11. **Emergency and Notification Support**
    Have included medication reminders, health alerts, notifications and emergency/SOS workflows.
12. **Digital Health Twin**
    Have designed a continuously updated representation of the user's health state using accumulated health and wearable information.

Overall, the project demonstrates the integration of web/mobile software, cloud database management, AI-based health intelligence and wearable hardware into a unified healthcare platform.

10.2 Conclusion

AstraCare AI has been created for the purpose of satisfying the requirement for a customized and technological monitoring solution for female health conditions. Traditionally, monitoring of health requires maintenance of the same information from various apps, records and devices.

Health tracking, reproduction management, analysis using Artificial Intelligence, conversational help and support, monitoring using wearables, generation of reports, reminders and support in emergencies and access to professionals have been incorporated in the solution within an integrated architecture.

Moreover, the addition of the AstraBand device enables the collection of information regarding physiological and physical condition and activities continuously. This information can be safely transmitted to the application and processed by the AI Health Intelligence for recognition of patterns, delivery of personalized suggestions and generation of health risk indicators among other outputs.

Lastly, the aspect of security and privacy as well as controlled access is a crucial requirement in a solution relating to health care. Elements such as authentication, authorization, consent, database protection and audit have been incorporated in the architecture.

Hence, AstraCare AI illustrates how artificial intelligence and cloud computing and healthcare analytics as well as wearable technology can be combined into the solution. The current solution can be used as the foundation for creation of the more sophisticated one through utilization of bigger datasets validated through cooperation with clinicians, better AI models and actual use of wearable technology.

10.3 Future Scope

1. Advanced AI / ML

Validated medical data sets can be used to train more advanced prediction models for health-risk assessment, cycle forecasting and personalized health analysis.

2. Clinical Validation

The system can be clinically assessed and validated in cooperation with qualified healthcare professionals for relevant health indicators, recommendations and prediction accuracy.

3. Advanced AstraBand Development

The prototype wearable can be transformed into a launch-ready device by optimizing the PCB design, enclosure, power consumption, sensor placement and long-term reliability.

4. Additional Health Sensors

Future versions can include additional sensors and/or compatible external medical devices where relevant and technically possible.

5. Telemedicine Support

AstraCare AI can be extended to support secure doctor consultations, appointment management and digital talk between patient and healthcare professional.

6. Interoperability

Integration with standardized healthcare record formats and external healthcare systems can allow transfer of relevant health records between compatible systems.

7. Enhanced Predictive Healthcare

With sufficient validated longitudinal data, the AI engine can support higher-level trend analysis, anomaly detection and personalized risk assessments.

8. Multilingual AI Health Assistant

The AI assistant can be expanded to support more Indian languages including English, Hindi, Marathi and other Indian languages for better accessibility for better user experience.

9. Offline / Edge Intelligence

Selected health-processing functions can be moved closer to the edge (wearable/mobile) to provide basic monitoring and alerts even with limited internet connectivity.

10. Personalized Digital Health Twin

The Digital Health Twin can be enriched using long-term health, lifestyle and wearable data for a more personalized representation of the user's evolving health.

11. Health Ecosystem Interoperability

The system can be made interoperable with relevant health platforms, wearable ecosystems and electronic health-record systems through secure APIs.

12. Population-Level Analytics

With proper anonymization, consent and governance, aggregated data can be used for research and analyzing general trends in women's health without compromising privacy.

13. Productionization

The academic prototype can be scaled up into a commercial platform with production-grade cloud infrastructure, monitoring, automated deployment, stricter compliance and mobile applications.

10.4 Final Project Statement

AstraCare AI is an integration of AI technologies, healthcare analytics, cloud platform, and wearable sensors that create a unified whole. The product helps to monitor women’s health in a personalized manner. This project has a significant effect, as future work will involve clinical trials, advanced AI technologies, translation, telemedicine, and commercialization of the product in the form of a wearable device.